Adjuvant cytokine-induced killer cell immunotherapy in hepatocellular carcinoma: Extended follow-up of a randomized controlled trial and post-treatment immune cell profiling.

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Background: Notwithstanding the pressing need for adjuvant therapy for hepatocellular carcinoma (HCC), most adjuvant therapies, including atezolizumab/bevacizumab, have failed. Meanwhile, adjuvant immunotherapy utilizing autologous cytokine-induced killer (CIK) cells for HCC improved recurrence-free survival (RFS) in a previously reported randomized controlled trial (RCT) and real-world data. This study aimed to assess the longer-term outcomes of the RCT and elucidate the underlying mechanisms of sustained effects of CIK cell treatment. **Methods:** This study comprised two parts: a long-term follow-up of the preceding RCT and an analysis of immune cells in patients who received adjuvant CIK cell treatment. In the original RCT, 226 patients who underwent curative treatment for stage I or II HCC were randomly allocated to either the CIK (n=114, 16 injections of 6.4×10^9 CIK cells over an 11-month period) or control group (n=112). The follow-up period was extended to 9 years after the enrollment of the last patient. The primary endpoint was recurrence-free survival (RFS). The secondary endpoints included cancer-specific survival (CSS) and overall survival (OS). Parallelly, a prospective study was conducted to investigate post-treatment changes of immune cells in peripheral blood of 7 patients, who received repeated transfer of CIK cells after curative treatment for HCC, using flow cytometry. **Results:** In the extended follow-up of the RCT (median follow-up=115.7 months, interquartile range=74.2–130.5 months), the CIK group sustained a significantly prolonged RFS (median=43.5 vs 27.4 months; hazard ratio [HR] =0.74, 95% confidence interval [CI]=0.55–0.99, *P*=0.045) and CSS (median=unreached; HR=0.49, 95% CI=0.25–0.95, P=0.04) compared to the control group. Adjuvant CIK cell therapy reduced the risk of overall death by 30%, although it did not achieve statistical significance (median=unreached; HR=0.70, 95% CI=0.44–1.11, P=0.1). A preliminary analysis of peripheral blood immune cells revealed that the CIK cell treatment tended to increase the frequencies of $CD8^+$ and $CD4^+$ classical memory cells. **Conclusions:** Adjuvant CIK cell treatment demonstrated significantly enhanced RFS and CSS in the prolonged follow-up of the RCT extending to 9 years in patients who received curative treatment for HCC. The CIK group also demonstrated a consistent trend of improved OS. The increase in memory T cell populations might be linked to the sustained off-treatment anti-tumor efficacy of CIK cell treatment. Research Sponsor: None.

Adjuvant chemotherapy or chemo-radiation in gallbladder cancer: A phase III randomized controlled study (ACCELERATE trial).

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Background: Role of adjuvant therapy in gallbladder cancer is still evolving and no prospective trial has compared chemotherapy (ChT) alone to combination of chemotherapy and chemo-radiation (CRT). We designed this trial to answer whether adding CRT to ChT improves relapse free survival. **Methods:** In this open-label, multicentric, phase 3, non-inferiority academic trial, operated gallbladder cancers (R0 or R1) patients were randomized to physicians choice of ChT alone (either 6 cycles of mGemOx- Gemcitabine 900 mg/m2 and oxaliplatin 80 mg/m2 IVI days 1 and 8 every 3 weeks or GemCis- Gemcitabine 1000 mg/m2 and cisplatin 25 mg/m2 1 and 8 every 3 weeks) or 3 cycles of physicians choice of ChT (as above) followed by CRT (radiation 45Gyin 25 fractions in 5 weeks with concurrent oral capecitabine in the dose of 825 mg/m2 twice a day on days of radiation and further 2-3 cycles of mGemOx or GemCis. The primary endpoint was the relapse free survival. Planned sample size was 100 subjects in each arm. **Results:** Between April 2018 and January 2021, 137 patients were screened, and 94 eligible patients were randomized, 49 (52.1%) in chemotherapy alone arm (standard-arm1) and 45 (47.9%) in chemotherapy plus chemo-radiation arm (experimental-arm 2). Slow accrual led to premature closure of trial. COVID pandemic might have contributed to that. Baseline characteristics were well balanced (like, sex, presenting symptoms, ECOG PS, duration of symptoms, comorbidities, location of tumour in GB, presence of gallstone disease, and tumour markers were well balance in both groups) except that a greater number of patients had deranged baseline LFTs in arm1 and there was a trend towards higher numbers of stage IIA in arm 1. The median age was 55 years (range 27-73 years). Females constituted 32 in each group. 43 and 45 patients had ECOG PS of 0-1 in arm 1 and arm 2 respectively. One patient in arm 1 was ineligible. All had R0 resection. Incidences of dose reductions and dose delays were similar. A greater number of patients in arm 1 experienced diarrheal episodes (p=0.021) and peripheral neuropathy (0.001). 42 (85.7%) patients in arm 1 and 28 (62.2%) patients in arm completed 5-6 cycles. 18 (36.73%) and 23 (51.11%) died till last follow up. 14 (28.57%) and 20(44.44%) died because of disease progression. One patient in each arm died of toxicity. The primary end point of study, relapse free survival was not estimable in arm 1 and was 34.39 months in arm 2 (p=0.202). Median overall survival was not estimable in arm 1 and was 34.56 months in arm 2 (p=0.123). Mean RFS was 51.96 Vs 43.99 months in arm 1 and arm 2 respectively. **Conclusions:** This trial suggests that addition of CRT to ChT doesn't improve outcome in resected gallbladder cancer compared to ChT alone. A larger trial is needed to address this issue. Clinical trial information: CTRI/2018/04/013218. Research Sponsor: None.

Nivolumab (NIVO) plus ipilimumab (IPI) vs lenvatinib (LEN) or sorafenib (SOR) as first-line (1L) therapy for unresectable hepatocellular carcinoma (uHCC): CheckMate 9DW expanded analyses.

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Background: In the phase 3 CheckMate 9DW study (NCT04039607), 1L NIVO + IPI demonstrated significant overall survival (OS) benefit vs LEN/SOR, higher objective response rate (ORR) with durable responses, and manageable safety in uHCC. We present efficacy by best overall response (BOR) subgroups and baseline characteristics, and additional safety analyses from the preplanned interim analysis. **Methods:** Patients (pts) with previously untreated HCC not eligible for curative surgical or locoregional therapies, Child-Pugh score 5 or 6, and ECOG performance status 0 or 1 were randomized 1:1 to receive NIVO 1 mg/kg + IPI 3 mg/kg Q3W (up to 4 cycles), then NIVO 480 mg Q4W or LEN 8 mg or 12 mg QD or SOR 400 mg BID until disease progression or unacceptable toxicity. NIVO was given for a maximum of 2 years. The primary endpoint was OS; secondary endpoints included ORR and duration of response (DOR) per blinded independent central review (BICR) using RECIST v1.1. **Results:** A total of 668 pts were randomized to NIVO + IPI (n = 335) or LEN/SOR (n = 333). At a median follow-up of 35.2 (range 26.8–48.9) months (mo), median OS (95% CI) was 23.7 (18.8–29.4) mo with NIVO + IPI vs 20.6 (17.5–22.5) mo with LEN/SOR (HR 0.79 [95% CI 0.65–0.96]; $P = 0.0180$). ORR (95% CI) per BICR was significantly higher with NIVO + IPI vs LEN/SOR (36% [31–42] vs 13% [10–17]; $P < 0.0001$); median DOR (95% CI) was 30.4 (21.2–not estimable [NE]) mo vs 12.9 (10.2–31.2) mo. Survival benefit of NIVO + IPI vs LEN/SOR was observed across BOR subgroups at the 24-week landmark timepoint (Table). In subgroup analyses, ORR (95% CI) per BICR was higher with NIVO + IPI vs LEN/SOR across HCC etiologies (uninfected: 35% [26–44] vs 8% [4–15]; HBV infected: 25% [17–34] vs 17% [10–25]; HCV infected: 50% [39–61] vs 16% [9–25]) and in pts with Barcelona Clinic Liver Cancer stage ≤B (33% [23–43] vs 13% [6–21]) or stage C (37% [31–44] vs 14% [10–19]). Safety data are shown in the Table. Additional exploratory analyses will be presented. **Conclusions:** These additional analyses from CheckMate 9DW demonstrate the efficacy and manageable safety of 1L NIVO + IPI in uHCC and further support its use as a potential standard-of-care treatment option in this setting. Clinical trial information: NCT04039607. Research Sponsor: Bristol Myers Squibb.

OS by BOR at week 24 landmark	NIVO + IPI			LEN/SOR		
BOR	CR + PR (n = 101)	SD[a] (n = 105)	PD (n = 47)	CR + PR (n = 28)	SD[a] (n = 212)	PD (n = 31)
Median OS (95% CI), mo	NR (44.4–NE)	30.0 (23.5–37.8)	16.0 (12.0–18.7)	28.3 (20.6–NE)	22.5 (20.5–24.8)	13.5 (8.7–25.3)

All treated pts	NIVO + IPI (n = 332)	LEN/SOR (n = 325)
Any-grade/grade 3–4 TRAEs, n (%)	278 (84)/137 (41)	297 (91)/138 (42)
Hepatobiliary	44 (13)/35 (11)	15 (5)/10 (3)
Cardiovascular	10 (3)/3 (< 1)	138 (42)/39 (12)
Hemorrhagic	2 (< 1)/1 (< 1)	20 (6)/5 (2)

[a]Includes non-CR/non-PD. CR, complete response; NR, not reached; PD, progressive disease; PR, partial response; SD, stable disease; TRAE, treatment-related adverse event.

521 Rapid Oral Abstract Session

A multicenter prospective study to evaluate the efficacy of resection for initially unresectable hepatocellular carcinoma after atezolizumab combined with bevacizumab (the RACB study): Short-term outcomes.

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Background: The IMbrave150 study showed that atezolizumab plus bevacizumab (atezo+bev) is superior to sorafenib in progression-free survival and overall survival, making it the recommended first-line systemic therapy for unresectable hepatocellular carcinoma (HCC). The RACB study was scheduled to evaluate the efficacy of atezo+bev in achieving conversion surgery for initially unresectable HCC. This time, we report the short-term outcomes of the RACB study. **Methods:** This prospective, single-arm, multicenter Phase II trial evaluates atezo+bev to achieve conversion surgery in patients with technically or oncologically unresectable HCC. Eligible patients had unresectable HCC without prior systemic chemotherapy and at least one target lesion based on RECIST ver1.1. Patients received atezolizumab (1200 mg) plus bevacizumab (15 mg/kg) every three weeks. If resectable on radiological assessment at 12 weeks, atezolizumab monotherapy was given followed by surgery. The primary endpoint was the progression-free survival. The secondary endpoints included the overall response rate, overall survival, resection rate, curative resection rate, on-protocol resection rate, and ICG retention rate at 15 min after atezo+bev therapy. The study is registered with the Japan Registry of Clinical Trials (jRCTs051210148). **Results:** Between March 2022 and March 2024, 55 patients from 17 centers were enrolled. Four patients were excluded due to ineligible lesions or other reasons. A total of 50 patients were included in the efficacy evaluation set. Macrovascular invasion and extrahepatic metastases were observed in 38 (76.0%) and 6 patients (12.0%), respectively. Two patients (4.0%) had metachronous extrahepatic metastases without intrahepatic tumors. The complete and partial response rates were 0% and 13% based on RECIST, and 2.2% and 26.1% based on mRECIST. The overall resection rate was 48%, corresponding to 24 cases. R0, R1, and R2 resections were performed in 21 (87.5%), 1 (4.2%), and 2 (8.3%) patients, respectively. Complications ≥Grade III according to the Clavien-Dindo classification included 3 bile leakages, 2 infections, 1 anastomotic leakage, and 1 hemoptysis. There were no surgery-related deaths. Of the 51 patients included in the safety analysis set, adverse events occurred in 34 cases (66.7%), with 16 Grade 3 or higher instances with one Grade 4 upper gastrointestinal bleeding, 1 Grade 4 encephalopathy, and 1 unexpected death after 1 cycle of atezo+bev. **Conclusions:** The RACB study's short-term outcomes demonstrated the feasibility of conversion surgery after atezo+bev in patients with initially unresectable HCC. The response rate of atezo+bev, as measured by mRECIST, was similar to IMbrave150, with a 48% resection rate, though careful monitoring of severe adverse events is essential. Clinical trial information: 051210148. Research Sponsor: CHUGAI PHARMACEUTICAL CO., LTD.

LBA522 **Rapid Oral Abstract Session**

Transarterial chemoembolization (TACE) combined with camrelizumab and apatinib versus TACE alone in the treatment of unresectable hepatocellular carcinoma eligible for embolization: A multicenter, open-label, randomized, phase 2 study (CAP-ACE).

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The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.

523 Poster Session

Racial disparities in secondary metastasis in patients with hepatocellular carcinoma: A 5-year National Inpatient Sample study.

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Background: Hepatocellular carcinoma (HCC), a primary liver malignancy, is one of the leading causes of cancer-related mortality worldwide. The incidence and outcomes of HCC vary significantly across racial groups. Secondary metastasis, a key factor in disease progression and prognosis, also appears to differ by race, though these disparities remain poorly understood. This study aims to explore how race influences the metastatic progression of HCC, providing valuable insights into potential racial differences in disease outcomes. **Methods:** A retrospective analysis of National Inpatient Sample (NIS) data from 2016 to 2020 was conducted, identifying adults with HCC using International Classification of Diseases, Tenth Revision (ICD-10) codes, and categorizing them based on the presence of metastasis. The primary outcome was racial differences in organ metastasis. Multivariable logistic regression models were used to adjust for potential confounders and to evaluate the impact of race on all metastatic patterns, with Caucasians as the reference category. Statistical significance was defined as p-values ≤ 0.05. **Results:** The study cohort included 236,985 adult patients with HCC, of whom 34,875 had secondary metastasis. The cohort was predominantly male (77.37%), with 22.63% female. The racial distribution was 52.61% Caucasian, 19.08% African American, 15.04% Hispanic, and 13.27% from other racial groups. After adjusting for confounders, African American patients had significantly higher odds of lung (aOR 1.42, $P<0.001$), bone (aOR 1.51, $P<0.001$), and brain metastasis (aOR 1.47, P=0.02), but lower odds of adrenal gland metastasis (aOR 0.74, P=0.003) compared to Caucasians. Asian/Pacific Islander patients demonstrated increased odds of lung (aOR 2.00, $P<0.001$) and brain metastasis (aOR 1.95, P=0.02). In contrast, Hispanic patients had lower odds of abdominal lymph node (aOR 0.69, P=0.008) and bone metastasis (aOR 0.83, P=0.007). **Conclusions:** This study highlights significant racial disparities in the metastatic patterns of hepatocellular carcinoma, emphasizing the need to consider these differences in clinical management and treatment strategies for HCC patients. Further research is required to investigate the underlying causes of these disparities and their impact on long-term survival. Research Sponsor: None.

Gender and ethnic disparities in hepatocellular carcinoma: Trends in incidence rates across U.S. populations from 2000 to 2021.

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Background: Hepatocellular carcinoma (HCC) is a primary malignancy of the liver strongly associated with liver cirrhosis and the etiologies producing cirrhosis. As such, racial and ethnic factors might affect incidence rates of HCC resulting in disparities. While disparities have been reported, the effect of gender within certain populations, particularly Hispanic and non-Hispanic White ancestry, have not been fully explored. This study looks at the changing incidence rates of HCC in the US amongst different populations with a particular focus on the impact of gender on incidence. **Methods:** Incidence rates of HCC between 2000 and 2021 among different sexes within racial and ethnic groups in the United States were analyzed utilizing delay-adjusted data from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program. Annual percent change (APC) was calculated by using Joinpoint Regression software. **Results:** Overall, HCC incidence rates since 2014 to 2021 show a slight decrease (APC, -0.06). While Hispanic men HCC incidence rates declined from 2012 to 2021 (APC, -0.44), Hispanic women incidence have had a steady increase since 2000 (APC, +2.52). Non-Hispanic Black men (APC, -3.07) rates vastly declined from 2015 to 2021 while non-Hispanic black women interestingly experienced a slight decrease in incidence from 2013 to 2021 (APC, -0.34). Non-Hispanic White men experienced stagnant rates of HCC since 2014 (APC, -0.05) while non-hispanic white women (APC, +3.59) showed increasing trends in HCC incidence since 2003. For comparison, non-Hispanic Asian/Pacific Islander men (APC, -2.19) and women (APC, -2.36), which both experienced a decline since 2007. This analysis was repeated without significant change. Further corroborations are necessary to validate these findings. **Conclusions:** These findings show that while the incidence rates of HCC overall have decreased, this benefit appears to be largely driven in male populations. In contrast, in women, and particularly Hispanic and Non-Hispanic White women, incidence rates have been increasing. This phenomenon likely reflects the changes in etiologies of cirrhosis and HCC. Surveillance strategies and screening might need to be optimized to reflect these trends.
Research Sponsor: None.

525 Poster Session

Mortality trends and racial disparities in hepatocellular carcinoma associated with metabolic syndrome in the United States and Texas: A 21-year analysis (1999-2020) using CDC WONDER database.

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Background: The prevalence of hepatocellular carcinoma(HCC) and metabolic syndrome (MetS) is increasing globally. This study aims to examine annual trends of mortality in MetS-associated HCC in the United States and its state of Texas, from 1999 to 2020, to evaluate public health initiatives and identify socioeconomic factors. **Methods:** Mortality trends for adults aged ≥25 years with MetS-associated HCC were analyzed using CDC WONDER database, identifying deaths through ICD-10 codes C22.9 "Liver Malignant neoplasms" and E88.9 "Metabolic Syndrome". Crude and age-adjusted mortality rates (AAMRs) per 100,000 people were extracted. Annual percent changes (APCs) in AAMRs, with 95% confidence intervals, were determined across various demographic (sex, race/ethnicity, age) and geographic (state, urban-rural, regional) subgroups. **Results:** Between 1999 and 2020, 271,046 documented deaths were attributed to HCC due to metabolic syndrome. The AAMR for MetS-related HCC mortality increased in the US from an adjusted rate (AR) (3.7) in 1999 to (4.5) in 2009 (APC: 2.10%; 95% CI: -0.13% to 3.10%), after which it increased to (7.6) in 2018 (APC:5.83%; 95% CI: 4.25% to 6.99%) and increased to 9.9 in 2020 (APC:13.19%; 95% CI:8.78% to 15.7%). In Texas, AAMR for MetS-related HCC mortality increased from AR (4.9) in 1999 to (8.9) in 2015 (APC: 3.95%; 95% CI:3.11% to 4.69%) after which it increased to (15.3) in 2020 (APC:10.64%; 95% CI: 8.18% to 17.33%). Men had consistently higher AAMRs than women (12.4 vs. 8). The AAMR in U.S. men increased from (5.1) in 1999 to (12.4) in 2020. The AAMR in U.S. women increased from (2.7) in 1999 to (8) in 2020. The non-Hispanic (NH) American Indian or Alaska Native population has the greatest AAMR (13.4), followed by the NH Black or African American(AA) (12.7) and the Hispanic or Latino population with AAMR (10.9). The low-risk population was NH White with AAMR (9.6) and NH Asian or Pacific Islander (6.6). The AAMR also varied by region (South:11.5; Midwest:9.9; West:9.4; Northeast:7.4) and non-metropolitan areas had higher AAMR (small metro:12.1; non-core areas:11; micropolitan areas:11.7) than metropolitan areas (large central metropolitan: 8.8; large fringe areas: 8.7). The states in the upper 90th percentile of MetS-related HCC AAMRs were Texas, Louisiana, Arkansas, Michigan, Nevada exhibited an approximately two-fold increase in AAMRs, compared to states falling in the lower 10th percentile Nebraska, Wisconsin, Washington, Maine. **Conclusions:** Mortality rates from hepatocellular carcinoma associated with metabolic syndrome have risen in the United States and Texas over the past two decades. However, significant demographic and geographic disparities in MetS-related HCC mortality continue, emphasizing the need for further investigation and the development of targeted treatment strategies. Research Sponsor: None.

Racial, gender and socioeconomic disparities in survival outcomes of hepatocellular carcinoma.

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Background: Hepatocellular carcinoma (HCC) is the sixth most commonly occurring cancer worldwide, with mortality rates increasing globally, particularly in the Western world. The 5-year survival rate of persons with HCC is less than 20%. It is, therefore, essential to investigate the extent to which social determinants of health influence survival outcomes. In this study, we explored the effect of gender, race/ethnicity and socioeconomic factors on survival outcomes among patients with HCC in the United States. **Methods:** We extracted data on patients with a histologic diagnosis of hepatocellular carcinoma from the Surveillance, Epidemiology and End Result (SEER) Registry from 2000-2021. The primary outcome of interest was the cause-specific survival of these patients. According to the International Classification of Diseases for Oncology, Third Edition (ICD-O-3) system, the histological code of HCC is 8170/3. We excluded patients of missing or unknown race, income, location, and patients outside the years of study. Cox proportional hazard regression (SPSS) was utilized to determine the association between patients' race/ethnicity, gender, annual median income, location and cause-specific survival. **Results:** 80,445 met the criteria and were included in our study. Gender proved to be a predictor of survival; compared to females, males had worse outcomes, with males HR 1.040 (95% CI; 1.020-1.060, p = <0.001). Race was a predictor of survival outcomes. Using American Indian/Alaska Native as our reference, Asian/Pacific Islander HR 0.847 (95% CI; 0.790-0.90, p = <0.001), Black HR 1.019 (95% CI; 0.950-1.093, p = 0.603), White HR 0.955 (95% CI; 0.893-1.023, p = 0.186). We also analyzed the effects of socioeconomic factors on survival. Patients with higher annual median income had better outcomes. >$120000 HR 0.817 (95% CI; 0.783-0.851, p = <0.001) while patients with lower annual median income had worse outcomes < $40000 HR 1.217 (95% CI; 1.134-1.305, p < 0.001). The effect of location on survival was not significant. Compared to non-metropolitan areas, metropolitan areas HR 0.988 (95% CI; 0.971–1.004, p = 0.138). **Conclusions:** Race/ethnicity, income and gender disparities all played a significant role in cause-specific survival in patients with HCC, with improved survival observed in females, Asians and patients with higher annual median income. More efforts are needed to bridge these survival disparities. Research Sponsor: None.

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Ethnic and socioeconomic disparities in cancer-free and all-cause survival in cholangiocarcinoma patients under 60: A population-based study.

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Background: Cholangiocarcinoma in individuals under 60 years of age is rare, and survival outcomes can vary based on ethnic and socioeconomic factors. While previous studies have examined these disparities in other cancers, little is known about their impact on cancer-free and all-cause survival in younger cholangiocarcinoma patients. This study investigates the role of race and income in survival outcomes for this population. **Methods:** Data for individuals under 60 diagnosed with cholangiocarcinoma were extracted from the Surveillance, Epidemiology, and End Results (SEER) 22 registry. Kaplan-Meier curves were constructed to assess overall and cancer-free survival. Median survival times were calculated for each racial and income group, and log-rank tests were used to compare survival differences across the groups. **Results:** A total of 11,214 individuals under 60 years of age diagnosed with cholangiocarcinoma were included. Median all-cause survival was 12.0 months for Asian patients, 11.0 months for White patients, and 9.0 months for both Black and Hispanic patients. Significant differences were observed between Asian vs White ($p < 0.005$), Asian vs Black ($p < 0.005$), Asian vs Hispanic ($p < 0.005$), White vs Black ($p < 0.005$), and White vs Hispanic ($p < 0.005$), with no significant difference between Black and Hispanic patients ($p = 0.859$). For cancer-free survival, the median was 13.0 months for Asians, 12.0 months for Whites, 11.0 months for Blacks, and 10.0 months for Hispanics, with significant differences between Asian vs Black ($p < 0.005$), Asian vs Hispanic ($p < 0.005$), and White vs Black ($p = 0.017$). The difference between Asian vs White was not statistically significant ($p = 0.052$), and no significant difference was observed between White vs Hispanic ($p = 0.418$) or Black vs Hispanic ($p = 0.418$). By income, patients with an income >$75,000 had significantly longer all-cause survival (12.0 months vs 9.0 months, $p < 0.005$) and cancer-free survival (13.0 months vs 11.0 months, $p < 0.005$) compared to those with an income of $0-$75,000. **Conclusions:** Significant disparities in cancer-free and all-cause survival were found based on ethnicity and income. Asian and White patients had longer survival compared to Black and Hispanic patients, and higher-income patients experienced better outcomes. These findings highlight the need for targeted efforts to address healthcare inequalities in cholangiocarcinoma care. Research Sponsor: None.

Disparities in liver cancer risk and outcomes in the LGBTQ community: A literature review.

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Background: Liver cancer continues to be a significant public health concern, particularly among high-risk groups such as individuals with hepatitis infections, alcohol use disorders, and metabolic diseases. While liver cancer disparities have been well-documented among racial and ethnic minority groups, emerging research highlights significant health inequities within the LGBTQ community, including higher rates of risk factors and limited access to screening and care. This literature review aims to assess the extent of liver cancer disparities within the LGBTQ community and identify gaps in prevention, screening, and treatment. **Methods:** We performed a systematic search across PubMed, Scopus, and Google Scholar for peer-reviewed studies published between 2000 and 2024. The inclusion criteria focused on studies addressing liver cancer risk, prevalence, or outcomes in LGBTQ populations. Data on key liver cancer risk factors such as hepatitis B and C prevalence, alcohol use, obesity, and access to healthcare services were extracted and synthesized. Additionally, we reviewed public health reports and clinical guidelines relevant to LGBTQ health disparities. **Results:** The review identified significant disparities in liver cancer incidence and mortality among LGBTQ individuals. Hepatitis B infection rates were about 1.5 times higher, and hepatitis C rates up to 1.7 times higher among LGBTQ populations, particularly among gay and bisexual men. Alcohol use disorder was about 30-40% more prevalent in LGBTQ individuals, which further increases the risk of liver cancer. Screening for liver cancer in LGBTQ populations is underutilized, with rates approximately 20% lower than the general population. There were reported delays in diagnosis in LGBTQ community due to stigma, discrimination, and lack of culturally competent healthcare services. While some interventions targeting LGBTQ health have shown promise, the overall lack of targeted liver cancer prevention and treatment strategies remains a significant barrier. **Conclusions:** LGBTQ individuals face increased liver cancer risk due to a higher prevalence of key risk factors and inadequate healthcare access. Current screening and preventive strategies are insufficient for addressing these disparities. Targeted interventions, including enhanced screening efforts, hepatitis vaccination and treatment programs, and the development of LGBTQ-specific healthcare protocols, are needed to reduce the liver cancer burden in this underserved population. Policymakers and healthcare providers must address the unique needs of LGBTQ individuals to achieve equitable cancer outcomes. Research Sponsor: None.

Prevalence of liver cancer risk factors in LGBTQ vs. general population.

Risk Factor	LGBTQ Population (%)	General Population (%)
Hepatitis B Infection	15	10
Hepatitis C Infection	18	12
Alcohol Use Disorder	25	16
Metabolic Syndrome	22	18
Screening Rate	40	60

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Hepatocellular carcinoma: A 30-year trend analysis of global burden stratified by risk factors, and socio-demographic indexes.

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Background: Hepatocellular carcinoma (HCC) remains a leading cause of cancer-related morbidity and mortality worldwide, driven primarily by chronic infections with Hepatitis B, Hepatitis C, and alcohol consumption. This study systematically examines global trends of HCC from 1991 to 2021, focusing on age-standardized mortality rates (ASMR) and Disability-Adjusted Life Years (DALYs) according to risk factors across Socio-Demographic Index (SDI) regions. By highlighting these patterns, the study provides insights into the interplay of socio-economic disparities and healthcare advancements on HCC trends. **Methods:** We utilized data from the Global Burden of Disease database to extract ASMR, and age standardized DALYs for HCC between 1991 and 2021. The data was stratified by etiology (Hepatitis B, Hepatitis C, alcohol consumption) and stratified across five SDI regions: low, low-middle, middle, high-middle, and high SDI. **Results:** The global HCC ASMR decreased by 4% from 1991 to 2021, with the largest reduction (20.4%) in low SDI regions. Hepatitis B-related HCC ASMR dropped by 16.3%, with reductions across all SDI groups. ASMR of Hepatitis C-related HCC remained stable globally but fell by 15.1% in low SDI regions. Alcohol-related HCC ASMR increased by 10% globally, with the highest rise in high SDI regions (24.5%), and NASH-related HCC ASMR increased by 26.7% globally, with a 42% rise in high SDI regions. Hepatitis B-related HCC DALYs decreased by 22.2%, with the sharpest declines recorded in high (31.4%) and low SDI regions (29%). Hepatitis C-related HCC DALYs saw a 8.8% global reduction, which was consistent across all groups. Alcohol-related HCC DALYs rose by 4.6% globally, with increases of 14.4% in high SDI and 20.5% in low-middle SDI regions. NASH-related HCC DALYs increased by 18.7%, with a 34.7% rise in low-middle SDI regions. **Conclusions:** The study reveals substantial progress in reducing the global burden of Hepatitis B- and C-related HCC, particularly in low SDI regions. This progress likely reflects advances in vaccination, antiviral treatments, and screening programs. However, the rising mortality and disability associated with alcohol- and NASH-related HCC, especially in high SDI regions, emphasize the growing impact of lifestyle-related risk factors. These findings underscore the need for tailored public health strategies to mitigate the increasing effects of these preventable risk factors on global HCC outcomes. Research Sponsor: None.

Non-contrast MRI as a surveillance tool for recurrent hepatocellular carcinoma after curative treatment: A prospective multicenter intra-individual comparison trial.

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Background: Hepatocellular carcinoma (HCC) often recurs following curative treatment, significantly impacting long-term survival. Contrast-enhanced multiphasic computed tomography (CECT) is widely utilized for HCC recurrence surveillance but carries risks such as radiation exposure and adverse reactions to contrast agents. This prospective multicenter trial aimed to compare the accuracy of non-contrast abbreviated magnetic resonance imaging (NC-MRI) with CECT in detecting late recurrent HCC. **Methods:** This prospective multicenter intra-individual head-to-head comparison trial (study identifier: NCT05690451, KCT0006395) enrolled participants who had undergone curative treatment for HCC and remained recurrence-free for over two years. Participants underwent three rounds of surveillance at 2–6-month intervals using both CECT and NC-MRI. The primary outcome was the detection accuracy of each modality, compared using the McNemar test. Secondary outcomes included detection sensitivity and specificity. **Results:** A total of 203 participants underwent 528 rounds of paired CECT and NC-MRI, detecting recurrent HCC in 22 participants (10.8%). Among these cases, 21 showed intrahepatic recurrent HCC with a median tumor size of 1.3 cm, and one exhibited aortocaval lymph node metastasis. NC-MRI demonstrated a detection accuracy of 96.6% (196/203), significantly higher than CECT's 91.6% (186/203) (P=0.006). NC-MRI also exhibited significantly higher sensitivity in detecting recurrent HCC compared to CECT (77.3% [17/22] vs. 36.4% [8/22]; P=0.012), while specificity did not differ significantly between NC-MRI and CECT (98.9% [179/181] vs. 98.3% [178/181]; P=0.999). **Conclusions:** NC-MRI showed significantly higher sensitivity and accuracy in detecting recurrent HCC among patients who had been disease-free for more than two years after curative treatment, establishing it as the preferred surveillance modality for late recurrence. Clinical trial information: NCT05690451.
Research Sponsor: the Ministry of Health & Welfare, Republic of Korea; HA21C0163.

	Contrast enhanced CT		Non-contrast MRI		Difference, % (95% CI)	P-value
	Estimates, % (n/N)	95% CI, %	Estimates, % (n/N)	95% CI, %		
Accuracy	91.6% (186/203)	87.8-95.5	96.6% (196/203)	94.0-99.1	4.9 (1.7, 8.2)	0.006
Sensitivity	36.4% (8/22)	14.5-58.2	77.3% (17/22)	58.3-96.3	40.9 (16.8, 65.0)	0.012
Specificity	98.3% (178/181)	96.5-100	98.9% (179/181)	97.4-100	0.6 (-0.5, 1.6)	0.999

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MRI imaging and machine learning based radiomics for detection of mixed HCC and CCA tumors: Still a need for liver biopsy?

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Background: Primary liver cancer (PLC), comprising hepatocellular carcinoma (HCC) and cholangiocarcinoma (CCA), is a leading cause of cancer mortality globally. The combined hepatocellular cholangiocarcinoma (cHCC-CC) subtype may be less common but is relevant to treatment efficacy. We therefore evaluated the diagnostic accuracy of various approaches in distinguishing these liver cancers. **Methods:** Patients diagnosed with HCC, CCA, and cHCC-CC at Beijing University Cancer Hospital and Institute, China were included. Radiologists of varying expertise independently assessed MRI scans, and we measured their diagnostic consistency. Radiomic features were extracted from MRI scans, and machine learning was applied to differentiate the cancer types. **Results:** Standard imaging was insufficient to reliably characterize cHCC-CC. Abdominal imaging experts (AIEs) had a higher mean sensitivity for HCC and CCA, 88% and 84% respectively, while non-experts (NIEs) had a lower sensitivity of 50% for HCC and 38% for CCA (HCC: p=0.03, CCA: p=0.008). Radiomic analysis found 'Sphericity' and 'ClusterShade' as the most relevant features. However, radiomics algorithms were also not sufficient to distinguish cHCC-CC from either HCC or CCA. Regarding sensitivity, the radiomic-based model was not better than radiologists for any of the three classes (p=0.065 for HCC, p=0.426 for CCA, and p=1.0 for cHCC-CC). The random forest algorithm yielded an accuracy of 76% in the test set, since it correctly classified most HCC and CCA, while only one quarter of cHCC-CC tumors. **Conclusions:** Until improved diagnostic tools are available, biopsy of liver cancer remains critical to the detection, diagnosis, and effective treatment of these cancers.
Research Sponsor: Sino-German Center for Research Promotion; GZ-1546 and C-0012; State Ministry of BadenWuerttemberg for Sciences, Research and Arts supporting the Clinical Co-operation Unit Healthy Metabolism at the Center for Preventive Medicine and Digital Health; CCU Healthy Metabolism; Foundation for Biomedical Alcohol Research; Baden-Wuerttemberg Center for Digital Early Disease Detection and Prevention; BW- ZDFP; Federal Ministry of Science and Education of Germany; Q-HCC, 01KD2214.

First line atezolizumab + bevacizumab (atezo/bev) or durvalumab ± tremelimumab (durva±treme) in unresectable hepatocellular carcinoma (HCC): A real world, multi-institutional retrospective cohort study.

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Background: HCC accounts for 80% of primary liver cancers, and worldwide, is the third common cause of cancer-related death. Two FDA approved 1st line systemic treatment options include atezo/bev and durva ± treme. Herein we report outcomes of these regimens in the 1st line setting via a real-world, multi-institutional retrospective cohort study. **Methods:** We reviewed clinical outcomes of patients (pts) who initiated 1st line atezo/bev or durva ± treme for unresectable HCC between March 2017 and February 2024, across 5 institutions (Mayo Clinic, University of Alabama, Cedars-Sinai Medical Center, University Hospitals Seidman Cancer Center and University of Michigan). Time to treatment failure (TTF) and overall survival (OS) were analyzed using the Kaplan-Meier method. Best (objective) response was tabulated. Multivariable analyses were performed using Cox models and logistic regression for time-to-event and binary outcomes, respectively, while adjusting for age, sex, race, HCC etiology, Child-Pugh class, and ALBI grade. **Results:** A total of 433 HCC pts were included; atezo/bev (336), durva ± treme (97). Median age at the start of therapy was 68 years; 77% were male, 82% were Caucasian. Etiologies of HCC were viral hepatitis (38%), MASLD (20%), alcohol (10%), non-viral multifactorial (10%), or unknown (22%). Pts receiving durva ± treme were less likely to have cirrhosis (11% vs. 28%, $p<0.01$) but similar in terms of macrovascular invasion (63% vs. 58%, $p=0.35$) compared to pts receiving atezo/bev. Reasons for treatment discontinuation were similar between the two groups, with the most common being disease progression - atezo/bev (56%) and durva ± treme (47%). The differences in clinical outcomes were not statistically significantly and remained consistent after multivariable adjustment. (Table, atezo/bev as the reference group). Outcomes were significantly different between Child-Pugh classes (CP: A,B,C) respectively, OS: 19.0 m, 6.4 m, 1.9 m ($p<0.001$); TTF: 6.1 m, 2.7 m, 1.3 m ($p<0.001$). **Conclusions:** In this real-world retrospective cohort study of unresectable HCC, the differences in overall survival, objective response, and time to treatment failure were not statistically significant between atezo/bev and durva ± treme in the 1st line setting. Research Sponsor: None.

Outcome	Atezo/bev	Durva±treme	Univariate p-value	Multivariable adjusted hazard ratio (95% CI)	Multivariable adjusted odds ratio (95% CI)
Overall Survival, Median (month)	14.0	14.6	0.77	0.78 (0.54-1.14)	–
Time to Treatment Failure, Median (month)	4.9	4.1	0.71	0.95 (0.72-1.26)	–
Objective response, %	26.3%	21.1%	0.31	–	0.78 (0.43-1.39)

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Development and validation of the SPEAR scoring model and predicting overall survival in unresectable hepatocellular carcinoma patients treated with TACE, molecular targeted therapies, and immune checkpoint inhibitors.

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Background: To develop and validate a predictive model to identify hepatocellular carcinoma (HCC) patients likely to benefit from combined treatment of transarterial chemoembolization (TACE), molecular targeted therapies (MTTs), and immune checkpoint inhibitors (ICIs). **Methods:** Data from 500 patients with unresectable HCC treated with TACE, MTTs, and ICIs (2019-2023) were analyzed. Patients were divided into a training set (241 from Nanfang Hospital) and a validation set (259 from various hospitals). Key prognostic variables were identified using random forest survival analysis, and the top five were used to construct a Cox proportional hazards model. Model performance was assessed using time-dependent AUC values, calibration curves, clinical decision curves (DCA), and net reclassification improvement (NRI) indices. Risk scores and optimal cut-off values for risk stratification were determined and validated. **Results:** Five key survival variables were identified: serum alpha-fetoprotein (AFP), red cell distribution width coefficient of variation (RDW-CV), aspartate aminotransferase (AST), platelet-to-lymphocyte ratio (PLR), and extrahepatic metastasis (EHM). The Cox model had a C-index of 0.694 in the training and 0.691 in the validation set. The training set's 1-year, 2-year, 2.5-year, and 3-year AUC values were 0.77, 0.73, 0.73, and 0.72, respectively, and 0.74, 0.74, 0.76, and 0.66 in the validation set. The ARAPE model demonstrated a more significant net benefit than existing models. Median overall survival (mOS) was 33.9 months for low-risk, 20.4 months for intermediate-risk, and 14.2 months for high-risk groups in the training set; in the validation set, mOS was 30.8, 18.7, and 10.5 months, respectively. **Conclusions:** The SPEAR model (Serum AFP, PLR, EHM, AST, and RDW-CV) model effectively stratifies risk for HCC patients receiving TACE combined with MTTs and ICIs, aiding in personalized treatment decisions. Research Sponsor: National Key Research and Development Program of China; National Natural Science Foundation of China; 82102879; National Natural Science Foundation of China; Natural Science Foundation of Guangdong Province; Science and Technology Projects in Guangzhou.

Characterization of the real-world first line (1L) use of single dose tremelimumab regular interval durvalumab (STRIDE), for treatment of unresectable hepatocellular carcinoma (uHCC), using two electronic medical record (EMR) databases.

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Background: In October 2022, the US FDA approved the STRIDE (Single Dose Tremelimumab Regular Interval Durvalumab) regimen, comprising tremelimumab 300 mg for 1 dose plus durvalumab 1500 mg every 4 weeks, for the 1L treatment of uHCC in adults aged ≥18 years. Thus far there is limited data characterizing STRIDE's 1L use in the real world. This observational study aims to describe patient characteristics, and clinical profiles of a 1L STRIDE cohort. **Methods:** A retrospective cohort study was conducted using EMR data from two community oncology practice networks, Florida Cancer Specialists and American Oncology Network, from January 1, 2018, to March 31, 2024. Adults with HCC who received ≥1 administration of the STRIDE regimen as 1L treatment following diagnosis were included. Patients with evidence of other cancers or systemic therapies pre-index date were excluded, to establish a cancer-naïve cohort at index. Index date was first administration of STRIDE. The analysis described the patient characteristics and clinical profile at index date. Patient demographics, severity indicators (cancer stage, albumin-bilirubin index (ALBI), alpha fetoprotein (AFP), Eastern Cooperative Oncology Group (ECOG)) and lab data were assessed on or closest to index date. **Results:** 79 patients initiated the 1L STRIDE regimen. The median (P25, P75) age at index was 70 (63.2-76.4) years. 72.2% were males, 64.6% were white, and 69.6% were overweight or obese patients. 95% had their first HCC diagnosis between 2022-2024 and the mean time from diagnosis to index date was 3.8 months (SD: 10.2). Among patients with reported cancer stage (n=57), most (82.5%, n=47) were diagnosed with stages IIIA to IVB. 18% had ECOG scores ≥2. 86% had ALBI grades between 2-3; 42.1% had AFP level ≥400 ng/ml; 30.4% had cirrhosis (among those that reported comorbidities) and 24.1% received any loco-regional therapy. **Conclusions:** Our study provides a real-world profile of patients on 1L STRIDE. Compared to the HIMALAYA trial, these findings suggest that patients in this real-world study tend to be older, diagnosed at late-stage, have a poor liver profile, and potentially worse prognosis over time. Further research is required to explore treatment pathways after index date, over a long follow-up, which may provide more insight into the continuity and effectiveness of IL STRIDE in the real-world. Research Sponsor: AstraZeneca.

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The impact of protein-energy malnutrition on outcomes in patients with cholangiocarcinoma admitted with sepsis: Insights from a national database.

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Background: Patients with cholangiocarcinoma (CCA) are prone to infection and related complications. Protein-energy malnutrition (PEM) is common among these patients, arising from either the malignancy or its treatments. PEM is linked to worse medical and surgical outcomes and longer hospital stays across various cancers. The impact of PEM on CCA patients with sepsis, however, remains underexplored. This study aims to fill this gap by assessing how PEM affects in-hospital outcomes in CCA patients admitted with sepsis. **Methods:** We queried the Healthcare Cost and Utilization Project National Inpatient Sample and identified adult patients with CCA admitted with the principal diagnosis of sepsis from 2018 to 2021, using the ICD-10 code. We categorized the cohort based on the presence or absence of PEM and applied complex sampling weights for national representativeness. We compared socio-demographic characteristics and comorbidities between CCA patients with sepsis who have PEM and those who do not. The primary outcome studied was all-cause in-hospital mortality. Secondary outcomes examined were acute kidney injury (AKI), renal replacement therapy (RRT), respiratory failure (RF), invasive ventilation (IV), septic shock, vasopressor requirement (VP), mean length of hospital stay (LOS), and mean total hospitalization charges (THC). Multivariate regression models assessed outcome disparities between groups, adjusting for patient characteristics and comorbidities, with statistical significance set at $p < 0.05$. **Results:** We identified a total of 22,680 adults with CCA who were admitted with the principal diagnosis of sepsis from 2018 to 2021. Among these, 7,245 patients (31.9%) had a concurrent diagnosis of PEM. The PEM group had a higher proportion of female (47.3% vs 44.1%, p=0.04) and non-white (41.7% vs 37.2%, $p<0.001$) patients and patients with Charlson Comorbidity Index > 3 (86.5% vs 83.3%, $p<0.001$). Patients with PEM had higher odds of all-cause in-hospital mortality with an adjusted odds ratio (aOR) of 1.26 (95% confidence interval (CI) 1.05-1.51). They also had higher odds of AKI with an aOR of 1.32 (95% CI 1.16-1.50), RF with an aOR of 1.30 (95% CI 1.09-1.51), and septic shock with an aOR of 1.22 (95% CI 1.06-1.40). Patients with PEM also had a higher LOS (9.0 vs 6.3 days) with an adjusted incidence rate ratio (aIRR) of 1.41 (95% CI 1.33-1.49) and THC ($115,971 vs $83,176) with aIRR of 1.35 (95% CI 1.25-1.45). **Conclusions:** Our study revealed that CCA patients with PEM admitted with sepsis had significantly higher odds of in-hospital mortality, AKI, RF, and septic shock compared to those without PEM. They also experienced longer hospital stays and higher hospitalization charges. These findings highlight the urgent need to address PEM in CCA patients and tailor inpatient management to reduce complications in this high-risk group. Research Sponsor: None.

Real-world impact of matched targeted therapy on survival in advanced biliary tract cancer: An international collaborative study.

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Background: The prognosis of advanced BTC remains poor. Although targeted agents have expanded late-line options, many patients (pts) become ineligible for these therapies after progressing on first-line treatment. Herein, we examined the real-world impact of targeted therapy on survival in advanced BTC. **Methods:** In this international collaborative study between the Mayo Clinic (US) and the National Cancer Center East (Japan), we included pts with advanced BTC with comprehensive tumor molecular profiling who had received at least 30 days of systemic therapy. Pts were categorized into 3 groups: 1) non-actionable, 2) pts with alterations who received matched targeted therapy at any point during their treatment (matched group), and 3) those with mutations who did not receive matched therapy (unmatched group). OS was calculated from the start date of first-line therapy until death and compared among three subgroups using Cox regression models. **Results:** Of 932 pts, 366 (39.3%) had an actionable alteration, and 135 of these received matched therapy. 566 (61.7%) did not have an actionable alteration (Table). The most common alterations were somatic *BRCA1/2* (N=106, 11.4%), *HER2* amplification (N=65, 7%), *FGFR2* fusion (N=61, 6.5%), *KRAS G12C/D* (N=46, 4.9%), and *IDH1* (N=38, 4.1%). 78.7% of *FGFR2* fusions were found in Caucasian pts, whereas *BRCA1/2* (97.5%), *HER2* (78.5%), and *KRAS G12C/D* (67.4%) alterations were more prevalent in Asian pts. The median OS was significantly longer in the matched group (21.4 months; 95% CI 18.4-27.9) compared to the unmatched group (14.6 months; 95% CI 12.8-17.6) and the non-actionable group (17.2 months; 95% CI 15.5-19.0). The 36-month OS rate was 33% in the matched group vs. 7% in the unmatched group ($p < 0.0001$). After adjusting for age, gender, prior surgical resection, primary tumor location, and country of origin, matched targeted therapy remained a strong independent predictor for improved OS (HR 0.61; 95% CI 0.48-0.79). **Conclusions:** These real-world findings suggest that matched targeted therapies significantly improve survival in advanced BTC, underscoring the need to incorporate them earlier in the treatment course and address barriers to treatment access. The regional prevalence of actionable alterations should also be considered when developing new targeted therapies. Research Sponsor: None.

Patient baseline characteristics stratified by actionable alterations and targeted therapy.

	Non-actionable (N=566)	Unmatched group (N=231)	Matched group (N=135)
Site, n (%)			
Japan	409 (72%)	166 (72%)	50 (37%)
US	157 (28%)	65 (28%)	85 (63%)
Age at Dx > 65, n (%)	324 (57%)	122 (53%)	54 (40%)
Female, n (%)	235 (42%)	92 (40%)	77 (57%)
Primary tumor location (category), n (%)			
Intrahepatic	217 (43%)	115 (55%)	99 (76%)
Extrahepatic	150 (30%)	39 (19%)	17 (13%)
Gallbladder	142 (28%)	54 (26%)	14 (11%)
Prior surgical resection, n (%)	220 (42%)	74 (34%)	48 (37%)

537 Poster Session

Real-world (RW) experience with atezolizumab + bevacizumab (A+B) for the treatment of unresectable HCC (uHCC): A multicenter study.

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Background: A+B is the preferred first-line (1L) standard of care for uHCC, with emerging RW evidence supporting its use in a broad patient (pt) cohort. We evaluated pt characteristics, clinical and treatment outcomes in pts treated at five U.S. institutions: Mayo Clinic, Houston Methodist, Moffitt Cancer Center, Mays Cancer Center, and University of Arizona. **Methods:** This is a retrospective study of uHCC pts who initiated A+B after January 1, 2019. De-identified pt data were extracted by treating oncologists. Overall survival (OS) and real-world progression-free survival (rwPFS) were assessed using Kaplan-Meier methods for all pts and for a "trial-like" subgroup with characteristics similar to those in the IMbrave150 trial (ECOG performance status [PS] 0-1, Child-Pugh [CP] class A, albumin-bilirubin [ALBI] grade 1-2). HCC-related hospitalizations were evaluated within 1 year of A+B initiation. **Results:** A total of 300 eligible pts were treated with 1L A+B (median age 68 years, 79% male, 79% White, 82% BCLC C, 44% viral etiology,12% ECOG PS >1). Liver function was CP A in 73% (36% A5, 31% A6, 6% not reported), CP B in 26% pts (15% B7, 7% B8, 5% B9), and CP C in 2 (<1%) pts. Compared with CP A, more CP B pts had characteristics of compromised liver function (CP A vs B: cirrhosis: 61% vs 91%; ascites: 7% vs 46%; encephalopathy: 4% vs 17%; esophageal varices: 14% vs 32%), bile duct invasion (2% vs 11%), and ECOG PS>1 (8% vs 22%), all Ps<0.001. Over a median follow-up of 8.7 mos, 244 (81%) pts discontinued A+B, primarily due to disease progression. Toxicity-related treatment discontinuation and hospitalizations due to treatment-related adverse events (TRAE) were similar between the two groups (Table). Median OS (mOS) was 14.4 mos (95% CI: 12.3, 18.2) and median rwPFS was 6.8 mos (95% CI: 5.8, 8.4). In the trial-like subgroup (n=194), mOS was 19.5 mos (95%CI: 14.6, 24.7) and median rwPFS was 8.8 mos (95% CI: 7.6, 12.1). In the CP B subgroup, mOS was 5.6 mos (95% CI: 4.6, 11.3) and median rwPFS was 3.1 mos (95% CI: 2.4, 5.8). **Conclusions:** This study demonstrates the RW outcomes of 1L A+B in a diverse pt cohort. Results from the "trial-like" pts further confirm the reproducible efficacy of A+B in clinical practice. Although pts with CP-B had higher hospitalization rates due to disease progression or symptoms, consistent with their underlying liver complications, they had similar rates of toxicity-related treatment discontinuation and TRAE-related hospitalizations compared to CP-A. Further characterization of safety for A+B in CP B pts within a prospective, controlled trial is warranted. Research Sponsor: F. Hoffman La-Roche Ltd.

N (%)	CP A (n=219)	CP B (n=79)
Treatment discontinuation*	167 (76)	75 (95)
Atezolizumab-related toxicity	20 (12)	13 (17)
Bevacizumab-related toxicity	26 (16)	16 (21)
Hospitalization*	87 (40)	58 (73)
Symptom-related	42 (48)	35 (60)
Progression	8 (9)	11 (19)
TRAE	7 (8)	6 (10)

*P-value comparing CP A vs CP B <0.001.

Financial toxicity in patients with newly diagnosed hepatocellular carcinoma.

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Background: Hepatocellular carcinoma (HCC) remains a leading cause of cancer-related deaths and has been associated with significant patient (pt) financial liability, however no study has yet examined pt-reported measures of financial toxicity (FT) in this population. We sought to assess pt-reported FT in newly diagnosed HCC and to identify pt-, disease-, and treatment-related factors associated with FT. **Methods:** Pts with HCC were recruited prior to treatment initiation during their initial visit at a multidisciplinary liver cancer clinic at a tertiary care center between January 2023-May 2024. The Comprehensive Score for Financial Toxicity-Functional Assessment of Chronic Illness Therapy (COST-FACIT), a validated survey for assessing financial distress in pts with cancer (range, 0-44; lower scores reflect worse financial wellness), was completed by participants during their initial visit. Sociodemographic (age, sex, race, ethnicity, primary language, insurance status, employment status, marital status, highest education completed, household income, living arrangement, home zip code), clinical (Barcelona Clinic Liver Cancer stage, performance status, age-adjusted Charlson comorbidity index, albumin-bilirubin score, presence and etiology of chronic liver disease, Child-Pugh grade, MELD-Na score), and planned treatment (surgery, liver-directed therapy, systemic therapy, clinical trial enrollment) characteristics were evaluated by questionnaires or medical record review. The EORTC QLQ-HCC18 was used to assess pt-reported quality of life (QOL). Area Deprivation Indices were calculated using pts' home zip codes. Mean COST-FACIT scores were compared using two-sided t test for binary variables and one-way ANOVA for non-binary variables. **Results:** Among 47 pts enrolled, 35 (74%) completed the COST-FACIT survey. Younger age (<65 vs ≥65 yrs; mean, 20.3 vs 27.8; p=0.05), non-English primary language (non-English vs English; mean, 16.2 vs 26.7; p=0.04), less completed education (high school or earlier vs college or beyond; mean, 23.3 vs 32.0; p=0.02), and worse self-reported QOL (EORTC QLQ-HCC18 score ≥cohort median vs <cohort median; mean, 21.4 vs 29.2; p=0.02) were significantly associated with worse FT. Although not statistically significant, Hispanic ethnicity (Hispanic vs not Hispanic; mean, 16.1 vs 26.4; p=0.07), lack of employment (not employed vs retired vs employed; mean, 15.5 vs 29.4 vs 24.3; p=0.07), and no plan for liver-directed therapy (no vs yes; mean, 17.9 vs 26.7; p=0.07) trended towards worse pt-reported FT. **Conclusions:** Among pts with newly diagnosed HCC, significantly higher pt-reported FT is associated with younger age, non-English primary language, less completed education, and worse QOL. Efforts are underway to longitudinally follow our study cohort to assess the course of self-reported FT over 1-year follow-up as well as its associations with treatment response and survival. Research Sponsor: None.

Efficacy and safety of durvalumab plus tremelimumab for unresectable hepatocellular carcinoma in Thailand: A real-world multicenter observational study.

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Background: Durvalumab plus tremelimumab (Durva/Treme) andatezolizumab plus bevacizumab are preferable first-line therapy for unresectable hepatocellular carcinoma (uHCC). This study aimed to evaluate the efficacy and safety of Durva/Treme in real world setting. **Methods:** Fifty patients with uHCC who enrolled in the expanded access program (EAP) for Durva/Treme as the first-line treatment from 12 centers in Thailand were included. Analysis was assessed for objective response rate (ORR), survival, and adverse events. We also compared data with the HIMALAYA study. **Results:** Median follow-up time was 9.8 months. There were 24 patients alive and 26 patients deceased. The median age was 62 years. Majority of the patients were male (80%). Hepatitis B, C and non-viral cause were identified in 44%, 30%, and 26%, respectively. Ninety-two percent of patients had Child-Pugh A, while 8% had Child-Pugh B(7). Macrovascular invasion was found in 24% of cases. The full data on efficacy and safety were summarized in table below. **Conclusions:** To date,this is the largest real-world data of Durva/Treme in the first-line treatment of uHCC. Our study demonstrated that Durva/Treme was effective but had lower ORR and higher liver toxicities than what reported in the HIMALAYA study. Research Sponsor: None.

	EAP (N=50)		HIMALAYA (N=393)	
Response by RECIST1.1, n (%)	2(4)		12(3.1)	
CR	4(8)		67(17)	
PR	24(48)		157(39.9)	
SD	18(36)		157(39.9)	
PD	2(4)		-	
Not evaluable				
ORR (%)	12		20.1	
DCR (%)	60		60.1	
mPFS (range), mo	5.36(0.2-12.5)		3.78(3.7-5.3)	
mOS (range), mo	NA		16.4 (14.2-19.6)	
Adverse events, n (%)	**Any grade**	**Grade > 3**	**Any grade**	**Grade > 3**
Any AE	37(74)	12(24)	378(97.4)	196(50.5)
Diarrhea	3(6)	1(2)	103(26.5)	17(4.4)
Rash	12(24)	0	87(22.4)	6(1.5)
AST elevation	23(46)	6(12)	48(12.4)	20(5.2)
ALT elevation	21(42)	7(14)	36(9.3)	10(2.6)
Increase bilirubin	8(16)	4(8)	20(5.2)	3(0.8)
Hypothyroidism	7(14)	0	47(12.1)	0
Myocarditis	1(2)	1(2)	-	2(0.5)
Myositis	1(2)	1(2)	-	3(0.8)

Prevalence and clinical characteristics of patients with brain metastases at diagnosis with advanced hepatocellular carcinoma in a retrospective registry.

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Background: Hepatocellular carcinoma (HCC) is a cancer with poor prognosis and rising incidence. The combination of atezolizumab plus bevacizumab (A+B) prolongs survival as 1st line therapy in advanced HCC (aHCC) but confers a rare risk of serious bleeding. HCC brain metastases (BM) have high hemorrhage rates and contraindicate the A+B regimen. Alternative immunotherapies also prolong survival without increased bleeding risk. We conducted a retrospective review of a large, diverse, cancer center registry to estimate the prevalence of and identify clinical characteristics associated with BM in aHCC. **Methods:** The University of California, San Francisco (UCSF) Helen Diller Family Comprehensive Cancer Center (HDFCCC) registry was queried for new cases of HCC between 2004 and 2022. AJCC stage III or IV were defined as aHCC. Cases with central nervous system (CNS) metastases (including brain as well as skull, face, or orbit as adjacent high-risk sites) at diagnosis were identified. Key clinical covariates including demographics, liver disease etiology, and symptomatology were described. **Results:** Among 4002 new HCC cases, 832 were classified as aHCC. 12/832 (1.4%, 95% CI: 0.8, 2.5) had synchronous BM. Key demographics are displayed in Table. 10/12 (83%) patients with BM were symptomatic, including headache (25%), neurologic deficits (67%), and/or syncope (8%). Bleeding complications and/or hemorrhagic features were reported in 17%. **Conclusions:** BM are rare at diagnosis with aHCC, present in only 1.4% of this registry dataset. The majority (83%) were symptomatic, suggesting that routine CNS imaging may not have clinical utility in asymptomatic patients. A high proportion with BM at diagnosis had viral etiology. Limitations of this retrospective registry analysis include potential for underdiagnosis in asymptomatic patients and incomplete data for clinical covariates. Research Sponsor: None.

Clinical characteristics of aHCC cases with BM at diagnosis between 2004-2022 in UCSF HDFCCC Registry.

	Overall aHCC Cases (N=832)	Cases with BM (n=12)
Median age (range) (yrs.)	61 (8-92)	60 (48-65)
Race/ethnicity (%)		
Asian	29	25
Black	10	8
Caucasian	57	67
Hispanic (any race)	15	17
Gender (%)		
Male	79	75
Female	21	25
Etiology (%)		
Viral hepatitis	29	75
Unknown/negative	71	25
Symptomatic BM (%)	NA	83
Headache		25
Neurologic deficits		67
Syncope		8
Hemorrhagic BM (%)	NA	17

541 Poster Session

Regorafenib combined with immunotherapy and prognosis of unresectable hepatocellular carcinoma.

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Background: To investigate the prognosis and influencing factors of patients with unresectable hepatocellular carcinoma (uHCC) receiving regorafenib in different treatment modes. **Methods:** A total of 227 patients with uHCC who received at least 2 cycles of regorafenib treatment in the Affiliated Hospital of Qingdao University between June 2018 and August 2022 were retrospectively collected through the Hospital Information System (HIS). 204 patients were finally enrolled by inclusion and exclusion criteria after screening. **Results:** The median follow-up was 25.9 (4.5-69.7) months. The median overall survival (OS) of 204 patients with uHCC was 27.7 (95%CI 21.98-33.42) months. The 1, 2, 3 and 5-year OS were 80.9%, 55.9%, 41.4% and 26.7%, respectively. Univariate analysis showed that combination with immune checkpoint inhibitor (ICI) and metabolic-associated complications were independent factors for survival benefit of regorafenib, while the Eastern Cooperative Oncology Group Performance Status (ECOG-PS) scale = 2, undecreased Alpha-fetoprotein (AFP), macrovascular invasion and extrahepatic metastasis were independent risk factors for poor prognosis ($P<0.05$). Compared with 60 patients receiving regorafenib alone, 144 patients receiving regorafenib combined with ICI had a significant survival benefit (median OS, 16.9 vs. 31.5 months, respectively, P=0.006). The median OS of 153 patients receiving second-line therapy with regorafenib was 30.3 months. Among 51 patients receiving third or more lines of therapy with regorafenib, compared with 4 patients receiving regorafenib alone, 47 patients receiving regorafenib combined with ICI had a significant survival benefit (median OS, 9.4 vs. 24.1 months, respectively, P=0.000). Among 148 patients receiving local treatment, the median OS of patients with regorafenib combined with ICI (n=115) or without ICI (n=33) was 33.2 and 16.5 months, respectively (P=0.005). **Conclusions:** Regorafenib has a good prognosis in second-line treatment of uHCC. The OS of patients with regorafenib combined with ICI were significantly better than that of patients without ICI. The OS benefit of regorafenib combined with ICI was also equivalent in patients with third or more lines. The benefit of regorafenib combined with ICI was better for patients receiving local treatment. Research Sponsor: None.

Concurrent transarterial chemoembolization plus atezolizumab and bevacizumab in unresectable hepatocellular carcinoma: Interim analysis from a multicenter real-world study (CHANCE 023).

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Background: Combining transarterial chemoembolization (TACE) with systemic therapies such as atezolizumab and bevacizumab (T+A) has shown potential efficacy in patients with unresectable Hepatocellular carcinoma (HCC) This interim analysis aims to evaluate the efficacy and safety of this combination therapy when administered concurrently within a two-month period. **Methods:** This multicenter, real-world study included patients diagnosed with unresectable HCC who received TACE combined with T+A from January 2020 to June 2024, across 37 centers in mainland China. The primary endpoint was overall survival (OS). This trial is registered with ClinicalTrials.gov, NCT06024252. **Results:** Out of a cohort of 335 patients, 293 (259 males) met the inclusion criteria, and 37.5% (112/293) adopted this regimen as second-line therapy. The interim analysis showed a median OS of 29.97 months and a one-year survival rate of 76.50%. The median PFS was 19.23 months, and the objective response rate (ORR) was 41.6%, with 26 complete responses and 96 partial responses. The disease control rate (DCR) was 84.6%. Common adverse events included bone marrow suppression, decreased white blood cell or platelet counts, among others. Serious adverse events occurred in 9.2% of patients (27/293) and were manageable with standard interventions. **Conclusions:** This study is a real-world analysis to highlight the potential benefits of combining TACE with T+A in this patient population. However, longer follow-up is required to better assess both overall survival and the long-term safety profile of this treatment strategy. Clinical trial information: NCT06024252.
Research Sponsor: None.

Organ-specific responses to nivolumab plus ipilimumab in patients with advanced hepatocellular carcinoma: A multicenter, retrospective study.

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Background: The Check-Mate 9DW trial demonstrated that combining ipilimumab with nivolumab (Ipi/Nivo) improved survival outcomes compared with sorafenib or lenvatinib in advanced hepatocellular carcinoma (aHCC). Accumulated evidence suggests that immune checkpoint inhibitor (ICI) monotherapy elicits limited intrahepatic responses in patients with aHCC. Here, we aimed to examine the organ-specific objective response rate (OSORR) of Ipi/Nivo compared with that of nivolumab monotherapy (Nivo) and to evaluate the OSORR of Ipi/Nivo based on the prior use of ICI treatment. **Methods:** We conducted a retrospective assessment of 69 Ipi/Nivo-treated patients and 164 nivolumab-treated patients for aHCC at five referral hospitals in Korea. Only patients classified as Child-Pugh Class A were included. Organ-specific response criteria were adopted from RECIST 1.1, according to the indicated primary sites, including the liver, lungs, lymph nodes and other sites of metastasis. **Results:** Baseline characteristics of the Ipi/Nivo and Nivo groups were comparable. However, 65.2% of patients in the Ipi/Nivo group had prior ICI exposure, compared with only 4.3% in the Nivo group. The objective response rate (ORR) of Ipi/Nivo was 29.0%. The Ipi/Nivo combination therapy elicited OSORRs for 204 individual lesions: 18.0% in the liver, 17.7% in the lungs, 30.0% in the lymph nodes, and 12.5% in other sites. Within the Ipi/Nivo group, the ORR was 45.8% for patients without prior ICI exposure and 20.0% for those with prior ICI exposure. Patients without prior ICI exposure demonstrated OSORRs for 72 individual lesions: 29.0% in the liver, 31.3% in the lungs, 33.3% in the lymph nodes, and 23.1% at other sites. Conversely, patients with prior ICI exposure exhibited OSORRs for 132 individual lesions: 11.5% in the liver, 11.4% in the lungs, 27.8% in the lymph nodes, and 7.4% at other sites. Patients who exhibited a response in one organ were more likely to demonstrate responses in other organs irrespective of the organ type, with only two patients exhibiting variable responses across different organs. Furthermore, these responders had improved survival outcomes, including longer progression-free survival and overall survival, compared with non-responders. In terms of nivolumab monotherapy, the ORR was 21.3%. Nivo yielded OSORRs for 305 individual lesions: 13.5% in the liver, 25.3% in the lungs, 39.3% in the lymph nodes, and 18.4% at other sites. **Conclusions:** In contrast to lesions of patients who received Nivo, intrahepatic lesions of Ipi/Nivo-treated patients without prior ICI exposure exhibited favorable responses, comparable with treatment responses of extrahepatic lesions. Conversely, Ipi/Nivo-treated patients with prior ICI exposure had reduced organ-specific responses across all organs when compared with those without prior ICI exposure. Research Sponsor: None.

Real-world assessment of MDM2 amplification and survival among patients with advanced or metastatic biliary tract cancer in the United States.

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Background: There are few real-world studies on biliary tract cancer (BTC), especially those related to biomarkers for potential novel targeted therapies. The aim of this study was to provide additional information of the prevalence of biomarkers, treatment patterns, and clinical outcomes of patients with BTC in the US, with a focus on patients with mouse double minute 2 homolog (*MDM2*) amplified, tumor protein 53 wild type (TP53 WT) BTC. **Methods:** This was an observational cohort study using deidentified data from the Flatiron Health-Foundation Medicine BTC Clinico-Genomic Database, which included patients with advanced or metastatic BTC who underwent genomic analysis and received antineoplastic therapy in the US between January 1, 2013 and June 30, 2023. Three cohorts were created: 1) 1L+ cohort that included all patients that initiated 1st line (1L) therapy during the study period, 2) 2L+ cohort that included all patients that initiated 2nd line (2L) therapy during the study period, and 3) *MDM2* amp + TP53 WT cohort that included patients in the 1L+ cohort that had *MDM2* amplified ($\geq$ 8 copies), TP53 WT. The prevalence of select biomarkers was assessed, as well as treatment patterns, and clinical outcomes including time to next therapy (TTNT) and overall survival (OS). **Results:** A total of 1,548 patients met the selection criteria for the 1L+ cohort, of whom, 875 initiated 2L therapy (2L+ cohort), and 55 had *MDM2* amplified, TP53 WT BTC (*MDM2* amp + TP53 WT cohort). In the 1L+ cohort, 66 (4.3%) had *MDM2* amplification and 872 (56.3%) had TP53 WT. The most common treatment regimens were cisplatin/gemcitabine (44.5%, 50.7%, 41.8% of patients in the 1L+, 2L+, and *MDM2* amp + TP53 WT cohorts, respectively) at 1L and FOLFOX (12.9%, 22.9%, 16.4% of patients, respectively) at 2L. The median (95% CI) OS from start of 1L was 10.7 (10.0 – 11.3) months for the 1L+ cohort and 11.9 (10.4 – 17.7) months for the *MDM2* amp + TP53 WT cohort. The median OS from the start of 2L was 8.0 (7.2 – 8.8) months for 2L+ cohort. Furthermore, the median TTNT was similar between 1L+ cohort and *MDM2* amp + TP53 WT cohort, 5.3 (5.0 – 5.7) and 4.8 (4.0 – 7.6) months, respectively. The median TTNT from the start of 2L in the 2L+ cohort was 4.3 (3.8 – 4.6) months. We will present co-mutation data in *MDM2* amp + TP53 WT cohort. **Conclusions:** The *MDM2* amplification and TP53 WT status does not appear to be a prognostic biomarker in advanced/metastatic BTC with similar median TTNT and OS in the 1L+ and *MDM2* amp + TP53 WT cohorts. Targeted treatments may help improve OS among these patients and thus further research into these treatments and how they impact survival is warranted. Research Sponsor: Boehringer Ingelheim.

Real-world insights into immune-related adverse events (irAE) in gastrointestinal (GI) malignancies.

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Background: This retrospective review aimed to examine the incidence and baseline characteristics (BLC) that irAE risk in patients with gastrointestinal (GI) cancers undergoing immune checkpoint inhibitor (ICI) therapy at our institution. **Methods:** GI cancer patients who received at least one dose ICI-only between 1/1/2017 and 7/31/2021 at the Ohio State University were included in the study. The BLC including hematological and chemistry labs were extracted with immune-related adverse events (irAE) details. Statistical analyses included descriptive statistics, chi-square test, logistic regression, and matched pairs t-tests were performed. **Results:** Our analysis included 198 patients (14 anal (Al), 65 hepatobiliary (HPB), 46 lower (LGI) and 74 upper (UGI)) with irAEs reported in 37 patients (19%). Median age of irAE-group was 67 years (range: 26 to 88) with 54% males and 86% Caucasians. Most of them (n=36) in had single agent ICI. Pneumonitis was most frequently (n=7) noted followed by dermatitis (n=), colitis (n=5), hypophysitis (n=5), and hepatitis (n=4). Relatively rare ones were hypothyroidism (n=3), nephritis (n=2), encephalitis (n=2), diabetes mellitus (n=1), neutropenia (n=1), and myasthenia gravis (n=1). In irAE-group, 18 (49%) required hospitalization ($\geq$ grade 3) and 5 (14%) had irAE-related deaths. Complete recovery from irAEs was reported in 20 patients (54%), with 13 patients (35%) restarting ICI therapy. The irAE incidence in HPB group was significantly higher than other GI malignancies (HPB vs. LGI vs. Al vs. UGI = 46% vs. 24% vs. 16% vs. 14%, p=0.004). History of asthma (41% vs. 16% (p=0.004) and female gender (26% vs. 15%, p=0.04) increased irAE-risk; presence of liver (9 % 22%, p=0.04) and $>$ 3 lung (8 vs. 21%, p=0.04) metastasis reduced the risk. Compared to non-irAE group, the irAE-group had significantly lower baseline (mean) whole blood cell count (WBC: 6.3 vs. 7.6 K/uL, p=0.02), neutrophil count (NC: 4.2 vs. 6.8 K/uL, p=0.04), and neutrophil-lymphocyte-count (NLR: 5.1 vs. 9.3, p=0.03), and lower albumin (3.7 vs. 3.5 mg/dL, p=0.02). A significant rise in NC (2.1 K/uL, p=0.0008), NLR (by 2.5, p=0.004), WBC (by 2.1 K/uL, p=0.002) from baseline was noted at irAE-incidence. Heterogeneity in the primary tumors and treatments received is a notable limitation of this analysis. **Conclusions:** Overall, the risk of irAEs in patients with GI cancers is relatively low, and certain BLC can predict their incidence. Research Sponsor: None.

Zanidatamab dose optimization in patients with HER2-positive biliary tract cancer (BTC).

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Background: Zanidatamab (zani) is a dual HER2-targeted bispecific antibody that binds 2 distinct HER2 domains, resulting in receptor crosslinking, clustering, and internalization as well as leading to reductions in cell-surface levels of HER2, phosphorylation of HER family members (EGFR, HER2, and HER3), downstream signaling, and ligand-dependent and -independent proliferation. Zani also induces complement-dependent cytotoxicity as well as antibody-dependent cellular cytotoxicity and phagocytosis. In the phase 2 HERIZON-BTC-01 study (Study 203), zani 20 mg/kg Q2W demonstrated a 41.3% confirmed objective response rate (cORR) and a manageable safety profile in patients (pts) with previously treated HER2+ BTC (IHC3+/2+ and ISH+). The objective of this analysis is to report optimal dose selection for zanidatamab in patients with HER2+ BTC through pharmacokinetic modeling. **Methods:** Source data included the phase 1 ZW25-101 study (Study 101; NCT02892123), Study 203 (NCT04466891), and *in vitro* study for ligand-dependent growth inhibition (LDGI) in HER2-expressing human cancer cell lines. In the LDGI study, cancer cells were cultured in media containing epidermal growth factor and were treated with different zani concentrations. The IC90 value for LDGI was obtained by nonlinear regression using the logistic dose-response model. To assess target saturation, the population pharmacokinetic model was used to simulate clearance at steady state following 5-30 mg/kg Q2W. The clinical utility of zani was evaluated based on the correlation of zani concentrations with both efficacy and safety. **Results:** Study 101 included 192 patients with HER2-expressing solid tumors (dose ranged from 5-30 mg/kg including QW, Q2W, and Q3W). The Ctrough values following the first zani 5 mg/kg QW dose were below the IC90 (25.0 μg/mL) for LDGI, and no confirmed responses were observed at this dose. The Ctrough values following the first zani dose of 10 mg/kg QW or 20 mg/kg Q2W were above IC90 and the 2 dose levels demonstrated a cORR of 25% and 28.6%, respectively. The simulated clearance suggests that the target-mediated elimination pathway was saturated at the dose level of 20 mg/kg Q2W, since clearance was comparable to that of the higher dose of 30 mg/kg Q2W. The efficacy and safety of 20 mg/kg Q2W was further evaluated in 87 pts with HER2-amplified BTC enrolled in Study 203. The clinical utility analysis based on the data from Study 203 demonstrated that the majority of pts had an exposure range on the plateau part of the efficacy curve. **Conclusions:** The zani dose of 20 mg/kg Q2W is expected to maximize the clinical benefit for pts with HER2+ BTC based on: reaching the desired target exposure (IC90 for LDGI), saturating target-mediated elimination pathway, and the clinical utility analysis of safety and efficacy. Higher exposures are not expected to result in higher responses. Clinical trial information: NCT02892123, NCT04466891. Research Sponsor: Jazz Pharmaceuticals.

Real-world treatment patterns among U.S. adults with unresectable hepatocellular carcinoma (uHCC) eligible for locoregional therapy (LRT).

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Background: Treatment (tx) in uHCC has mostly relied on locoregional liver-directed tx, such as transarterial chemoembolization (TACE) or radioembolization (TARE), but recent pharmacotherapy advances have provided more tx options. This study aims to describe current real-world tx patterns and clinical outcomes of uHCC eligible for LRT. **Methods:** uHCC patients enrolled in TARGET-HCC (an ongoing, 5-year, longitudinal, observational cohort of HCC patients receiving usual care at academic and community sites in the U.S.), who initiated tx between 6/1/2020-6/1/2023 were included and followed until 12/31/2023 (data cutoff). Patients receiving curative HCC tx (ablation, resection, or transplant) or with metastatic HCC, portal vein involvement, or Child-Pugh-C liver disease at diagnosis (dx) were excluded. Patient and disease characteristics, and txs data were abstracted from electronic health records. Index tx was defined as first tx after dx, with a 30-day grace period for combination txs. Subsequent tx was defined as first tx after the grace period; for index systemic therapies, the subsequent tx differed from the index tx. Real-world progression-free survival (rwPFS) was defined as time from index tx to first event: radiological progression (including new lesions, extrahepatic involvement, vascular invasion, or >20% increase in tumor burden from baseline), death, or hospice entry. Real-world time to progression (rwTTP) was defined as time from index tx to first radiological progression. **Results:** A total of 115 patients were included; mean age was 65.1 years and 77% were male. At dx, 41% of patients were BCLC B, 14% were non-metastatic BCLC C, and 38% were BCLC A with lesion >5 cm, rapidly progressive disease, or contraindications to LRT. Tx initiation with TACE/TARE was common, either alone (n=59, 51%) or in combination with systemic tx (n=5, 4%). Among patients initiating systemic tx (n=46 ,40%), 30 received IO-IO (immuno-oncology) combinations, 12 received IO only, and 4 received a TKI only (tyrosine kinase inhibitor). Following index tx, 21 (18%) patients died receiving no subsequent tx and 27 (23%) were censored at data cutoff. Among those who received subsequent tx (n=67, 58%), there was heterogeneity in tx choice: TACE/TARE (n=32), systemic tx (n=16) other LRT (n=10), multimodal therapy (n=5), and transplant (n=4). We observed 60 progression events and 38 deaths corresponding to a median rwPFS and rwTTP of 7.9 (95% CI: 5.0, 13.3) months and 11.3 (95% CI: 6.7, 28.9) months, respectively. **Conclusions:** Our results showed high variability in management of uHCC eligible for LRT with TACE/TARE commonly utilized. Tx choice may be influenced by factors beyond staging criteria. Prognosis for uHCC eligible for LRT remains poor, and further research on adaptive, multimodal tx strategies is needed to inform tx guidelines and improve outcomes. Research Sponsor: Merck.

Real-world survival differences in advanced biliary tract cancer patients with ctDNA detected IDH1 mutations and FGFR2 fusions receiving first-line gemcitabine-cisplatin with and without immunotherapy.

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Background: Comprehensive genomic profiling (CGP) is recommended for patients (pts) with metastatic BTC (mBTC) given its aggressive, heterogeneous disease profile. Liquid biopsy is a rapid, non-invasive option with high concordance to tissue-based CGP. First-line (1L) therapy in mBTC is gemcitabine-cisplatin (GemCis) +/- durvalumab or pembrolizumab (IO) regardless of CGP results. However, data suggests pts with targetable alterations may have worse outcomes with addition of IO compared to all comers. We examined outcomes for mBTC pts receiving 1L GemCis vs. GemCis+IO following ctDNA-detected *IDH1* mutation (*IDH1*+) or *FGFR2* fusion (*FGFR2*+). **Methods:** Real-world data was sourced from GuardantINFORM, which comprises aggregated commercial payer health claims and de-identified records from pts with clinical ctDNA testing via Guardant360 (G360). Pts with mBTC and >1 treatment claim after G360 results, and IDH+ or FGFR2+ between April 2019 and June 2024 were analyzed. Only pts treated with 1L GemCis +/- IO were included. Outcomes were assessed via real-world time to treatment discontinuation (rwTTD), real-world time to next treatment (rwTTNT), and real-world overall survival (rwOS), all in months with 95% confidence intervals. Log-rank test was used to compare Kaplan-Meier survival curves. **Results:** 412 pts were *IDH1*+ and 154 were *FGFR2*+. In the *IDH1*+ cohort, 184 pts were treated with GemCis and 106 pts with GemCis+IO. Pts receiving GemCis demonstrated improved rwOS vs. GemCis+IO [27.2 (23.6-37.8) vs. 16 (13.6-19.9) mo.; HR=0.4 (95CI 0.27-0.59), p<0.001]; there was no difference in rwTTNT [13 (10.8-14.3) vs. 10 (8.4-12.3); HR=0.85(95CI 0.6-1.22), p=0.4] or rwTTD [4.8 (4.1-5.5) vs. 6.6 (4.7-8.1); HR=1.17 (95CI 0.9-1.5), p=0.25]. Similarly, in the *FGFR2*+ cohort, pts receiving GemCis + IO (n=44) had worse rwOS than pts on GemCis (n=70) [NR (15.6-NR) vs. 43 (34.2-2.1); HR=0.37(95CI 0.18-0.74), p=0.005], with no difference in rwTTNT [8.9 (4.89-NR) vs. 13.78 (7.42-NR); HR=0.63 (95CI 0.36-1.1), p=0.1] or rwTTD [5.0 (3.1-7.8) vs. 4.2 (3.5-5.5); HR=1.08 (95CI 0.72-1.6), p=0.7]. Among all mBTC pts receiving GemCis+IO (n=1113), non-*IDH1*+ pts (n=1007) had numerically better rwOS compared to *IDH1*+ pts (n=106) [18.6 (17.2-21.3) vs. 16.0 (13.6-19.);HR=0.76 (95CI 0.6-1.03), p=0.075]. No survival difference was observed between *IDH1*+ (n=184) and non-*IDH1*+ (n=2333) pts receiving GemCis [27.2 (23.6-37.8) vs. 26.2 (24.5-28.2); HR=1.08 (95CI 0.88-1.32), p=0.4]. **Conclusions:** Addition of IO to GemCis in *IDH1*+ and *FGFR2*+ mBTC pts resulted in decreased rwOS. This data supports ctDNA CGP to inform clinical decision-making prior to 1L as well as at progression. Further studies in clinical cohorts are needed to confirm these findings. Research Sponsor: None.

549 Poster Session

Real-world experience with atezolizumab plus bevacizumab (A+/-B) in Hispanic patients with advanced hepatocellular carcinoma (HCC) at a Hispanic-majority cancer center.

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Background: Combination of atezolizumab (A), a programmed death-ligand 1 (PD-L1) inhibitor, and bevacizumab (B), a vascular endothelial growth factor (VEGF) inhibitor, has demonstrated efficacy for first-line treatment for HCC. However, real-world data on the outcomes of this combination therapy in diverse patient populations, particularly in Hispanic U.S. patients, remains limited. This study aims to evaluate the clinical outcomes of patients intended to treat with A+/-B at a Hispanic-majority U.S. cancer center. **Methods:** This retrospective study of patients diagnosed with advanced unresectable HCC who received treatment with A+B at a Hispanic-serving NCI-designated cancer center. Patients were included if they received at least one dose of A, either as monotherapy or in combination with B, between January 2019 and April 2022. Demographic and clinical data, including age, gender, ethnicity, liver disease etiology, Child-Pugh (CP) classification, comorbidities, and treatment details, were collected. The primary endpoints were the overall response rate (ORR) and complete response (CR) rate. Statistical analyses were conducted to evaluate the association between baseline characteristics and clinical outcomes. The median overall survival (mOS) evaluated by Kaplan-Meier Method. **Results:** 38 patients received at least A (84% A+B, 6 patients had B held due to bleeding risk after consent). Median age 65 years (45-90). 32 (84%) patients were male. 22 (58%) patients were Hispanic. 45% had HCV-related liver disease. 32% had CP B with no statistically significant association between CP Class and ethnicity. 53% had concurrent diabetes. 26% patients had obesity. The median time on treatment was 7.5 mo (1- 24 mo) with no statistical difference between Hispanics and non-Hispanics (9 vs 6 mo, p 0.70). CR 37% with ORR of 42%. Using Fisher's exact test, no association was noted between obesity, ethnicity, treatment option or CP class to occurrence of complete or partial response. mOS was 19 mo. There was no statistical difference in mOS between CP A and CP B (19 vs 10 mo, p value 0.6) and no difference between Hispanics (14 mo) vs non-Hispanics (19 mo) (p 0.5), using log rank test. **Conclusions:** This real-world analysis of a Hispanic-majority U.S. cohort of patients with advanced HCC shows 42% ORR and 37% CR. The findings suggest that this regimen is effective across a diverse patient population, including those with comorbid conditions, such as liver dysfunction (CP B 32%), HCV (45%), diabetes (53%, and obesity (26%). Limitations are the small study population and retrospective nature of the study. Evaluation of toxicity data is ongoing. Further prospective studies are needed to identify biomarkers of outcomes in Hispanic patients with HCC, which is ongoing at our cancer center (NCT03894917). Research Sponsor: None.

HCC in the Rio Grande Valley: Outcomes in a predominant Latino community.

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Background: Hepatocellular carcinoma (HCC) has slowly increased in the U.S. in recent years, with a higher incidence among racial and ethnic minorities, particularly Latinos. This retrospective study examines the different etiologies, presentation of diagnosis, management, and outcomes of patients presented in a multidisciplinary-liver tumor board in the Rio Grande Valley, home of 1.3 million Latinos. **Methods:** A retrospective chart review was conducted on patients with HCC diagnosis who were presented at the Liver Tumor Board of DHR Health Transplant Institute between 2017 and 2023. The data collected was extracted from the EHR, including medical history, laboratory results, and radiological studies required for staging, transplant eligibility assessments, clinical characteristics, and treatments. **Results:** A total of 250 cases were reviewed, and 70 patients with complete clinical data and a diagnosis of HCC were included in the study. Of those, 98.5% were identified as White/ Hispanic. The median age at diagnosis was 64.4 ± 9.2 years, with 72.9% male and 11.4% uninsured. The average BMI was 31.5 kg/m^2, metabolic-associated liver disease (MALD) was present in 28.6% of patients, 25.7% were alcohol-related, and 38.6% presented multiple etiologies. A descriptive analysis of this population is found in the table. Ablative therapy was used in 68 (97.1%) of the patients. The number of patients sent for liver transplant evaluation was 31 (44%), and only 3 (4.2%) patients received a liver transplant. Barriers to transplant included lack of transportation, inadequate social support, underinsurance, and fear of the transplant itself. **Conclusions:** Our data by staging aligns with the national distribution of incidence cases from 2012-2022 per SEER program. The rate of transplantation at our institution is below the national average, highlighting the need to bridge this disparity and address race-specific barriers to transplant within the Latino community. Research Sponsor: None.

Sociodemographic data.

Parameter	Stage IA N=15 (21.4%)	Stage IB N=20 (28.6%)	Stage II N=10 (14.3%)	Stage IIIA N=8 (11.4%)	Stage IIIB N=9 (12.9%)	Stage IVA N=5 (7.1%)	Stage IVB N=3 (4.3%)	All Stages N=70 (100%)
Child Pugh A	7 (46.7%)	8 (40.0%)	5 (50.0%)	4 (50.0%)	2 (22.2%)	2 (40.0%)	1 (33.3%)	29 (41.4%)
Child Pugh B	6 (40.0%)	6 (30.0%)	5 (50.0%)	2 (25.0%)	5 (55.6%)	2 (40.0%)	2 (66.7%)	28 (40.0%)
Child Pugh C	2 (13.3%)	6 (30.0%)	0 (0.0%)	2 (25.0%)	2 (22.2%)	1 (20.0%)	0 (0.0%)	13 (18.6%)
MELD-Na mean, (min-max)	12.8 (7.0-21.0)	13.6 (6.0-23.0)	12.8 (6.0-20.0)	12.3 (5.0-17.0)	13.6 (8.0-26.0)	13.0 (10.0-19.0)	11.3 (7.0-14.0)	13.6 (6.0-26.0)
Therapy*: Ablative (TACE, Y90, Radiation, other)	15	21	10	6	8	6	2	
Surgery (including transplant)**	1	1	2	2	2	0	0	
Systemic	2	6	5	4	6	3	1	
Overall Survival (months)	29.5	26.8	34.7	17.5	24.1	17.3	14.9	25.3

*Several patients received one or more types of therapy.
**All transplant patients received ablation therapy.

Nivolumab plus ipilimumab in advanced hepatocellular carcinoma with Child-Pugh B: A multicenter retrospective study.

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Background: The phase 3 CheckMate 9DW trial proposed that nivolumab plus ipilimumab (Nivo/Ipi) significantly improved the survival of patients with advanced hepatocellular carcinoma (aHCC) compared with sorafenib or lenvatinib as first-line treatment, but it only included patients with Child-Pugh A. Given the limited treatment options for patients with poor liver function, we aimed to assess the efficacy and safety of Nivo/Ipi in aHCC patients, stratified by liver function. **Methods:** This study included patients with aHCC who received Nivo/Ipi treatment at three referral hospitals between March 2020 and September 2023. Patients received nivolumab (1 mg/kg) plus ipilimumab (3 mg/kg) every 3 weeks (four doses), followed by nivolumab (240 mg) every 2 weeks. **Results:** Of 106 patients treated with Nivo/Ipi, 96 evaluable patients were included in the final analysis. Among them, 76 patients had Child-Pugh A, while 20 had Child-Pugh B. The baseline characteristics were generally comparable between the two groups, except for a significantly higher rate of extrahepatic metastasis in patients with Child-Pugh A (86.8% vs. 65.0%, p = 0.043) and a higher incidence of macrovascular invasion in patients with Child-Pugh B (55.0% vs. 27.6%, p = 0.032). The objective response rate (ORR) was 28.9% (22/76, 4 complete responses [CR] and 18 partial responses [PR]) in the Child-Pugh A group and 10.0% (2/20; 0 CR and 2 PR) in the Child-Pugh B group. Disease control rates were 42.1% in the Child-Pugh A group and 25.0% in the Child-Pugh B group. In a median follow-up duration of 29.3 months (95% confidence interval [CI], 28.8-29.8), the median overall survival (OS) was 8.9 months (95% CI, 5.5-17.6) in the Child-Pugh A group and 3.9 months (95% CI, 2.1-4.8) in the Child-Pugh B group ($p < 0.001$). The median progression-free survival (PFS) was 1.6 months (95% CI, 1.2-3.5) in the Child-Pugh A group and 1.2 months (95% CI, 0.8-2.1) in the Child-Pugh B group (p = 0.003). Adverse events of any grade occurred at a comparable rate between the two groups, although immune-related adverse events were significantly more frequent in the Child-Pugh A group. Multivariable analysis indicated that Child-Pugh B was consistently associated with a worse PFS and OS, while prior exposure to immune checkpoint inhibitors (ICIs) was associated with a poorer PFS. Notably, both responders with Child-Pugh B were both ICI naïve. **Conclusions:** While Nivo/Ipi treatment showed reduced efficacy in patients with Child-Pugh B, identifying potential responders beyond the current clinical trial criteria may offer opportunities to improve outcomes in this challenging population. Research Sponsor: None.

Outcomes of durvalumab (D) with or without tremelimumab (T) in routine clinical practice according to HIMALAYA trial eligibility: Preliminary results of the international DT-real study.

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Background: HIMALAYA showed that D+T and D are effective options for unresectable hepatocellular carcinoma (uHCC). However, data on outcomes according to the adherence to HIMALAYA inclusion criteria in routine clinical practice are lacking. **Methods:** In the context of a prospectively maintained database including 1293 patients (pts) with uHCC treated with immunotherapy, we analysed pts treated with D+T or D across 8 centres in USA, Asia and Europe. Pts who met >1 key exclusion criterion of HIMALAYA (prior systemic therapy, Child-Pugh class B-C, Vp4 thrombosis) were defined HIMALAYA-OUT and compared with HIMALAYA-IN pts for overall survival (OS), progression-free survival (PFS), objective response rate (ORR) and disease control rate (DCR) by RECIST 1.1 and treatment-related adverse events (TRAEs) per CTCAE v.5.0. **Results:** Up to February 2024, 108 pts (mean age 66 years, male sex 81%) started D+T (n=69, 64%) or D (n=39, 36%). 62 pts (57%) were treated in 1° line and 46 (43%) in >2° line. Child-Pugh class was A in 67 pts (62%). Vp4 was present in 17 pts (16%). 31 pts (29%) were HIMALAYA-IN and 19/31 (61%) received D+T. After a median follow-up of 4.3 months (m, 95%CI 3.3-4.9), median OS (mOS) was 11.5 m and 12-m OS rate was 42%. mOS was not reached in HIMALAYA-IN pts (12-m OS rate 62%) and 8.9 m (95%CI 6.0-12.1) in HIMALAYA-OUT pts. Survival hazard ratio (HR) for HIMALAYA IN vs OUT was 0.28 (95%CI 0.09-0.93, p=0.037). Median PFS was 2.6 m (95% CI 2.2-5.2) overall, 4.6 m (95%CI 2.1-8.5) in HIMALAYA-IN and 2.6 m (95%CI 1.9-5.2) in HIMALAYA-OUT pts (HR 0.70, 95%CI 0.38-1.30, p=0.266). ORR and DCR (evaluable in 53 pts, 49%) were 15.1% (95%CI 6.5-29.7%) and 43.4% (95%CI 27.5-65.1) (Table). Any grade TRAEs occurred in 31.5% (95% 21.8-44.0%), grade 3-4 TRAEs in 8.3% (95%CI 3.8-15.8%), TRAEs requiring systemic corticosteroids in 8.3% (95%CI 3.8-15.8%) and discontinuation due to toxicity in 3.7% (95%CI 1.0-9.5%). **Conclusions:** Preliminary observational data from DT-Real study suggest a reproducible efficacy and safety of D+T and D in pts with uHCC fitting the inclusion criteria of HIMALAYA in routine clinical practice. Research Sponsor: None.

		HIMALAYA IN			HIMALAYA OUT	
	HIMALAYA IN (n=31)	D+T (n=19)	D (n=12)	HIMALAYA OUT (n=77)	D+T (n=50)	D (n=27)
mOS (m, 95% CI)	NR	NR	NR	8.9 (6.0-12.2)	11.2 (6.6-13.2)	4.9 (2.6-12.2)
12-m OS (%)	61.8	63	88.9	37.2	37.4	44.4
mPFS (m, 95% CI)	4.6 (2.1-8.4)	8.5 (2.1-8.5)	2.4 (1.6-2.5)	2.6 (1.9-5.2)	2.4 (1.8-6.7)	2.6 (1.8-5.2)
ORR (%,95%CI) (N=53)	23.1 (4.8-67.4)	25.0 (5.1-73.1)	0	12.5 (4.1-29.2)	13.8 (3.8-35.3)	9.1 (0.2-50.6)
DCR (%, 95%CI)	46.2 (16.9-100)	50.0 (18.3-100)	0	42.5 (24.8-68.1)	41.4 (21.4-72.2)	45.4 (14.8-100)
Any grade TRAEs (%,95%CI)	32.3 (15.5-59.3)	31.6 (11.6-68.7)	33.3 (9.1-85.4)	31.2 (20.0-46.4)	28.0 (15.3-47.0)	37.1 (17.8-68.1)
Grade 3-4 TRAEs (%, 95%CI)	9.7 (2.0-28.3)	15.8 (3.3-46.1)	0	7.8 (2.9-17.0)	4.0 (0.5-14.4)	14.8 (4.0-38.0)

553 Poster Session

Comparison of efficacy and safety of PD-1 and PD-L1 antibodies in combination with hepatic arterial infusion chemotherapy and target therapy for unresectable intrahepatic cholangiocarcinoma: A multicenter retrospective study.

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Background: Immune checkpoint inhibitors (ICIs) in combination with hepatic arterial infusion chemotherapy (HAIC) and target therapy have shown promising antitumor activity in unresectable intrahepatic cholangiocarcinoma (iCCA). The present study aimed to compare the efficacy and safety of programmed cell death protein 1 (PD-1) inhibitors and programmed cell death ligand 1 (PD-L1) inhibitors in this setting. **Methods:** In this multicenter retrospective study, we included patients with iCCA who received ICIs + HAIC + target therapy from June 2018 to June 2023. Propensity score matching (PSM) was performed to reduce confounding factors. Patients were divided into two groups: those treated with PD-L1 antibody + HAIC + target therapy (Group A) and those treated with PD-1 antibody + HAIC + target therapy (Group B). Progression-free survival (PFS), overall survival (OS), and objective response rate (ORR) were compared between the two groups. Adverse events were assessed according to CTCAE v5.0. **Results:** A total of 336 patients met the inclusion criteria and were enrolled for analysis. There were no significant differences in OS, PFS, or ORR between the two groups (median OS: 17.7 vs. 25.3 months, P=0.139; median PFS: 8.3 vs. 10.0 months, P=0.534; ORR: 48.4% vs. 31.1%, P=0.051). Following PSM (ratio = 3, caliper = 0.2), 30 patients in Group A and 83 patients in Group B were included. The ORR, according to the modified RECIST criteria, was significantly higher in Group A compared to Group B (50% vs. 28.9%, P=0.037). However, no significant difference in PFS (median 10.0 vs. 9.0 months, P=0.29) or OS (median 17.6 vs. 22.8 months, P=0.293) was observed between the two groups. Adverse events (AEs) occurred in all 113 patients (100%), wherein grade 3–4 AEs were reported for 12 (10.6%) patients in group A and 45 (39.8%) in group B. There was a tendency for Group A to have a lower incidence of grade 3–4 AEs compared to Group B (40.0% vs. 54.2%, P=0.182),although this difference was not statistically significant. **Conclusions:** This study found that in the treatment combining ICIs with HAIC and target therapy for unresectable iCCA, Group A demonstrated a superior tumor response and a more favorable safety profile. Research Sponsor: None.

Effects of disease aetiology on outcomes for unresectable HCC treated with lenvatanib or atezobev: A real world retrospective study.

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Background: Hepatocellular Carcinoma (HCC) is the most common primary liver cancer and a leading cause of cancer-death worldwide.[1] Standard of care first line treatment for unresectable HCC is Atezolizumab plus Bevacizumab (AtezoBev)[1] while the multi-kinase inhibitor Lenvatinib[2] is also available in the first line setting. No randomised studies have compared these treatments directly. Real world data suggests similar overall survival (OS) with AtezoBev possibly superior in patients with a viral hepatitis aetiology and Lenvatinib superior in NAFLD.[3,4,5,6,7] **Methods:** We evaluated patients treated with Lenvatinib or AtezoBev in a retrospective cohort of patients who commenced first-line treatment in the West of Scotland from 04/12/2015 to 28/11/2023. 102 received Lenvatinib and 52 AtezoBev. We collected data on progression free survival (PFS), OS, demographics and disease aetiology- classified primarily as viral hepatitis, acholic liver disease (ALD) or NAFLD. **Results:** AtezoBev patients had improved mOS compared with Lenvatinib (13.8 vs 9. 1 months, p = 0.04). No statistically significant differences were observed in mPFS (5.5 vs 3.5 months, p = 0.31). Of the three aetiologies considered, the only statistically significant ($p<0.05$) difference between the treatments was OS in the NAFLD group (24.1 vs 9.1 months, p = 0.04). When considering patient demographic and disease factors, Child-Pugh score was most significant. Patients with Childs=5 liver disease had much longer OS and PFS compared with Childs≥6 disease. Similarly, patients with ECOG PS=0 had improved survival than PS≥1. There were no statistically significant ($p<0.05$) differences between AtezoBev and Lenvatinib. Age, sex and social deprivation (SIMD) score had minimal effects on survival although high levels of social deprivation and male sex were over-represented in our population. **Conclusions:** In contrast to other real world analyses, our patients demonstrated improved mOS with Atezo/Bev vs Lenvatinib.[3] However, mPFS was similar for the two regimes. The difference in OS may be explained by differences in patient selection, with the Lenvatinib group disproportionately representing patients with a poorer performance status (PS>0) and higher Childs-Pugh scores (CP>5). Furthermore, patients in the AtezoBev group who progressed were often offered Lenvatinib second-line, possibly improving mOS, whereas Lenvatinib patients who progressed were offered either Sorafenib, a clinical trial or best supportive care. However, we cannot rule out a direct survival benefit from the use of AtezoBev vs Lenvatinib in the first line setting. The choice of first line therapy for HCC remains complex and must account for patient preferences such as convenience and side effect profile in addition to observed trial outcomes and the increasing body of real world evidence. 1. Finn RS et. al. Atezolizumab plus Bevacizumab in Unresectable Hepatocellular Carcinoma. N. Engl. J. Med. 2020 May 14;382(20):1894-1905. 2. Kudo, Masatoshi et. al. Lenvatinib versus sorafenib in first-line treatment of patients with unresectable hepatocellular carcinoma: a randomised phase 3 non-inferiority trial. Lancet. 2018. Volume 391, Issue 10126, 1163 – 1173. 3. Wang, BC. et. al. Lenvatinib Versus Atezolizumab Plus Bevacizumab in the First-Line Treatment for Unresectable Hepatocellular Carcinoma: A Meta-Analysis of Real-World Studies. Targ. Oncol. 2024. 19, 203–212. 4. Casadei-Gardini A et. al. Atezolizumab plus bevacizumab versus lenvatinib for unresectable hepatocellular carcinoma: a large real-life worldwide population. Eur. J. Cancer. 2023; 180:9-20. 5. Rimini M et. al. Atezolizumab plus bevacizumab versus lenvatinib or sorafenib in non-viral unresectable hepatocellular carcinoma: an international propensity score matching analysis. ESMO Open. 2022 Dec;7(6):100591. 6. Su CW et. al. Similar efficacy and safety between lenvatinib versus atezolizumab plus bevacizumab as the first-line treatment for unresectable hepatocellular carcinoma. Cancer Med. 2023. 12(6):7077-7089. 7. Rimini, M., et al. Survival outcomes from atezolizumab plus bevacizumab versus Lenvatinib in Child Pugh B unresectable hepatocellular carcinoma patients. J. Cancer. Res. Clin. Oncol. 2023. 149, 7565–7577. Research Sponsor: None.

555 Poster Session

Molecular and clinical determinants of targeted therapy (TT) outcomes in biliary tract cancer (BTC): Analysis of a prospectively maintained next generation sequencing (NGS) biorepository.

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Background: BTCs are genomically diverse. TT have established antitumor activity in a subset of pts that harbour actionable alterations; however, most patients (pts) do not respond to or experience acquired resistance on TT. We sought to explore genomic mechanisms of response across the spectrum of BTCs. **Methods:** We leveraged a prospectively maintained cohort of pts with histologically confirmed BTC who underwent molecular profiling using an FDA authorized, targeted, NGS assay. Additional clinicopathologic information was retrospectively extracted with an aim to describe the genomic profile and clinical actionability using OncoKB, to explore outcomes for pts treated with TT, and to nominate innate and acquired genomic mechanisms of resistance. **Results:** N = 1254 pts had molecular profiling: 61.2% intrahepatic (iCCA), 16.8% extrahepatic cholangiocarcinoma (eCCA), 22.1% gallbladder cancer (GBC). MSI_{high} was observed in 2.1% and the median TMB was 3.3 (range: 0-74.6 muts/mb). Among MSS tumors, the most frequently altered genes were *TP53* (36%), *CDKN2A* (21%), *ARID1A* (18%), *KRAS* (17%), and *IDH1/2* (16%). Genomic profiles differed significantly based on anatomic subsite. Oncogenic drivers were mostly clonal except for *ERBB2* amplifications and mutations, which were subclonal in 19%, and 28% of cases respectively. OncoKB level 1/2 events observed in 32.2% (iCCA 40%, eCCA 15%, and GBC 22%). Progression-free survival (PFS) for pts with these oncogenic drivers: *IDH*, *FGFR2*, *ERBB2, BRAF V600E*, MSI_{high}, and TMB_{high} from start of matched TT (N=164) is listed in the table. Clinical factors or co-occurring genomic alterations did not impact outcomes. Profiling of paired pre- and post-progression samples (N=25) nominated pathway reactivation in *FGFR2*, *BRAF*, *NTRK,* and *ERBB2* driven tumors with emergence of recurrent oncogenic *FGFR2*, *RAS*, *MEK*, and *MET* alterations, respectively, as well as off target *MYC*amplification and *CDKN2A* loss. Loss of the oncogenic driver at progression was only observed for *ERBB2* driven tumors. **Conclusions:** Acknowledging the limitations of the study design, matched TT in a real-world, prospectively maintained cohort offers meaningful PFS, most notably in those pts with MSI_{high} BTC treated with immunotherapy. In those with acquired resistance to TT, alterations predicted to reactivate the primary oncogenic pathway were identified. ERBB2 heterogeneity and loss is predicted to be a mechanism of resistance to HER2 targeted therapy. Research Sponsor: None.

Outcomes for BTC treated with matched TT.

Target	Median PFS, months (95% CI)
IDH1/2 **(N = 72)**	4.2 (2.9-6.8)
FGFR2 **(N=37)**	7.4 (6.5-9.4)
ERBB2 **(N =22)**	8.2 (2.95-16.7)
MSI $_{high}$ **(N = 14)**	NR (10.15-NR)
TMB_{high} **(N=12)**	2.69 (1.64-NR)
BRAF **V600E (N=7)**	16.6 (3.7-NR)

Clinical and genomic characteristics of hepatocellular carcinoma-cholangiocarcinoma: Insights from real-world data.

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Background: Combined hepatocellular carcinoma-cholangiocarcinoma (cHCC-CCA) is a rare and complex liver cancer with features of both hepatocellular carcinoma (HCC) and cholangiocarcinoma (CCA), making its management challenging. This study aims to analyze the clinical characteristics, genomic profiles, and treatment outcomes of cHCC-CCA patients in a real-world setting. **Methods:** We conducted a retrospective analysis of cHCC-CCA patients across three first-line treatment groups: chemotherapy alone(Gemcitabine/Cisplatin, FOLFOX), HCC-directed therapies (atezolizumab/bevacizumab, lenvatinib, nivolumab, durvalumab, sorafenib), and chemoimmunotherapy (Gemcitabine/Cisplatin combined with immunotherapy). This international study, involving Mayo Clinic and the National Cancer Center East (Japan), assessed overall survival (OS) using Kaplan-Meier and Cox regression models. **Results:** A total of 68 patients were included, with a mean age of 66 years; 74% were male and 66% were White Non-Hispanics. Cirrhosis was present in 74% of patients at diagnosis, and Hepatitis B and C were common etiologies in 57%. Prior primary resection and regional therapies (Transarterial chemoembolization (TACE), Transarterial radioembolization (TARE)) were recorded in 39% and 50% of patients, respectively. Advanced/metastatic disease was present in 52% of the cohort. The distribution of first-line treatments was as follows: 47% received chemotherapy alone, 39% received HCC-directed therapy, and 14% received chemoimmunotherapy. Genomic alterations were detected in 44% patients, with *TP53* (28%), *ARID1A* (25%), and *TERT* (22%) being the most frequent. The HCC-directed therapy group demonstrated the best overall response rate (52%) and had a median OS of 15.8 months (mo), compared to 11.8 mo for chemotherapy and 4.7 mo for chemoimmunotherapy (p=0.8). Patients who received regional treatments had a longer median OS of 15.8 mo compared to those who did not (10.2 mo) (p=0.4). **Conclusions:** HCC-directed therapy showed a trend towards better survival outcomes compared to other treatments, though not statistically significant. A considerable proportion of patients had genomic alterations, which may have influenced survival outcomes. Ongoing research with larger cohorts is underway to further elucidate the impact of genomic variations on treatment efficacy and refine therapeutic strategies for this challenging malignancy. Research Sponsor: None.

Baseline characteristics of patients with combined hepatocellular carcinoma-cholangiocarcinoma (cHCC-CCA).

	Total (N=68)
Mean age at diagnosis, years (SD)	66(6.4)
Gender, n (%)	
Male	50 (74%)
Ethnicity, n (%)	
White, Non-Hispanic	45 (66%)
Cirrhosis at diagnosis, n (%)	50 (74%)
Etiology of Liver disease, n (%)	
Viral (Hepatitis B and C)	39 (57%)
History of Primary resection, n (%)	27 (39%)
Regional treatment, n (%)	34 (50%)
Afp, n (%)	
< 200	23 (64%)
ca19_9, n (%)	
< 50	27 (82%)

557 Poster Session

Efficacy and safety of endoscopic retrograde cholangiopancreatography (ERCP) vs percutaneous transhepatic biliary drainage (PTBD) for cholangiocarcinoma biliary obstruction: An updated systematic review and meta analysis.

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Background: Perioperative and/or palliative biliary drainage for the management of cholangiocarcinoma biliary obstruction has been part of the backbone of the multidisciplinary approach that this condition requires. For both intrahepatic and perihilar cholangiocarcinoma, ERCP and PTBD are the two main techniques available. There has yet to be consensus favoring one approach over the other. **Methods:** Following the PRISMA guidelines, a systematic review and meta-analysis were performed to determine if there was a difference in efficacy and safety of ERCP compared to PTBD for Cholangiocarcinoma Biliary Obstruction. Primary outcomes were successful biliary drainage, mean number of needed procedures, length of stay and overall adverse events rate. Data were analyzed using R version 4.2.2. The risk of bias was assessed by the Robins-I and RoB 2 tool. The quality of evidence was graded using the GRADE scale and Newcastle-Ottawa guidelines. **Results:** Three randomized clinical trials and two observational studies described the successful biliary drainage rate of ERCP vs PTBD for cholangiocarcinoma associated biliary obstruction, totaling 340 patients. Meta-analysis test for overall effect, showed no statistical difference between groups (RR = 0.88; 95% CI = [0.69; 1.13]; p = 0.23; Fig. 1). Similarly, in terms of the mean number of needed procedures per patient, a total of three randomized clinical trials and two observational studies, totaling 340 patients were included, no statistically significant difference was found (SMD = 0.64; 95% CI = [- 9.36; 10.63]; p = 0.57; Fig. 2). Likewise, one randomized clinical trial and two observational studies showed no statistically significant difference on the length of stay (days) mean between groups (SMD = 0.06; 95% CI = [- 1.57; 1.68]; p = 0.73; Fig. 3). Although, the three clinical trials and three observational studies showed a mild decreased adverse event and death (within 3 months) rate in the ERCP group, after undergoing statistical analysis, meta-analysis test for overall effect, showed no statistical difference (RR = 0.99; 95% CI = [0.55; 1.77]; p = 0.97; Fig. 4 and RR = 0.61; 95% CI = [0.10; 3.87]; p = 0.22; Fig. 5, respectively). The risk of bias was low with moderate-to-high quality of evidence amongst the included studies. **Conclusions:** Our updated systematic review and meta-analysis demonstrated no statistically significant difference in the successful biliary drainage rate, mean number of needed procedures, length of stay, death within 3 months and adverse event rate between ERCP vs PTBD for management of cholangiocarcinoma biliary obstruction. Further randomized studies are needed to confirm these findings. Research Sponsor: None.

Risk of proteinuria with atezolizumab plus bevacizumab versus lenvatinib in first-line systemic treatment for hepatocellular carcinoma.

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Background: Proteinuria presents a challenging complication during systemic therapy for hepatocellular carcinoma (HCC). This study aims to identify risk factors for proteinuria in patients with HCC treated with atezolizumab plus bevacizumab (Atezo/Bev) or lenvatinib (LEN) as first-line systemic treatment. **Methods:** A retrospective analysis was conducted on 622 consecutive patients with unresectable HCC who received Atezo/Bev or LEN as first-line systemic treatment between October 2013 and October 2022. Cumulative incidence of proteinuria was estimated using Kaplan–Meier curves and compared using log-rank tests. Risk factors for proteinuria were identified using Cox proportional-hazard models, along with propensity score-matched and subgroup analyses. **Results:** Among 367 patients treated with Atezo/Bev and 255 with LEN, the cumulative incidence of proteinuria at 12 months was 27.5%. In the multivariable analysis, Atezo/Bev treatment (HR, 1.57; 95% CI, 1.03–2.42), diabetes (HR, 1.64; 95% CI, 1.03–2.61), Child–Pugh class B (HR, 3.43; 95% CI, 1.34–8.78), macrovascular invasion (MVI; HR, 1.58; 95% CI, 1.04–2.38), and an estimated glomerular filtration rate ≤60 mL/min/1.73 m^2 (HR, 3.21; 95% CI, 1.84–5.62) were identified as risk factors for proteinuria. A higher risk of proteinuria in Atezo/Bev patients compared with LEN was consistently observed in the PS-matched cohort, particularly pronounced in subgroups with MVI (HR, 2.84; 95% CI, 1.23–6.54) compared with those without MVI (HR, 1.31; 95% CI, 0.69–2.47). **Conclusions:** Patients treated with Atezo/Bev as first-line systemic treatmentfor HCC exhibited a higher risk of proteinuria compared with those with LEN, particularly when accompanied by MVI. Research Sponsor: None.

559 **Poster Session**

Germline pathogenic variants in a cohort of individuals with biliary tract cancer.

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Background: Biliary tract cancers (BTC) are rare but have been implicated in hereditary cancer conditions such as Lynch syndrome (LS) and *BRCA1* and *BRCA2.* Data from paired somatic and germline genetic testing on BTC have shown that ~5% may be related to germline pathogenic variants in *BRCA1* and *BRCA2.* Identification of these germline pathogenic variants (PV) often provide patients with an opportunity for personalized treatment including platinum-based chemotherapies, PARP inhibitors and/or immunotherapy. Germline testing also provides crucial data to relatives regarding cancer risk and access to increased cancer surveillance and risk-reducing surgeries. Current guidelines recommend germline genetic testing when the likelihood of identifying a PV is >5%. The contribution of germline PV in BTC has not been described in a diverse patient population. **Methods:** In this prospective clinic-based study, individuals with BTC were referred by their oncologist for genetic counselling and testing. The majority of individuals were enrolled based on participation in the Legresley Biliary Registry, which recruits all individuals with BTC. Blood was sent for germline multi-gene panel testing. **Results:** Germline genetic testing was performed for 139 individuals. Average age of diagnosis was 57.8 years (range 28-84). The majority of individuals were diagnosed with intrahepatic cholangiocarcinoma (CCA) (54.7%), followed by extrahepatic CCA (20.9%) and gallbladder cancer (10.1%). The majority of individuals were male (56.1%). The population was ethnically diverse with 55% European, 24% Asian, 9% Middle Eastern/North African and 6% African. A minority of individuals had a previous cancer (18%) and 24.5% met current provincial eligibility criteria. Germline testing was complete on 136 individuals. 25/136 (14.7%) PV were identified in 20 individuals. 8/136 (5.9%) of the cohort carried PV in genes known to increase the risk for BTC (*BRCA1*, *BRCA2* and LS genes), 9/136 (6.6%) carried PV in other genes with clinically actionability (e.g. *PALB2*, *ATM*), and 8/136 (5.9%) carried heterozygous PV in genes for recessive diseases. Variants of uncertain significance were identified in 42/136 (30.9%) and negative results were identified in 74/136 (54.4%). Personal and/or family history was not suggestive of the associated cancer condition in 9/16 (56%) of the PV cohort. Tumor profiling by whole genome sequencing on some individuals with PV found corresponding somatic mutational signatures, consistent with variant pathogenicity. **Conclusions:** Detection rates for PV in a diverse BTC cohort was up to 14.7%, including 5.9% among genes known to increase the risk for BTC. The majority of PV were found in individuals lacking personal and/or family history suggestive of the associated hereditary cancer condition. These results suggest that guidelines should be updated to recommend universal germline genetic testing for individuals with BTC.
Research Sponsor: None.

End-of-life (EOL) outcomes and healthcare utilization for patients with hepatocellular carcinoma (HCC) who received immune checkpoint inhibition (ICI).

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Background: ICI is standard of care treatment for advanced HCC. EOL outcomes for patients with HCC who receive ICI prior to death are unknown. We examine relationships between EOL outcomes, healthcare utilization, and receipt of ICI for patients with advanced HCC referred to our tertiary center. **Methods:** Patients with advanced HCC evaluated at our center on or after January 1, 2020 and who died by March 29, 2024 were included. Patients were identified using diagnostic codes for liver cancer and HCC, and subsequently verified by manual review of the electronic health record. Demographic data, Child-Pugh (CP) status, and treatment history were collected. Primary EOL outcomes include: documentation of advance directives and goals of care (GOC) conversations, location of death, palliative care referral, hospice referral, days in hospice, and code status. Secondary healthcare utilization outcomes include: systemic therapy receipt, emergency department (ED) visits, hospitalization, and intensive care unit (ICU) admissions within 14, 30, and 90 days of death. Outcomes were stratified by ICI or non-ICI as last therapy received and p-values were derived using Pearson's chi-square test for equality of proportions. **Results:** Of 221 evaluated patients, 71 died and met criteria for analysis. Baseline characteristics include mean age 64.1 years, 70.8% male, 71.8% received systemic treatment (median 1 line, range [1-7]), 54.1% CPA, 37.5% CPB, and 8.3% CPC at the time of last treatment. No statistically significant differences in primary EOL outcomes were detected when stratified by receipt of ICI or non-ICI as last therapy. Median days enrolled in hospice were not statistically significant between groups (24.5 days [non-ICI] vs 10 days [ICI]; p = 0.39). A higher proportion of patients who received ICI as last therapy had increased healthcare utilization across all outcomes (see Table). **Conclusions:** Patients with advanced HCC receiving ICI as their last treatment before death compared to those receiving non-ICI had similar EOL outcomes but higher healthcare utilization. This may be due to increased real-world use of ICI in CPB patients. Further investigation into risk stratification to predict high healthcare utilizers could guide decision-making and conversations around ongoing use of ICI, particularly near the EOL. Research Sponsor: U.S. National Institutes of Health; 2T32CA217834-07.

Healthcare utilization outcomes stratified by receipt of ICI.

Outcome Before Death *n*= 71	90 days	30 days	14 days
Therapy within			
Total:	33/71 (46.5%)	18/71 (25.4%)	6/71 (8.5%)
Last therapy ICI	69.7%	66.7%	33.3%
ED visit			
Total:	29/71 (40.8%)	27/71 (38.0%)	22/71 (30.1%)
Last therapy ICI	58.6%	55.6%	59.1%
Hospitalization			
Total:	32/71 (45.1%)	30/71 (42.3%)	29/71 (40.8%)
Last therapy ICI	59.4%	56.6%	58.6%
ICU admission			
Total:	6/71 (8.5%)	6/71 (8.5%)	6/71 (8.5%)
Last therapy ICI	66.7%	66.7%	66.7%

The impact of adherence to NCCN and BCLC treatment guidelines on survival in HCC: A single center study.

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Background: Hepatocellular carcinoma (HCC) is a prevalent and lethal disease in Saudi Arabia, and globally. Guidelines such as Barcelona Clinic Liver Cancer (BCLC) and National Comprehensive Cancer Network (NCCN) are used to guide therapy for this disease. Adherence to these guidelines should lead to better outcomes. We aim to examine the adherence to guidelines and its impact on survival in a real world setting. **Methods:** This retrospective cross-sectional study was conducted at the Ministry of National Guard for Health Affairs, Riyadh, Saudi Arabia. Study included all adult patients with HCC who were diagnosed between 2020 and 2023, and presented at our HCC multidisciplinary tumor board (MDT). Data extracted included demographics, clinical and disease characteristics, treatment modalities, and survival status. Chi-square tests, and Kaplan-Meier survival analysis were used to explore relationships between tumor characteristics, adherence to NCCN and BCLC guidelines, and survival outcomes. Adherence to guidelines were assessed by two independent physicians. **Results:** The study included 208 patients with HCC, the majority were males (73%) with a mean age of 73 years. Less than half of the patients (41.7%) had an Eastern Cooperative Oncology Group (ECOG) performance status of 1. Tumor staging showed that 35% of the patients had BCLC stage D. Child pugh class A and B were equally present in 45% of the patients. Moreover, 24% of the patients had distant metastases or vascular invasion (in portal vein) placing them in BCLC stage C. It is noteworthy that Alpha-Fetoprotein (AFP) was abnormal (>8ng/ml) in 58% of the patients, 18% of them had AFP of more than 400ng/ml. Additionally, the prevalence of comorbid conditions were remarkable for liver cirrhosis being present in 85% of patients, while diabetes mellitus and hypertension each present in 63% of patients. Finally, survival analysis showed a median survival of 16.8 months in all patients with HCC (range 0-47.8 months). Adherence to guidelines were high, with 97.6% (n=203) following NCCN guidelines and 70.7% (n=147) following BCLC guidelines. Following both guidelines showed better survival with mean of 37 months (SD=1.841) in patients with child pugh class A (n=53), and mean survival of 24 months (SD=2.251) in patients with child pugh class B (n=74). **Conclusions:** Adherence to NCCN guidelines was high in our MDT presented patients compared to BCLC guidelines adherence which was lower, but with comparable survival outcomes. Research Sponsor: None.

A global comparative study on MDT practices in the management of hepatocellular carcinoma in high-income countries (HICs) and low- and middle-income countries (LMICs).

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Background: As the sixth most common malignancy worldwide, hepatocellular carcinoma (HCC) accounted for 4.7% of all new cancer diagnoses in 2022. Low- and middle-income countries seem to bear a disproportionate burden of the disease, with over 50% of new HCC cases estimated to occur in China and Africa. Research on multidisciplinary team (MDT) practices in these regions is limited. The objective of this study is to compare MDT practices in the management of HCC in HICs and LMICs. **Methods:** Data on MDT practices from hospitals that manage HCC in Italy, Spain, Switzerland, Germany, Denmark, Canada and the USA, representing HICs, and in China, South Africa and Egypt, representing LMICs were collected through virtual semi-structured interviews and workshops. **Results:** Ten key areas of differences were identified (Table 1) that could be categorized as organizational/regulatory or clinical. In HIC institutions, the focus currently is on optimizing MDTs as standard of care approach accessible to all patients with HCC; MDT principles are often integrated into national policies. In LMICs, emphasis is on establishing MDTs and expanding their capacity; patient access to MDT care is not a mandatory requirement, nor regulated in national policy. Of note, LMICs are seeking early technology adoption to overcome some potential barriers to MDT interactions; online formats are more often explored to facilitate participation and AI-assisted diagnostic tools are explored to compensate for absence of radiologists. Absence of screening and surveillance for early detection in HCC is more common in LMICs, and therefore likely that patients more often present with advanced stage disease. Also, with limited diagnostic modalities, diagnostic work-up and staging may be sub-optimal. **Conclusions:** Significant variation exists in HCC MDT practices between HICs and LMICs. The ultimate goal of this study is to develop a format for clinical MDTs that will facilitate interaction between HICs and LMICs. Consideration for the differences highlighted above is crucial for the development of this concept. Research Sponsor: None.

10 key areas highlighting differences identified between HICs and LMICs.
MDT development direction: Standard of Care for all patients vs. specific session for complex cases
National/regional regulatory requirements Institutionalizing MDT principles MDT composition (specialists) Use of digital tools Reimbursement Access to therapeutic options Stage presentation of HCC patients Use of clinical staging Patient follow-up after MDT discussion

563 Poster Session

Leading the way: How leading Chinese institutions drive MDT excellence in hepatocellular carcinoma.

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Background: HCC MDTs involve specialists collaborating on patient cases to create personalized treatment plans. Implementing MDTs can be challenging in countries with limited resources and high disease burdens. In China, there were 368,000 new liver cancer cases in 2022 representative of 40% of global cases. Therefore, evaluating MDT practices at leading hospitals in China may offer valuable insights that could help improve HCC MDT practices worldwide. **Methods:** This study selected hospitals with distinct MDT models: one for all new patients and one for complex cases. Hospitals handling over 1,000 HCC cases annually were chosen. Desk research and virtual interviews with MDT members from three leading Chinese hospitals—The First Affiliated Hospital of USTC, The First Affiliated Hospital of SYSU, and Tongji Hospital—were conducted to analyze their MDTs and determine key best practices. **Results:** We identified 6 key best practice in HCC MDTs at these leading Chinese hospitals. 1) Quality Control on MDT process: Leading hospitals ensure the MDT's efficient execution and effective outcomes. E.g., USTC established a quality control team to audit the MDT process. 2) Real-World Evidence: Leading hospitals prioritize the construction of real-world databases linked to MDT-care to inform impact on outcomes. E.g., SYSU's preliminary data shows impact of MDT care on survival rates (From 2006 to 2010, 5,627 patients who underwent radical liver cancer resection with MDT care saw their five-year survival rate reach 60%, a notable improvement over historical averages) 3) Comprehensive Teams: These hospitals typically have robust teams, involving around 10 specialties. E.g., Tongji Hospital's MDTs involve ~10 specialties for holistic care. 4) Patient Journey Coverage: MDTs manage patients from diagnosis to rehabilitation, providing a holistic approach to patient management. 5) Guideline-based treatment decisions: They follow latest guidelines coupled with access to approved therapies to make treatment decisions 6) Training: Young physicians are invited to actively participate in MDT consultations (e.g., through case preparation and presentation, etc.). **Conclusions:** Evaluation of MDT practices at leading China institutes highlights key best practices that could be adopted to enhance care quality for patients with HCC. The ongoing efforts and successes of these institutions provide a model for broader adoption and standardization of MDT practices across the country. Research Sponsor: None.

Hepatotoxicity following external radiotherapy in patients with hepatocellular carcinoma treated with or without prior yttrium-90 radioembolization.

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Background: A variety of local therapy options exist to treat Hepatocellular Carcinoma (HCC) including external radiotherapy (EBRT). Prior yttrium-90 (Y90) delivered transarterially is often a contraindication to EBRT in clinical trials and guidelines due to concerns of cumulative hepatotoxicity, though clinical data to suggest higher rates of hepatotoxicity is limited. This study examines hepatotoxicity profiles after EBRT with and without prior Y90 treatment using the ALBI grading system. **Methods:** Patients treated for HCC with EBRT from 2016 to 2024 with sufficient laboratory follow-up were retrospectively reviewed. Patients were classified into three ALBI grades based on ALBI scores: 1 (< -2.6), 2 ($-2.6 < x < -1.39$), and 3 (> -1.39). The total EBRT dose was defined using the EQD2 method ($\alpha/\beta = 10$). The ALBI scores at 3- and 6-months post-EBRT were compared to baseline for patients with or without prior Y90 using paired t-tests. The main RT planning goal was ≥700 cc liver- (tumor volume+ Y90 volume) receiving ≤15 Gy. **Results:** 62 patients were treated with EBRT with (n=14) or without (n=48) prior Y90. At 6-months post-EBRT, 11 patients with prior Y90 and 39 patients without prior Y90 had appropriate laboratory follow-up. EBRT dose, technique, and baseline liver function are outlined in Table 1. The median time from Y90 to EBRT was 12.3 months (range 4-61 months). In patients with prior Y90, ALBI scores were similar at 3-mo (mean = -2.4 SD = 0.8, vs mean= -2.4 SD = 0.7, p= 0.60) and 6-mo (mean= -2.3 SD= 0.9, vs mean = -2.4 SD = 0.6, p= 0.653) post-EBRT compared to baseline, respectively. In patients without Y90, ALBI scores were clinically similar at 3-mo (mean = -2.2 SD = 0.7, vs mean= -2.3 SD = 0.6, p= 0.15) and 6-mo (mean= -2.1 SD= 0.8, vs mean = -2.3 SD = 0.7, p=0.021) post-EBRT compared to baseline, respectively. The proportion of patients with an increase in ALBI grade at 3- and 6-mo post-EBRT was similar regardless of prior Y90 treatment. In patients treated with Y90, 14.3% and 36.4% had an increase in ALBI grade at 3- and 6-months, respectively. In those treated without prior Y90, 21.3% and 35.9% had an increase in ALBI grade at 3- and 6-months, respectively. **Conclusions:** Patients with HCC treated with EBRT after Y90 appear to have a similar rate of hepatotoxicity defined by the ALBI grading system compared to patients treated without prior Y90. In well-selected patients, prior Y90 should not be an absolute contraindication for EBRT. Research Sponsor: None.

Demographics and ALBI changes post-EBRT.

	EBRT alone (n = 48)	EBRT after Y90 (n = 14)
Median Age Years (range)	67.6 (17.9-84.3)	65.6 (55.1-85.2)
Proton, IMRT/SBRT	22 (46%), 26 (54%)	9 (64%), 5 (36%)
Baseline Child Pugh Score (A, B, C)	34 (70.8%), 10 (20.8%), 4 (8.4%)	11 (78.6%), 3 (21.4%), 0 (0%)
Median EBRT EQD2 (range)	82 Gy (44-99)	82 Gy (67-86)
Baseline ALBI Grade (1, 2, 3)	23 (47.9%), 20 (41.7%), 5 (10.4%)	6 (42.9%), 7 (50%), 1 (7.1%)

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Clinical outcomes of stereotactic body radiation therapy to the liver.

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Background: Stereotactic body radiation therapy (SBRT) can accurately and effectively target liver lesions with an ablative dose. However, prior studies suggest that SBRT directed at tumors near the central hepatobiliary tract may lead to downstream toxicity including biliary stricture and impaired liver function. This study aims to evaluate clinical outcomes and hepatobiliary toxicities associated with SBRT to the liver. **Methods:** This retrospective study reviews 177 patients who received CT- or MR-guided SBRT to the liver at Dana-Farber Cancer Institute and Brigham and Women's Hospital from 2010 to 2024. Patient demographics, treatment characteristics, and clinical outcomes were extracted from electronic medical records. Local control, regional control, and overall survival were assessed using Kaplan-Meier methods. Toxicity was evaluated by tracking the incidence of biliary stricture requiring intervention and Child-Pugh (CP) scores pre- and post-SBRT. **Results:** Of the 177 patients, 47 (26.6%) had primary disease and 130 (73.4%) had metastatic disease, with a median follow-up of 58.7 months. Among those with primary disease, 9 (19.1%) had post-SBRT local progression (median time to progression not reached, 95% CI: 47.3, NA) and 26 (55.3%) had regional progression (median time to progression of 20.5 months, 95% CI: 6.5, 58.1). Of patients with metastatic disease, 44 (33.8%) had local progression (median time to progression of 40.2 months, 95% CI: 23.0, NA) and 96 (73.8%) had regional progression (median time to progression of 5.9 months, 95% CI: 4.8, 9.3). Median overall survival post-SBRT was 34.6 months for primary disease (95% CI: 20.2, 57.7) and 23.6 months for metastatic disease (95% CI: 19.0, 28.8). 18 patients (10.1%) later developed biliary stricture requiring intervention (12.8% of primary disease and 9.2% of metastatic disease). Of the 119 patients with CP scores available, 59 patients (49.6%) had a maximum post-SBRT CP score increase of 2 or more (55.0% of primary disease, 46.8% of metastatic disease). **Conclusions:** This study demonstrates that SBRT is an effective treatment option for local control in the liver, with better survival and progression outcomes for primary disease compared to metastatic disease. Among potential post-SBRT toxicities, initial analysis indicates frequent increase in CP score and low incidence of biliary stricture requiring intervention. An ongoing analysis will explore whether the dose to central bile ducts correlates with late strictures and toxicities. Research Sponsor: Tufts University School of Medicine.

566 **Poster Session**

Association between immune-related adverse events and prognosis after immune checkpoint inhibitor in hepatocellular carcinoma: A meta-analysis and systematic review.

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Background: Immune checkpoint inhibitors (ICIs) have become the preferred treatment for unresectable hepatocellular carcinoma (HCC). Conflicting data exist regarding the prognostic value of immune-related adverse events (irAEs) in these patients. We aimed to explore potential the association between irAEs and prognosis in HCC patients treated with ICIs. **Methods:** We searched PubMed, Scopus, Web of Science, and CENTRAL databases for articles published from inception to June 2024 using keywords including ICI, HCC, and irAEs. Two reviewers independently screened studies for eligibility and relevant data. A random effects model was used for statistical analysis, and subgroup analysis was performed by the grade of irAEs. **Results:** Of 3,028 studies, 25 (n=4,116 patients) met criteria for inclusion. Atezolizumab plus bevacizumab was the most common treatment regimen (n=10 studies). About half (49.5%) of patients experienced irAEs. irAEs were associated with increased objective response rate (ORR) (pooled OR: 1.72; 95% CI: 1.36 – 2.19, I^2=39%), higher complete response rate (pooled OR: 1.45; 95% CI: 1.21-1.74, I^2=74%), and longer progression-free survival (PFS) (pooled HR: 0.66; 95% CI: 0.52-0.84, I^2=71%). Although irAEs were associated with longer overall survival, this difference was not statistically significant (pooled HR: 0.83; 95% CI: 0.62 – 1.11, I^2=73%). Subgroup analysis showed a survival benefit for patients with grade 1-2 irAEs (pooled HR: 0.50; 95% CI: 0.36-0.67, I^2=0%) but not for those with grade 3-4 irAEs (pooled HR: 0.95; 95% CI: 0.24-3.72, I^2=84%). **Conclusions:** The development of irAEs is associated with a favorable prognosis in HCC, including improved PFS and higher ORR. A survival benefit was noted in patients with mild irAEs, but not in those with severe irAEs. Future studies are needed to determine if a differential survival benefit relates to the continuation vs. discontinuation of ICI therapy. Research Sponsor: None.

A multicenter, prospective observational study of chemotherapy in elderly patients with biliary tract cancer.

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Background: In addition to gemcitabine (GEM) plus cisplatin (GC), GEM plus S-1 (GS) and GC plus S-1 (GCS) have been used as standard treatments for advanced biliary tract cancer (aBTC) based on results from phase III trials in Japan prior to the introduction of immune checkpoint inhibitors. The aim of this study was to evaluate the efficacy and safety of these chemotherapies in elderly patients (pts) with aBTC. **Methods:** This was a multicenter, prospective observational study (UMIN000045156). Inclusion criteria were: age >=70 years; unresectable or recurrent BTC; and scheduled for chemotherapy with GCS, GC, GS, GEM, or S-1. Data were prospectively collected on treatment tolerability, safety and geriatric assessments (G8, IADL, CCI), and overall survival (OS), progression-free survival (PFS), and objective response rate (ORR) were evaluated. Inverse probability-weighted propensity score analyses were performed for comparisons of GCS versus GC and of GC versus GEM. **Results:** A total of 305 pts were enrolled from 53 Japanese hospitals between August 2021 and January 2023. Of these, 294 received chemotherapy (GCS, n=75; GC, n=131; GS, n=26; GEM, n=52; S-1, n=10). Median follow-up was 11.8 months for all pts. Median age was 76 years (range: 70-89 years). Adverse events are summarized in the Table 1. Rates of pts without G8 deterioration after 3 months were 58.7% for GCS, 55.7% for GC, 34.6% for GS, 42.3% for GEM, and 20% for S-1. Rates of pts without IADL deterioration after 3 months were 82.7% for GCS, 73.3% for GC, 42.3% for GS, 57.7% for GEM, and 40.0% for S-1. With inverse probability-weighted propensity score analyses, median OS was 16.4 months (95% confidence interval [CI], 14.7–20.6 months) in the GCS group and 13.3 months (95%CI, 11.7–21.9 months) in the GC group (hazard ratio [HR] 0.80, 95%CI 0.55–1.17; p=0.258). Median PFS was 11.8 months (95%CI, 9.7–14.5 months) in the GCS group and 7.7 months (95%CI, 6.0–10.2 months) in the GC group (HR 0.55, 95%CI 0.38–0.80; p=0.002). ORR was 31.0% in the GCS group and 15.5% in the GC group (p<0.001). Median OS was 13.3 months (95%CI 11.1–19.4 months) in the GC group and 15.5 months (95%CI 6.4–18.7 months) in the GEM group (HR 0.74, 95%CI 0.42–1.29; p=0.282). Median PFS was 6.9 months (95%CI 6.0–9.6 months) in the GC group and 5.1 months (95%CI 3.0–12.0 months) in the GEM group (HR 0.79, 95%CI 0.42–1.49; p=0.463). ORR was 14.1% in the GC group and 7.5% in the GEM group (p=0.305). **Conclusions:** GCS is an effective therapy compared to GC in elderly pts with aBTC without increasing toxicities or deteriorating activities of daily living. Clinical trial information: UMIN000045156. Research Sponsor: None.

Chemotherapy-related adverse events (grade 3 and 4).

Events, n (%)	GCS (n=75)	GC (n=131)	GS (n=26)	GEM (n=52)	S-1 (n=10)
Neutropenia	20 (26.7)	54 (41.2)	8 (30.8)	10 (19.2)	0
Thrombocytopenia	8 (10.7)	9 (6.9)	2 (7.7)	1 (1.9)	0
Anorexia	2 (2.7)	6 (4.6)	2 (7.7)	3 (5.8)	3 (30.0)
Malaise	0	8 (6.1)	4 (15.4)	4 (7.7)	3 (30.0)
Febrile neutropenia	1 (1.3)	1 (0.8)	2 (7.7)	1 (1.9)	0

Prompt initiation of durvalumab and tremelimumab treatment for unresectable hepatocellular carcinoma in patients with chronic active hepatitis B.

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Background: Chronic hepatitis B virus (HBV) infection is a common etiology of HCC, especially in East Asia. Clinical trials using immune checkpoint inhibitors (ICIs) usually exclude patients with chronic active hepatitis B (serum HBV viral load > 2000 IU/mL) or require the HBV viral load to be below a certain level using anti-HBV medications. This study examines the safety and efficacy of concurrently administering the STRIDE regimen (durvalumab with a single dose of tremelimumab) or durvalumab alone with anti-HBV medications in this patient population. **Methods:** We enrolled patients with advanced HCC, Child-Pugh A liver function reserve, and untreated chronic active hepatitis B. Initially, only patients naïve to ICIs were eligible, and patients received 1500mg of durvalumab intravenously every four weeks for up to two years except for disease progression or occurrence of unacceptable toxicities. After the positive results of the HIMALAYA study, we shifted to the STRIDE (single tremelimumab regular interval durvalumab) regimen, adding one dose of 300mg tremelimumab on the first day to regular durvalumab treatment, and allowed patients who had received prior programmed cell death-1 blockade therapy to be enrolled. Anti-HBV treatment with entecavir was initiated within seven days before starting ICI therapy. The primary endpoint was the rate of HBV reactivation (defined as a ≥2 log increase in serum HBV DNA from baseline) during the first six months. **Results:** Following the protocol design, 30 patients were enrolled. Two patients were female, and the mean age was 66.9 years. Ten patients received durvalumab alone, and 20 received the STRIDE regimen. Macrovascular invasion and extrahepatic spread were present in 16 (53.3%) and 20 (66.7%) patients, respectively; 18 (60%) patients had alpha-fetoprotein level ≥ 400 ng/mL. More than half (60.0%) of the patients had received prior systemic therapy for HCC, including 2 patients who had received PD-1 blockade therapy. Seven patients received the study treatment as the 3rd or later line of systemic therapy. The mean±standard deviation baseline HBV viral load was 771.0±354.1 KIU/mL. No patients experienced HBV reactivation or HBV-associated hepatitis. Hepatitis flare was noted in 8 (26.7%) patients, but none of them were associated with HBV reactivation. The objective tumor response rate was 10% and 25% for the durvalumab treatment alone and the STRIDE regimen, respectively. No unexpected toxicities were identified in this study. The most common treatment-related adverse events were skin rash and liver function test abnormalities. **Conclusions:** For patients with chronic active hepatitis B, ICI therapy could be promptly initiated as long as anti-HBV medications were administered simultaneously. This study was partially supported by AstraZeneca. Clinical trial information: NCT04294498. Research Sponsor: National Science and Technology Council, Taiwan; NSC-112-2314-B-002-267, NSC 113-2314-B-002 -001, and 113-2923-B-002-001-MY3; AstraZeneca.

569 **Poster Session**

Real-world Child-Pugh scores and outcomes of patients with hepatocellular carcinoma at a high-volume center.

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Background: Hepatocellular carcinoma (HCC) is the most common primary liver cancer and most often occurs in the setting of cirrhosis. Patients (pts) with advanced HCC are not eligible for curative intent therapies such as ablation, surgical resection, or liver transplantation. For the past five years, immune checkpoint inhibitor (ICI)-based systemic therapy regimens have been the mainstay of first-line treatment for pts with advanced HCC. Most trials assessing ICI-containing regimens, including IMBRAVE150 and HIMALAYA, excluded pts with Child-Pugh (CP) scores worse than A. Unfortunately, fewer than half of pts maintain a CP A score at the time of HCC diagnosis and referral to medical oncology. Real-world data on CP scores of patients undergoing ICI-based therapy for advanced HCC at the time of medical oncology referral and treatment start are lacking. **Methods:** Pts who initiated treatment with either atezolizumab plus bevacizumab (A+B) or durvalumab plus tremelimumab (D+T) in the Ochsner Health system between 1/1/20 and 12/31/23 were included in this analysis. CP scores were calculated at the time of their initial medical oncology visit and again at the time of their first systemic therapy cycle. Pt demographics and clinical outcomes were analyzed. **Results:** 200 pts were identified in this study, 159 of whom (79.5%) were male. Median age at treatment start was 66.8 years. 117 pts (58.5%) were Caucasian and 63 pts (31.5%) were African American (AA). 152 and 48 pts received A+B and D+T, respectively. At the time of first medical oncology visit, 58 pts (29%) were classified as CP B (n=51) or C (n=7). At the treatment start date, 82 pts (41%) were classified as CP B (n=73) or C (n=9). The median survival across all groups was 10.5 m. Median survival CP A, B, and C pts were 15.4, 7.6, and 1.6 months (m), respectively. 6-month survival rates for CP A, B, and C pts were 75.7%, 52.2%, and 0%, respectively. 12-month survival rates for CP A, B, and C pts were 58.2%, 35.1%, and 0%, respectively. There was no statistically significant difference in CP score at either time point between Caucasian and AA pts. **Conclusions:** Although pts with CP scores worse than A were not included in the pivotal studies that have established recent standard of care therapies for pts with advanced HCC, a significant proportion of pts being treated with ICI-based regimens have at least CP B liver disease, with a poorer prognosis in that subset of pts. Extremely poor survival rates for those with CP C scores argue against use of ICI-based treatment for these pts. In this analysis, no significant difference was observed in CP scores between Caucasian and AA pts. Research Sponsor: None.

Adjuvant chemoradiation combined with immunotherapy for patients with high-risk resectable extrahepatic cholangiocarcinoma and gallbladder cancer: A phase II, multicenter, randomized controlled trial (ACCORD trial).

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Background: Extrahepatic cholangiocarcinoma (ECC) and gallbladder cancer (GBC), as the majority of biliary tract cancer (BTC), has a markedly high risk of recurrence after surgery. However, adjuvant treatments specifically for resectable ECC and GBC patients are scare and adjuvant chemotherapy alone delivers limited efficacy. Immunotherapy and radiotherapy are potential effective treatments and both of them may synergize with chemotherapy. **Methods:** ACCORD was a multicenter, phase 2, randomized controlled trial to assess the efficacy and safety of chemoradiation with immunotherapy as an adjuvant treatment in resectable ECC/GBC, compared to observation. The primary endpoint was overall survival (OS) and the secondary endpoints included recurrence-free survival (RFS) and safety. Patients in the chemoradiation-anti-PD-1 group received Camrelizumab intravenously every 3 weeks after surgery. Camrelizumab therapy was not terminated until disease progression, unacceptable toxic effects occurred or informed consent withdrawal. After 2 courses of Camrelizumab treatment, chemoradiation was carried out simultaneously. Patients in the observation group received no anticancer treatment unless relapse was detected. **Results:** From March 2020 to June 2022, a total of 93 ECC/GBC patients after curative resection were randomized 1:1 into chemoradiation-anti-PD-1 group (n=46, Camrelizumab + concurrent Capecitabine and radiotherapy) and observation group (n=47). The 1-year, 2-year and 3-year OS rate were 95.7%, 71.4%, and 58.2% in the chemoradiation-anti-PD-1 group, and 80.9%, 52.9%, 30.5% in the observation group (Hazard ratio (HR) 0.43, 95% confidence interval 0.24-0.79; P=0.004). The 1-year, 2-year and 3-year RFS rate were 78.3%, 54.0%, and 40.3% in the chemoradiation-anti-PD-1 group, and 55.3%, 27.0%, 17.2% in the observation group, with a hazard ratio of 0.46 (95% CI, 0.28 to 0.76; $P < 0.001$). In the chemoradiation-anti-PD-1 group, the main adverse effect ≥ grade 3 were anemia (7 [15.2%]), dermatitis radiation (5 [10.9%]), and nausea (5 [10.9%]); and 100% of patients completed the whole treatment. **Conclusions:** Chemoradiation combined with immunotherapy as an adjuvant therapy demonstrated superior survival outcomes over observation in resectable ECC/GBC patients with a well-tolerable safety profile, supporting the potential of this combination treatment as effective adjuvant therapy for these high-risk patients. Clinical trial information: NCT04333927. Research Sponsor: National Natural Science Foundation of China; Research and Development Plan for Key Areas in Guangdong Province; Key Research and Development Plan of Guangzhou City.

571 Poster Session

Updated results of the PATH study: Adjuvant therapy with donafenib plus a PD-1 inhibitor for patients with hepatocellular carcinoma at high risk of recurrence after resection.

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Background: There is no universally accepted standard treatment option or strategy for post-surgery adjuvant therapy in hepatocellular carcinoma (HCC). Among BCLC A/B patients (pts) with MVI, adjuvant transarterial chemoembolization has been reported to provide a 1-year recurrence-free survival (RFS) rate of around 60-70%. We conducted a pilot study to explore the utility of donafenib combined with a PD-1 inhibitor as adjuvant therapy for pts at high risk of recurrence. Herein, we present the updated results with a median follow-up of 11.6 months. **Methods:** It was planned to enroll 30 HCC pts who had undergone radical surgery and presented with at least one of the following risk factors: a) Presence of microvascular invasion (MVI); b) Presence of satellite nodule(s); c) More than three tumor nodules; d) Portal vein tumor thrombus. Pts with invasion of the main portal venous system were excluded. Donafenib combined with a PD-1 inhibitor (i.e., toripalimab) was initiated at 4 to 8 weeks post-resection and was to be continued for up to six months, unless recurrence occurred or there was an intolerable adverse reaction. No other adjuvant therapies were permitted before enrollment and during the study, except for anti-virus treatment. The primary endpoint was the cumulative 1-year RFS rate. The secondary endpoints included overall survival, quality of life (QoL) measured with the FACT- Hepatobiliary questionnaire, and safety. **Results:** From June 2020 to November 2023, a total of 30 pts were recruited. 27 pts were included in the efficacy analysis. The median time from surgery to enrollment was 1.4 months. MVI was the most common risk factor (n=25, 92.6%). The majority of the pts had only one risk factor (n=23, 85.2%). As of the data cutoff on August 13, 2024, relapse had occurred in four pts and the median RFS was not reached. The 1-year RFS rate was 81.6% (90% CI, 67.8%-95.4%) in all evaluable pts. QoL showed a mild increase from baseline while on adjuvant therapy. No deaths were observed. In the 30 pts who had received at least one dose, the incidence of any grade and grade 3 treatment-related adverse events (TRAEs) were 90.0% (27/30) and 40.0% (12/30), respectively. No grade 4 or 5 TRAEs were reported. The grade 3 TRAEs occurring in at least two pts were palmar-plantar erythrodysesthesia syndrome (13.3%), rash (13.3%), increased alanine aminotransferase (10.0%), and increased aspartate aminotransferase (6.7%). **Conclusions:** The study's duration exceeded expectations due to slow recruitment. Nonetheless, it showed a promising outcome with a regimen of 6 months of adjuvant therapy with donafenib plus toripalimab in BCLC A/B pts at high risk of recurrence. An improvement in QoL was observed among patients during the adjuvant therapy. No new safety issues were identified. Clinical trial information: NCT04418401. Research Sponsor: None.

Patient (pt) characteristics, treatment patterns, and outcomes in unresectable hepatocellular carcinoma (uHCC) treated with first-line (1L) systemic therapy in the United States (US).

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Background: Guidelines recommend systemic therapy for uHCC ineligible for locoregional therapy. Until the approval of tremelimumab + durvalumab for uHCC, in Oct 2022, 1L systemic therapies in the US included multikinase inhibitors and atezolizumab (A) + bevacizumab (B). It is crucial to understand factors driving real-world treatment decisions and outcomes in uHCC. Characteristics, treatment patterns, and outcomes were assessed in US pts with uHCC treated with 1L systemic therapy. **Methods:** Data were collected from electronic medical records from US Mayo Clinic sites. Pts with uHCC who initiated an NCCN-recommended 1L systemic therapy between Jun 2020–Oct 2022 (A + B was the only approved 1L immunotherapy (IO)-based regimen) with ≥2 follow-up visits, were included. Index was the date of first 1L systemic therapy. Pts were assessed by risk of post-index gastrointestinal (GI) bleeding (pts with Child-Pugh class B or C, pre-index GI bleeding, uncontrolled hypertension [HTN] [≥2 anti-HTN drugs or ≥2 vital entries of SBP >140 mmHg or DBP >90 mmHg]), or significant varices and band ligation). Treatment patterns, overall survival (OS), and post-index GI bleeding were assessed. **Results:** A total of 186 pts met the inclusion criteria. Most common etiologies of liver disease were metabolic dysfunction-associated steatohepatitis (42.7%) and hepatitis C virus (40.2%). Pre-index GI bleeding was reported in 29.0% of pts, of which 46.3% occurred within 6 months (mo) pre-index; 62.4% of pts had EGD, of which 72.4% had varices. There were 128 (68.8%) pts with GI bleeding risk and 58 (31.2%) pts without GI bleeding risk. The most common 1L systemic therapies for pts with GI bleeding risk were A + B (72.7%) and A only (9.4%), while for pts without GI bleeding risk, A + B (29.3%), nivolumab + ipilimumab (13.8%), and B + chemotherapy (13.8%) were the most common. Median OS and OS rates at 12, 18, and 24 mo were lower, and post-index GI bleeding rate was higher in pts with vs without GI bleeding risk (Table). For pts treated with A + B, median OS and OS rates at 12, 18, and 24 mo were lower in pts with vs without GI bleeding risk (Table). **Conclusions:** In this network, prior to the approval of other 1L IO-based regimens for uHCC, many pts received A + B, despite risk of post-index GI bleeding. Median OS with A + B was shorter in pts with vs without GI bleeding risk. Data highlight the complexity of uHCC and the unmet need for guidance on characteristics-driven treatment decisions. Research Sponsor: AstraZeneca.

	Full cohort			A + B cohort		
	All pts N=186	GI bleeding risk N=128	No GI bleeding risk N=58	All pts N=110	GI bleeding risk N=93	No GI bleeding risk N=17
Median follow-up, mo	20.6	20.9	20.4	21.3	21.6	20.3
Median OS, mo	14.4	11.4	40.3	13.9	12.8	Not reached
OS rates, %						
12 mo	55.2	48.9	68.9	55.2	52.3	70.6
18 mo	45.3	38.1	61.1	44.1	41.6	57.8
24 mo	39.4	31.8	56.8	37.1	34.6	51.3
Post-index GI bleed, %	17.2	23.4	3.4	17.3	19.4	5.9

Clinical outcome after complete response with immunotherapy in hepatocellular carcinoma.

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Background: Immunotherapy has shown great efficacy in hepatocellular carcinoma (HCC). No clear consensus exists on when to stop the immune checkpoint inhibitor (ICI) after complete response (CR), and clinical course after stopping ICI remains unknown. **Methods:** We identified 138 HCC patients who received ICI at Cedars-Sinai Medical Center and Mayo Clinic Rochester from 2016-2023. Modified Response Evaluation Criteria In Solid Tumors (mRECIST) was used to assess treatment response. We further described the clinical course of patients who achieved CR after ICI discontinuation, including cancer recurrence, subsequent treatment, liver transplantation (LT) listing and outcomes. **Results:** The median age at ICI initiation was 68 years; 79% were male; 55.9% were white; 51.5% were BCLC stage C. 9.4% CRs, 25.4% partial responses, 39.9% stable diseases, and 25.4% progressive disease. Hepatitis C etiology was associated with higher likelihood of CR (P=0.005). Among 13 patients who achieved CR, 10 stopped ICI (60% for LT evaluation, 40% for toxicity). After stopping ICI, 40% had recurrence (Table). 3 patients underwent local therapy for recurrent HCC within Milan criteria and 1 patient restarted ICI for recurrence beyond Milan criteria. Among CR patients, 9 patients underwent LT evaluation after stopping ICI and 7 patients were waitlisted. After listing, 3 dropouts were observed, due to recurrent HCC beyond Milan criteria, sepsis leading to death, and personal preference. 4 patients underwent LT. 2 patients underwent local therapy before LT for recurrent HCC (Table). All transplanted patients had the last dose of ICI more than 1 year before LT (12, 21, 24, and 29 months). On explanted livers, 2 patients had no residual viable tumor and 2 patients had residual HCC within Milan criteria. 1 patient developed acute cellular rejection and was treated with immunosuppressants and maintained stable graft function. None had graft loss or evidence of HCC recurrence in the follow-up period at 8, 20, 28, and 51 months. **Conclusions:** Immunotherapy demonstrated excellent effects on a subset of HCC patients, including 9.4% who achieved CR. After stopping ICI, 40% had cancer recurrence but 3 out of 4 were within Milan criteria. 4 underwent successful LTs (2.9%) with excellent post LT outcome. Larger prospective data is needed to establish a protocol for ICI use, especially to determine the optimal timing for ICI discontinuation after CR. Research Sponsor: None.

	BCLC at ICI initiation	ICI cycles before stopping, ICI	Recurrence after last ICI (days)	Within Milan at recurrence	Treatment(s) received after recurrence and outcomes	OS (days)
Patient 1	B	7, Nivolumab	730	No	Nivolumab. Died from GI bleeding	1065
Patient 2	B	64, Atezo/Bev	149	Yes	TACE with CR, followed by LT without recurrence.	1203
Patient 3	A	8, Nivolumab	1393	Yes	Y90 with CR. No recurrence.	1984
Patient 4	C	NA, Pembrolizumab	407	Yes	TACE and MWA with CR, followed by LT without recurrence.	1722

Patterns of radiological progression in participants (pts) with embolization-eligible hepatocellular carcinoma (HCC) treated with durvalumab (D) + bevacizumab (B) + transarterial chemoembolization (TACE) and placebos + TACE: EMERALD-1 post hoc analysis.

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Background: EMERALD-1 (NCT03778957) met its primary endpoint, demonstrating improved progression-free survival (PFS) in pts with locoregional HCC treated with D + B + TACE versus placebos (PBO) + TACE (stratified Cox proportional hazards hazard ratio [HR], 0.77; 95% confidence interval [CI], 0.61–0.98 [Lencioni R, et al. *J Clin Oncol* 2024;42(suppl 3). Abs LBA432]). This post hoc analysis assessed outcomes by radiological progression pattern in pts treated with D + B + TACE or PBO + TACE. **Methods:** Pts included in this analysis received D (1500 mg) or PBO for D (Q4W) in combination with conventional (c)- or drug-eluting bead (DEB)-TACE (investigator choice, 1–4 TACE procedures within 16 weeks). Subsequently, pts received D (1120 mg) + B (15 mg/kg), or PBO for D + B (Q3W). This study analyzed radiological progression patterns at the time of first progressive disease (PD) as assessed by the investigator per modified Response Evaluation Criteria in Solid Tumors. A new lesion was classified as a new intrahepatic lesion (NIH) or new extrahepatic lesion (NEH); tumor growth of existing intrahepatic lesions (increase of ≥20% of an existing target lesion with at least >5 mm absolute increase or unequivocal PD with a non-target lesion) was classified as intrahepatic growth (IHG; categories were not mutually exclusive). Efficacy was assessed by time to progression (TTP). **Results:** In the D + B + TACE arm, 53.9% of pts had PD, and in the PBO + TACE arm 79.0% of pts had PD. The most common pattern of disease progression across both treatment arms was NIH, occurring in 73 (35.8%) and 107 (52.2%) pts in the D + B + TACE and PBO + TACE arms, respectively, with 31 (15.2%) and 34 (16.6%) pts exhibiting IHG, and 24 (11.8%) and 39 (19.0%) pts exhibiting NEH, respectively. Improved TTP was observed in pts treated with D + B + TACE versus PBO + TACE, regardless of progression pattern (Table). **Conclusions:** Overall, the rate of progression was lower with D + B + TACE compared with PBO + TACE. The pattern of disease progression observed with D + B + TACE and PBO + TACE was similar, with NIH the most common pattern of progression in both treatment arms. Consistent benefit in TTP was observed with D + B + TACE versus PBO + TACE, regardless of progression pattern. Clinical trial information: NCT03778957. Research Sponsor: AstraZeneca.

	NIH		IHG		NEH	
	D + B + TACE (n=73)	PBO + TACE (n=107)	D + B + TACE (n=31)	PBO + TACE (n=34)	D + B + TACE (n=24)	PBO + TACE (n=39)
Median (95% CI) TTP, months	13.7 (10.8–16.5)	8.8 (7.0–10.9)	5.1 (3.0–9.0)	4.3 (2.9–4.8)	6.7 (2.8–16.6)	4.6 (2.9–6.8)
HR (95% CI)*	0.78 (0.57–1.07)		0.61 (0.36–1.03)		0.66 (0.36–1.15)	

*HR and CI were estimated using a Cox proportional hazards model, the CI was calculated using a profile likelihood approach.

Outcomes by transarterial chemoembolization (TACE) modality from participants (pts) with embolization-eligible hepatocellular carcinoma (HCC) treated with durvalumab (D) + bevacizumab (B) + TACE and placebos (PBO) + TACE: EMERALD-1 subgroup analysis.

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Background: EMERALD-1 (NCT03778957) met its primary endpoint, showing improved progression-free survival (PFS) in pts with locoregional HCC treated with D + B + TACE versus PBO + TACE (hazard ratio [HR], 0.77; 95% confidence interval [CI], 0.61–0.98). This analysis assessed the impact of TACE modality (conventional TACE [cTACE] or drug-eluting bead [DEB]-TACE) on efficacy and safety outcomes. **Methods:** Pts in this analysis received D (1500 mg) or PBO for D (Q4W) plus cTACE or DEB-TACE (investigator choice; TACE modality was a stratification factor). After completing the last TACE, pts received D (1120 mg) + B (15 mg/kg) or PBO for D + B (Q3W). PFS, time to progression (TTP), and overall response rate (ORR; BICR per RECIST v1.1) with D + B + TACE and PBO + TACE (intent-to-treat population) are reported by TACE modality. Safety was assessed in the safety analysis set (pts received ≥1 dose of study treatment [tx], regardless of randomization). **Results:** Overall, 59.3% of pts received cTACE and 40.7% received DEB-TACE in the D + B + TACE arm; similarly, 58.5% of pts received cTACE and 41.5% received DEB-TACE in the PBO + TACE arm. Most pts received 1 or 2 TACE procedures in both cTACE (60% in D + B + TACE arm; 67.2% in PBO + TACE arm) and DEB-TACE groups (55.6% in D + B + TACE arm; 53.6% in PBO + TACE arm). Baseline characteristics in the cTACE and DEB-TACE groups were similar, with some differences in the relative distribution of BCLC, HAP, and tumor burden (BCLC Score A; 29.0% vs 17.9%: HAP Score A; 36.5% vs 25.0%: tumor burden within up-to-7 criteria; 56.4% vs 37.5%, respectively). Baseline characteristics were generally well balanced across tx arms within the cTACE and DEB-TACE groups. PFS, TTP, and ORR improved with D + B + TACE versus PBO + TACE, regardless of TACE modality (Table). In the D + B + TACE and PBO + TACE arms, max Grade 3–4 adverse events possibly related to study tx were reported by 25/100 (25.0%) and 3/116 (2.6%) pts in the cTACE group, and 16/54 (29.6%) and 9/84 (10.7%) pts in the DEB-TACE group, respectively. **Conclusions:** Overall, pts receiving D + B + TACE had improved PFS, TTP, and ORR versus PBO + TACE regardless of TACE modality. Safety was manageable with both cTACE and DEB-TACE. Clinical trial information: NCT03778957. Research Sponsor: AstraZeneca.

	D + B and cTACE (n=121)	PBO and cTACE (n=120)	D + B and DEB-TACE (n=83)	PBO and DEB-TACE (n=85)
Median (95% CI) PFS, months	19.4 (12.4–22.3)	11.1 (7.2–14.0)	11.1 (7.2–14.2)	6.7 (5.0–7.3)
PFS HR (95% CI)	0.80 (0.59–1.10)		0.74 (0.51–1.06)	
Median (95% CI) TTP, months	22.3 (19.4–27.7)	13.6 (9.2–16.6)	15.0 (11.1–22.4)	6.9 (5.1–11.1)
TTP HR (95% CI)	0.70 (0.49–0.98)		0.55 (0.36–0.83)	
Pts with measurable disease at baseline, n	119	119	83	84
ORR, n (%) pts with response*	62 (52.1)	43 (36.1)	26 (31.3)	17 (20.2)
ORR, odds ratio (95% CI)	1.93 (1.15–3.27)		1.80 (0.89–3.69)	

*Includes confirmed complete or partial response.

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First-line durvalumab with arterial chemotherapy in major portal invaded hepatocellular carcinoma: The phase 2 DurHope study with biomolecular analyses.

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Background: Current clinical trials for advanced hepatocellular carcinoma (HCC) significantly underrepresent patients (pts) with Vp4 portal vein tumor thrombus (PVTT), a poor prognostic factor, including only 0–15% despite their real-world prevalence of 24–49%. Previous FOHAIC-1 study indicated the favorable efficacy of arterial FOLFOX chemotherapy in this population. Preclinical evidence suggested that oxaliplatin in the FOLFOX regimen can induce immunogenic cell death. The DurHope study is a biomolecular exploratory, phase 2 trial of first-line durvalumab combined with arterial FOLFOX in pts with Vp4 PVTT. **Methods:** This trial enrolled 30 HCC pts with Vp4 (n = 24, 80.0%) or Vp3 (n = 6, 20.0%) PVTT. Arterial FOLFOX was given on day 1–3, plus durvalumab (1120 mg) intravenously on day 7 (± 2) of each 21-day cycle up to 8 cycles. Maintenance durvalumab (1500 mg) was administered every 28 days thereafter until disease progression, unacceptable toxicity, or death. Primary endpoint was 1-year OS rate. Secondary endpoints included progression-free survival (PFS), objective response rate (ORR) per RECIST v1.1, and safety. Single-cell RNA sequencing (scRNA-seq) was performed on pre-treatment biopsies and peripheral blood mononuclear cells (PBMCs) collected at baseline and during treatment from responders (CR/PR) and nonresponders (SD/PD). **Results:** With a median follow-up of 21.6 months, the 1-year OS rate was 63.3%, median OS was 13.9 months (95% CI, 10.7–NR), and median PFS was 9.0 months (95% CI, 4.5–12.2). ORR was 53.3% (CR: 5/30 [16.7%]; PR: 11/30 [36.7%]), and median duration of response was 9.1 months (> 6 months: 11/16 [68.8%]; > 12 months: 4/16 [25%]). Grade ≥ 3 TRAEs occurred in 43.3% of patients, with no unexpected safety events beyond those of each agent. ScRNA-seq of biopsies revealed enrichment of chemotherapy resistance and immune escape signatures in non-responders, while responders exhibited higher immune cell infiltration and stronger immune-malignant cell interaction. Dynamic scRNA-seq of serial PBMCs demonstrated expanded T cell subsets and increased expression of cytotoxicity-related genes (e.g., GZMB, GNLY, and IFNG) and proliferation marker (MKI67) after treatment, which was more pronounced in responders. This was accompanied by decreased non-classical monocyte frequency and reduced M2-directed monocyte polarization in responders. In vivo mouse models further showed enhanced efficacy of the combination of chemotherapy and PD-L1 blockade, indicating the synergistic effects of this therapeutic approach. **Conclusions:** Durvalumab plus arterial FOLFOX showed promising efficacy and manageable toxicities in advanced HCC with Vp4 PVTT. Biomolecular exploratory analyses revealed enhanced cellular immune responses triggered by treatment, along with novel molecular biomarkers that may help predict therapeutic response. Clinical trial information: NCT04945720. Research Sponsor: AstraZeneca.

The effect of TU-100 on hepatocarcinogenesis and colorectal liver metastasis in MASH model.

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Background: TU-100 is most commonly used herbal medicine in clinical setting. We have already reported that TU-100 suppressed dysbiosis (J Med Invest, 2013), bacterial translocation (Surg Today, 2017) and hepatic fibrosis (Surgery, 2016). This time we investigate the effect of TU-100 on hepatocellular carcinoma (HCC) and colorectal liver metastasis (CRLM) in MASH model. **Methods:** 1: Effect on HCC: Seven-week-old Tsumura Suzuki obese diabetes mice, a model that shows the spontaneous onset of MASH and HCC, were used. They were divided into a TU-100 treated group and a control group. Mice were sacrificed at 24 and 48 weeks to evaluate hepatic steatosis, fibrosis, carcinogenesis, cytokine expression, and microbiome abundance. 2: Effect on CRLM: Six-week-old male C57BL/6J mice were used. They were divided into the western diet (WD) group and the WD+TU-100 group. MC38 colon cancer cells were injected into the spleen at 16 weeks, and mice were killed at 2 weeks after injection to evaluate hepatic steatosis, fibrosis and mRNA expression in the liver. **Results:** 1: Effect on HCC: At 24 weeks, the TU-100 group showed significantly lower expression of IL6, IL1B, and ACTA2 mRNA in the liver ($P<0.05$). At 48 weeks, the TU-100 group showed a lower rate of MASH than the control group (28% vs 72%; $P=0.1$). Tumor diameter was significantly smaller in the TU-100 group compared with that in the control group ($P<0.05$). Regarding the intestinal microbiome, the genera Blautia and Ruminococcus were increased in the TU-100 group ($P<0.05$), whereas Dorea and Erysipelotrichaceae were decreased in the TU-100 group ($P<0.05$). 2: Effect on CRLM: Steatosis of the WD+TU-100 group was significantly suppressed compared with the WD group ($p<0.05$). The WD+TU-100 group showed significantly smaller tumor diameter than the WD group ($p<0.05$). In the WD+TU-100 group, expression of SAA1 and TIMP1 mRNA in the liver was significantly lower than in the WD group ($p<0.05$). **Conclusions:** TU-100 regulates the intestinal microbiome and may suppress hepatic steatosis and subsequent hepatocarcinogenesis in the MASH model. Furthermore, TU-100 may have effect on CRLM via suppression of SAA1/TIMP1. Research Sponsor: None.

Clinical outcomes of camrelizumab + rivoceranib vs sorafenib (CARES-310) as first-line treatment for patients with unresectable hepatocellular carcinoma (uHCC) of non-viral and viral etiology.

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Background: CARES-310 (NCT03764293) evaluated the combination of PD-1 inhibitor, camrelizumab (cam), and VEGFR-1-3 inhibitor, rivoceranib (rivo), compared to sorafenib (sor) for the treatment of uHCC. Cam + rivo significantly improved median overall survival (mOS) and median progression-free survival (mPFS) compared to sor (mOS, 23.8 months [mo] [95% CI 20.6, 27.2] vs 15.2 mo [95% CI 13.2, 18.5] hazard ratio [HR] 0.64 [95% CI 0.52, 0.79]; one-sided $p<0.0001$; mPFS, 5.6 mo [95% CI 5.5, 7.4] vs 3.7 mo [95% CI 3.1, 3.7]; HR 0.54 [95% CI 0.44, 0.67]; one-sided $p<0.0001$). The most common (≥10%) grade ≥3 treatment-related adverse events in the cam + rivo arm were hypertension (38.6%) and AST increased (20.2%). **Methods:** A post-hoc analysis of CARES-310 was performed, where mOS and mPFS were estimated using the Kaplan-Meier method and compared between the 3 etiology groups of non-viral, hepatitis C virus (HCV), and hepatitis B virus (HBV) using the log-rank test. **Results:** mOS was longer with camrelizumab plus rivoceranib compared with sorafenib in patients with non-viral (HR 0.68 [95% CI 0.39, 1.19]), HCV (HR 0.37 [95% CI 0.162, 0.84]), and HBV etiologies (HR 0.70 [95% CI 0.55, 0.89]) (Table). Similarly, mPFS was longer with camrelizumab plus rivoceranib compared with sorafenib in patients with non-viral (HR 0.55 [95% CI 0.34, 0.91]), HCV (HR 0.50 [95% CI 0.23, 1.06]), and HBV etiologies (HR 0.57 [95% CI 0.45, 0.72]) (Table). **Conclusions:** Cam + rivo in CARES-310 suggested clinically meaningful mOS benefit in non-viral and viral HCC vs sor and provides assurance of clinical benefit for first line treatment to patients with uHCC independent of etiology. Clinical trial information: NCT03764293. Research Sponsor: Elevar Therapeutics; Jiangsu Hengrui Pharmaceuticals.

Etiology Group	Camrelizumab + Rivoceranib (n=272)	Sorafenib (n=271)
Non-viral, n (%)	42 (15.4%)	45 (16.6)
mOS, mo (95% CI)	26.8 (10.3, NR)	15.2 (9.6, 23.2)
mPFS, mo (95% CI)	6.1 (4.1, 13.8)	3.7 (2.4, 5.5)
HCV, n (%)	22 (8.1)	29 (10.7)
mOS, mo (95% CI)	31.7 (10.8, NR)	13.3 (8.0, 21.5)
mPFS, mo (95% CI)	12.7 (3.7, NR)	3.7 (1.8, 9.1)
HBV, n (%)	208 (76.5)	197 (72.7)
mOS, mo (95% CI)	23.0 (18.9, 26.0)	15.6 (13.3, 19.2)
mPFS, mo (95% CI)	5.6 (5.5, 7.3)	3.7 (2.7, 3.7)

HCC, hepatocellular carcinoma; mOS, median overall survival; mPFS, median progression-free survival; mo, months; HR, hazard ratio; HCV, hepatitis C virus; HBV, hepatitis B virus, NR, not reached.
Results used Cox proportional hazard model.
Confidence Intervals (CI) used the Brookmeyer and Crowley method.

First-line iparomlimab and tuvonralimab (QL1706) or iparomlimab (QL1604) + bevacizumab (BEV) for unresectable hepatocellular carcinoma (HCC): Updated results from the phase Ib/II DUBHE-H-106 study.

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Background: Novel treatment options for unresectable HCC are needed. Iparomlimab and tuvonralimab are anti-PD-1 and anti-CTLA-4 antibodies, respectively. The DUBHE-H-106 study aims to assess safety and efficacy of first-line QL1706 or QL1604 + BEV for HCC. Preliminary data have been reported on 2023 ASCO Annual Meeting. Here, we report updated results. **Methods:** This study consists of three cohorts. Systemic therapy-naive adult patients (pts) with HCC, ≥ one measurable untreated lesion per RECIST v1.1, BCLC stage B–C, Child-Pugh score ≤ 7, not amenable to or progression after locoregional therapy, ECOG performance status of 0–1 were eligible. In Cohort A, six pts received QL1706 5 mg/kg + BEV 15 mg/kg Q3W. If ≤ two pts had dose-limiting toxicities (DLT), another six pts would be enrolled. If ≤ three of twelve pts had DLT, the safe dose of BEV would be determined, and eight more pts would be enrolled. Otherwise, enrollment of another dose group (QL1706 5 mg/kg + BEV 7.5 mg/kg Q3W) would initiate, using the same procedure. If number of DLT exceeded the criteria, further dose reduction of BEV or study termination would be discussed. Then 40–60 pts were randomized 1:1 to Cohort A or B. In Cohort B, pts received QL1604 200 mg + BEV (safe dose) Q3W. Enrollment of Cohort C would initiate according to the preliminary results of Cohort A and B. Pts received QL1706 7.5 mg/kg + BEV (safe dose) Q3W, using the same procedure in Cohort A. If ≤ three of twelve pts had DLT, 8–28 more pts would be enrolled. **Results:** Between Jun 2021 and Dec 2023, Cohort A, B, and C included 50, 26, and 40 pts, respectively. Baseline data were balanced in each cohort. All pts were in the safety set. No DLT was reported. Incidences of adverse events (AE) were similar in three cohorts. Efficacy evaluable set included 47, 26, and 37 pts in three cohorts. Numerically, QL1706 + BEV showed better efficacy compared to QL1604 + BEV, and higher response and 12-month progression-free survival (PFS) rates were found in Cohort C vs Cohort A. Detailed results were shown in Table. **Conclusions:** First-line QL1706 or QL1604 + BEV showed acceptable toxicities and promising efficacy for unresectable HCC. QL1706 7.5 mg/kg + BEV 15 mg/kg Q3W may have better anti-tumor activity and were recommended for trials in future. Clinical trial information: NCT05603039. Research Sponsor: Qilu Pharmaceutical Co., Ltd. (Jinan, China).

Endpoints	Cohort A	Cohort B	Cohort C
Treatment-related AE (TRAE)	43 (86%)	25 (96%)	37 (92%)
Grade ≥ 3 TRAE	24 (48%)	14 (54%)	16 (40%)
Serious TRAE	11 (22%)	9 (35%)	10 (25%)
Immune-related AE	27 (54%)	9 (35%)	22 (55%)
TRAE leading to dose interruption	23 (46%)	16 (62%)	16 (40%)
Objective response, n (%; 95% CI)	18 (38%; 25%–54%)	6 (23%; 9%–44%)	16 (43%; 27%–61%)
Disease control, n (%; 95% CI)	35 (74%; 60%–86%)	18 (69%; 48%–86%)	30 (81%; 65%–92%)
Median PFS (95% CI), months	7.0 (3.1–9.6)	5.4 (2.4–11.0)	7.0 (4.2–not evaluable)
12-month PFS rate (95% CI)	26.8 (14.7–40.4)	24.4 (9.9–42.1)	40.9 (24.3–56.9)

Lenvatinib plus hepatic arterial infusion chemotherapy of oxaliplatin, fluorouracil, and leucovorin versus lenvatinib alone for advanced hepatocellular carcinoma.

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Background: Lenvatinib stands out as a first-line therapy for individuals with advanced hepatocellular carcinoma (HCC). Concurrently, hepatic arterial infusion chemotherapy comprising oxaliplatin, fluorouracil, and leucovorin (FOLFOX-HAIC) has emerged as a potential option for those with advanced HCC. It is necessary to investigate the efficacy and safety of lenvatinib plus FOLFOX-HAIC (lenvaHAIC) for advanced HCC in real-world situations. **Methods:** In this retrospective analysis, 127 consecutive patients underwent lenvaHAIC, while 184 patients received lenvatinib alone as first-line treatment at six Chinese academic centers between January 2019 and June 2022. Following 1:1 propensity-score matching, we established paired cohorts (113 patients in each group) for evaluating survival. Overall survival (OS), progression-free survival (PFS), objective response rate (ORR) evaluated by RECIST 1.1 and mRECIST criteria, and safety profiles were compared between the two groups. **Results:** The lenvaHAIC group exhibited significantly prolonged median PFS and OS than the lenvatinib group (PFS: 12.3 vs. 6.2 months; OS: 25.6 vs. 12.3 months; $P < .001$ for each). In the propensity-score matched cohorts (113 pairs), both PFS and OS were notably extended in the lenvaHAIC group compared with those in the lenvatinib group ($P < .001$). Multivariate analysis identified lenvaHAIC treatment as an independent factor for improved PFS (hazard ratio [HR] 0.45; $P < .001$) and OS (HR 0.38; $P < .001$). Grade 3-4 adverse events, including nausea, vomiting, diarrhea, thrombocytopenia, and neutropenia, were more prevalent in the lenvaHAIC group. **Conclusions:** LenvaHAIC may be a promising treatment in patients with advanced hepatocellular carcinoma, demonstrating enhanced OS, PFS, and ORR and an acceptable safety profile.
Research Sponsor: None.

Selective internal radiation therapy with yttrium-90 resin microspheres followed by lenvatinib plus PD-1 inhibitor for large or huge advanced-stage hepatocellular carcinoma: A retrospective cohort study from China.

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Background: Yttrium-90 resin microspheres have been introduced into China in recent years, but the efficacy or safety on Chinese patients are rarely reported. This study aimed to evaluate the efficacy and safety of selective internal radiation therapy (SIRT) with yttrium-90 resin microspheres followed by lenvatinib and PD-1 inhibitor in Chinese patients with large (5.1–10.0 cm in diameter) or huge (>10.0 cm) advanced-stage hepatocellular carcinoma (HCC). **Methods:** Data of large or huge Barcelona Clinic Liver Cancer stage C HCC patients treated with SIRT followed by lenvatinib plus PD-1 inhibitor from November 2022 to October 2023 were prospectively collected and retrospectively analyzed. Key endpoints included overall survival (OS), progression-free survival (PFS), complete response rate (CRR), objective response rate (ORR), disease control rate (DCR) based on modified Response Evaluation Criteria in Solid Tumors, and adverse events (AEs) graded based on Common Terminology Criteria for Adverse Events v5.0. Risk factors affecting PFS were analyzed using univariate and multivariate Cox regression models. **Results:** A total of 30 patients (27 males, 3 females; mean age 54 ± 11 years) were included. The mean longest diameter of the tumor was 9.6 ± 4.2 cm (range: 5.2–17.6 cm). Nineteen (63.3%) patients had more than three intrahepatic lesions, 15 (50.0%) had vascular invasion, and 13 (43.3%) had extrahepatic metastasis. Sixteen (53.3%) patients had undergone prior anticancer therapies. Two patients received two SIRT procedures, and the rest had one. Lenvatinib and PD-1 inhibitor treatment was initiated 3–7 days after the initial SIRT procedure. CRR was 20.0%, ORR was 73.3%, and DCR was 90.0%. Median PFS was 8.0 months (95% CI, 5.6–10.4), and median OS was not reached. OS rates at 6, 12, and 18 months were 93%, 74%, and 74%, respectively. Univariate and multivariate analysis indicated that baseline alpha-fetoprotein levels, baseline eosinophil count, and neutrophil-to-lymphocyte ratio at 1 month after SIRT were independent risk factors for PFS. AEs of any grade occurred in 86.7% of patients, with grade 3 AEs in 23.3%, and no grade 4/5 AEs were recorded. **Conclusions:** SIRT followed by lenvatinib and PD-1 inhibitor demonstrates promising efficacy and a favorable safety profile in Chinese patients with large or huge advanced-stage HCC. Research Sponsor: None.

IMMULAB: A phase II trial of immunotherapy with pembrolizumab in combination with local ablation for patients with early stage hepatocellular carcinoma (HCC)—The IKF-IMMULAB trial.

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Background: Ablation of neoplastic tissue through radiofrequency ablation (RFA), microwave ablation (MWA), or brachytherapy is a potentially curative option for patients with early stage hepatocellular carcinoma (HCC); however, recurrence rates are high. While updated results from IMbrave050 do not support the use of adjuvant therapy following resection or local therapy, the potential of peri-interventional systemic treatment as a potentially more effective approach to increase recurrence free and long-term survival remains to be determined. We propose that peri-interventional treatment with pembrolizumab may improve outcomes following local ablative therapies. **Methods:** This single arm phase II trial investigates peri-interventional treatment with pembrolizumab combined with RFA/MWA or brachytherapy, or - as recommended for tumors ≥ 3 cm - by combination with TACE in early stage HCC with maintained liver function. 200 mg of pembrolizumab (q3w) was administered intravenously for 2 cycles followed by radiologic imaging and local therapy. Pembrolizumab was continued for up to 12 months. Primary endpoint was radiological response rate according to RECIST 1.1 prior to local therapy. **Results:** 30 pts (ECOG 0 or 1) were enrolled in 9 centers in Germany with a median age of 70 years (73.3% males, median tumor size 25 mm with minimum of 10 mm and maximum of 72 mm, 60% solitary lesions). All pts received at least 1 dose of study treatment and the median number of cycles was 13. ORR was 13.3% with 6.7% CR and 6.7% PR after two cycles. After a median follow-up of 34.5 months, median time to recurrence (TTR) was 16.43 months, with 8 pts remaining free of recurrence. mOS was not reached, with a 2y OS rate of 76% and a 3y OS rate of 66%. No new safety signs were observed. Subsequent local ablation was performed in 13 pts, and 10 pts started on systemic therapies (5 TKI, 3 ICI and 2 pts received TKI and ICI). **Conclusions:** In the IKF-IMMULAB trial the expected objective response rate of 30%, according to RECIST1.1., prior to local therapy was not reached. However, with recent evidence that radiologic response underestimates pathological response in ICI-treated HCC patients and supported by a 2y OS rate of 76% and a 3y OS rate of 66%, there is evidence for the efficacy of peri-interventional treatment with pembrolizumab without new safety signals. Clinical trial information: NCT03753659. Research Sponsor: MSD SHARP & DOHME GMBH.

TYRA-430: First reversible FGFR4/3 inhibitor designed to overcome current challenges in FGF19-driven hepatocellular carcinoma treatment.

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Background: Hepatocellular carcinoma (HCC) is an aggressive malignancy that accounts for approximately 80-90% of all primary liver cancers and is projected to become the third leading cause of cancer-related mortality by 2030. Previous work demonstrated that human Fibroblast Growth Factor 19 (FGF19) was overexpressed in 20-30% of HCC, thereby exerting an oncogenic effect via signaling through its cognate receptors FGFR4, FGFR3, and the co-receptor Klotho b (KLB). Multiple selective covalent FGFR4 inhibitors have been clinically evaluated in FGF19-overexpressing HCC. However, selective inhibition of FGFR4 alone was insufficient and resulted in low response rates and limited duration, likely due to redundant signaling through FGFR3. Here we describe the first report on the preclinical profile of TYRA-430, a first-in-class reversible FGFR4/3 inhibitor. **Methods:** The preclinical profile of TYRA-430 was assessed *in vitro* in HCC cellular assays and *in vivo* in mouse xenograft models. The *in vitro* potency was assessed by measuring cell viability using Cell Titer-Glo 2.0 Luminescent assay. Percentage of tumor growth inhibition (TGI) was used as a measure to evaluate *in vivo* efficacy of TYRA-430. **Results:** TYRA-430 exhibited potency against Ba/F3 cells dependent on wildtype FGFR3 and FGFR4, as well as the known Cys552 and Val550 gatekeeper (GK) resistance mutations in FGFR4. *In vitro* data showed that TYRA-430 displayed superior potency over the reversible multi-kinase inhibitors sorafenib and lenvatinib, and the covalent FGFR4 inhibitors fisogatinib (BLU-554) and roblitinib (FGF401) in KLB/FGF-19/FGFR3/4 driven models of HCC (Hep3B, HuH-7, and JHH-7 cells). Furthermore, in a human HuH-7 HCC xenograft model in nu/nu mice, TYRA-430 achieved 96% tumor growth inhibition (TGI), compared to 75% TGI for lenvatinib and 86% TGI for FGF401. Additionally, in the GA180 FGF19-driven gastric cancer PDX model, TYRA-430 demonstrated superior efficacy with 93% tumor growth inhibition (TGI), compared to 56% TGI achieved by roblitinib. **Conclusions:** TYRA-430 was active and potent in multiple KLB/FGF-19/FGFR3/4 models of human hepatocellular carcinoma *in vitro* and *in vivo*. Moreover, TYRA-430 displayed potent activity against known FGFR4 resistance mutations compared to covalent FGFR4-specific inhibitors in the clinic. TYRA-430 will be investigated in a Phase 1 clinical trial in hepatocellular carcinoma and other advanced solid tumors. Research Sponsor: Tyra Biosciences.

Trastuzumab emtansine as second-line therapy in HER2 positive advanced biliary tract cancers: Single arm prospective phase II clinical trial (TAB-3 trial).

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Background: HER2 overexpression or amplification is a therapeutic target of interest in advanced BTC, especially gallbladder cancers (GBC). Limited studies evaluate the role of T-DM1 in these groups of cancers. **Methods:** This study was an investigator-initiated, open-label, single-arm, phase II trial in patients (pts) aged 18 years or older with HER2-positive (defined as IHC 3+ or IHC 2+, and FISH positive), advanced BTCs who had previously received systemic therapy (irrespective of prior additional HER2 targeted therapy). Pts received T-DM1 at 3.6mg/kg q 3 -weekly till disease progression (PD), unacceptable toxicities, or patient choice. The primary end-point of the study was an improvement in 3-month progression-free survival (PFS) from 50% (historical cohort) to 70% in the study arm. Secondary endpoints included overall response rates (ORR), disease control rate (DCR), overall survival (OS), and incidence of grade 3 and 4 treatment-related adverse events (TRAE). **Results:** From July 2023 to July 2024, '52' pts with HER2-positive BTC were screened, and 40 enrolled in the study. A majority of patients had GBC (95%) and most commonly received combination chemotherapy with gemcitabine-cisplatin, with or without nab-Paclitaxel (78%). Six patients (15%) had previously received trastuzumab in combination with chemotherapy. A majority of patients (n=24; 60%) had rapid progression (within 3 months) on prior systemic therapy. With a median follow-up of 7.1 months [95% confidence interval (CI): 5.1-9.1] , 24 patients had disease progression with a 3-month PFS of 51.2% (95% CI: 33.4 - 69) and median PFS of 3.1 months (95% CI: 2.3-3.8); median OS was 7.1 months (95% CI: 5.1-9.1). Partial responses (PR) were seen in 5 patients (12.5 %) and 8 had stable disease (20 %) for a disease control rate of 32.5 %. Patients without rapid disease progression on prior therapy (n=16; 40%) had a 3-month PFS of 70% (95% CI: 44.8 - 95.2) and median PFS of 4.9 months (95% CI: 2.9-6.9). Grade 3 and grade 4 TRAE in the overall cohort were noted in 11 patients (28%), with the commonest being grade 2 fatigue in 6 patients and grade 3 thrombocytopenia in 2 patients. **Conclusions:** T-DM1 was well tolerated in pre-treated HER2 positive advanced BTCs, but did not statistically improve PFS in comparison to historical data. Patients who did not have rapid progression on prior systemic therapy appeared to have a more favorable survival in comparison to rapid progressors and the role of T-DM1 in such favorable cohorts can be explored in larger studies. Clinical trial information: CTRI/2023/07/055785. Research Sponsor: ZYDUS PHARMA PRIVATE LIMITED; TRACK (TATA MEMORIAL CENTRE RESEARCH FUNDING COMITTEE).

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Current status of physicians' perception of HCC conversion/downstaging therapy: A nationwide survey in China.

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Background: Conversion or downstaging therapy for intermediate and advanced unresectable hepatocellular carcinoma (HCC) has emerged as a significant exploring area of clinical practice and research in recent years. Nonetheless, there exists considerable variability in both the selection criteria for patients undergoing conversion therapy and the regimens selected for conversion across different regions and medical institutions. The present study aims to elucidate the current clinical application of conversion therapy in China, as well as to identify the considerations that physicians take into account when selecting eligible patients for conversion therapy and determining the appropriate conversion modality. **Methods:** Physicians meeting predefined inclusion criteria were invited to complete an online questionnaire from January to July, 2024. The collected data were subsequently pooled and analyzed descriptively. **Results:** A total of 120 valid questionnaires were retrieved, mainly form surgical (69.2%, n=83) and interventional (30.8%, n=37) departments. Generally, the study showed that approximately 51% of stage Ib-IIIa patients will be selected for the treatment goal of conversion or downstaging, and the overall successful rate as meeting criteria for surgical resection after treatment was 36%. Three primary factors were prioritized by physicians for determination of conversion therapy: the type of portal vein tumor thrombus (PVTT) (97%, 116/120), the future liver volume (90%, 108/120), and Child-Pugh classification (90%,108/120). In the specific physician group who selected the following attributes, patients classified as Child-Pugh A (82%, 89/108) or Child-Pugh B7 (66%, 71/108), those with tumor diameter exceeding 5 cm (92%, 98/106) and Eastern Cooperative Oncology Group (ECOG) performance score of 0 to 1 (89%, 76/85) were most frequently selected for conversion therapy. Additionally, study showed that patients presenting with Vp1-2 (72%, 83/116) and Vp3 (71%, 82/116) could also be considered for conversion therapy. Currently, the prevalent treatment approach of conversion therapy involves a combination of local and systemic therapies, which is utilized in approximately 73% of cases. Among systemic treatment methodologies, the combination of Lenvatinib and immunotherapy is the most widely adopted by physicians. Besides, higher ORR was the foremost consideration for 83% (99/120) of physicians, followed by rapid response (68%, 82/120), adherence to guidelines and consensus (63%, 76/120), and lower tumor progression rate (58%, 70/120). **Conclusions:** This study elucidates the current status of conversion therapy for HCC in China. The findings underscore the necessity for optimizing conversion modalities and advancing standardized practices in the application of conversion therapy. Research Sponsor: Beijing Kangwei institute of Biological Health Sciences.

Phase II study of modulation of sorafenib (SOR)-induced autophagy using hydroxychloroquine (HCQ) in advanced hepatocellular cancer (HCC).

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Background: SOR is the first systemic therapy approved for advanced HCC, but has shown only modest improvements in survival. Resistance to SOR in pre-clinical models has been attributed to autophagy induction. Autophagy inhibition with HCQ enhanced SOR-induced cell death and apoptosis in early pre-clinic and clinical studies. Data from the phase I study of SOR plus HCQ in advanced solid tumors at showed clinical safety and efficacy. Therefore, we conducted a prospective study to evaluate efficacy of SOR and HCQ in advanced HCC patients (pts) (NCT03037437) and report the final efficacy analysis. **Methods:** Prospective phase II study of SOR 400 mg po BID + HCQ 400 mg daily in pts with advanced HCC (CP A-B8 cirrhosis). Cohort 1: first-line SOR/HCQ. Cohort 2: add HCQ upon progressing on SOR. CP B pts started at 200 mg BID, with dose escalation as tolerated. Cycle = 4 weeks. Primary endpoint: mTTP. Secondary endpoints: mOS, objective response rate (ORR) by RECIST; AEs (NCI-CTCAEv3.0); PD analysis for markers of autophagy and immunity. Pts evaluable for efficacy if completed C1. Historically, the mTTP for patients treated with sorafenib is 5.5 mo (SHARP trial). We predicted the addition of HCQ will improve mTTP by 50% to 8.2 months. **Results:** N=33 completed C1. Median age 64.1 years SD 8.6 (46- 80). 88% Male; 58% Hispanics. ECOG 0-1: 100%. CP B cirrhosis: 35%. Etiology of cirrhosis: HCV 68%, ETOH 32%, NASH 11%. BCLC B 19%, C 81%. AFP>400: 43%, PVT: 27%, extrahepatic/metastases: 68%, post-transplant: 27%. N=5 (14%) were in cohort 2 (prior SOR). Reason off study: PD (n=21), toxicity (n=5), lost to f/u (n=1), withdrew (n=3). mTTP is 8.0 months (95% CI: 4.0-19.0). mOS 10.0 months (95% CI: 6.0-22.0). ORR (CR+PR): 27%. Best response: CR n=1 (3%), PR n=8 (24%), SD n=14 (42%). 4+ cycles: n=19 (58%). AE on treatment: grade 1 (68%), 2 (24%), 3 (7%), 4 (0.6%), 5 (0.3%). No Gr 4/5 related to SOR or HCQ. Dose interruption (20%), dose reduction (10%). **Conclusions:** SOR/HCQ had a better mTTP (8.0 months) than historical control (mTTP=5.5 months, p=0.03), and the trial met its primary endpoint. SOR/HCQ ORR (27%) was higher than historically SOR alone (ORR 2%, SHARP) in pts with advanced HCC, predominantly BCLC C, with CP A and B cirrhosis (35%), and was tolerable. While immune checkpoint inhibitors (ICIs) are taking the forefront in advanced HCC, SOR/HCQ may still have a role in patients with CP B cirrhosis, post-transplant, or contraindications to ICIs. Further, analysis of predictive markers of response is ongoing. Clinical trial information: NCT03037437. Research Sponsor: Mays Cancer Center, Pilot Award; NIH CA054174.

A multicenter, phase 2 study of atezolizumab plus bevacizumab (atezo+bev) combination therapy in patients with unresectable hepatocellular carcinoma (HCC) and Child–Pugh class B cirrhosis: CHALLENGE trial final results.

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Background: The Phase 2 CHALLENGE trial (jRCTs031210355) evaluated the safety and efficacy of atezo+bev combination therapy in patients with advanced HCC with Child–Pugh class B cirrhosis. Results of the trial's main analysis demonstrated that atezo+bev was well-tolerated. Here, we present the trial's final safety and efficacy outcomes. **Methods:** This multicenter, open-label, single-arm, Phase 2 study enrolled patients with advanced HCC and a Child–Pugh score of 7 or 8 who had not received prior systemic therapy. Patients were administered atezolizumab 1200 mg plus bevacizumab 15 mg/kg every 3 weeks. The primary endpoint was the frequency of severe adverse events (SAEs). Secondary endpoints included the objective response rate (ORR), progression-free survival (PFS), time to progression (TTP), overall survival (OS), and frequency of adverse events (AEs). **Results:** A total of 31 patients were enrolled between December 2021 and April 2023. Child–Pugh score was 7 points in 25 patients (80.6%) and 8 points in 6 patients (19.4%). Fourteen patients (45.2%) were classified as having Barcelona Clinic Liver Cancer stage C. We set 30 eligible patients as the population for analysis. The final analysis performed with a median follow-up duration of 517days showed that the frequency of SAEs was 30.0% (95% confidence interval [CI] 14.7–49.4%). ORRs according to RECIST 1.1 and modified RECIST were 40.0% and 46.7%, respectively. Median PFS, TTP, and OS were 240 days (95% CI: 176–526 days), 240 days (95% CI: 176–526 days), and 470 days (95% CI: 256–576 days), respectively. Frequent Grade ≥3 SAEs were blood bilirubin increased (n=3), proteinuria (n=3), hypoalbuminemia (n=2), and hypertension (n=2). Grade ≥3 severe liver-related AEs occurred in 7 patients (23.3%), including blood bilirubin increased (n=3) and esophageal varices hemorrhage (n=2). Grade ≥3 severe immune-related AEs occurred in 2 patients – these were mucositis oral (n=1) and malaise (n=1). The mean Child–Pugh score (±standard deviation) at the start of treatment was 7.2±0.4, and was 7.0±1.0, 7.2±0.9, and 7.6±1.7 at the third and fifth cycles and at the withdrawal study, respectively. **Conclusions:** Atezo+bev was well-tolerated with demonstrable anti-tumor effects in patients with unresectable HCC and Child–Pugh class B cirrhosis. Additional studies are warranted to confirm the efficacy results of atezo+bev in this patient group. Clinical trial information: jRCTs031210355.
Research Sponsor: None.

Updated results from ALTER-H004 study: Anlotinib combined with TACE as adjuvant therapy in patients with hepatocellular carcinoma (HCC) at high risk of recurrence after surgery—A single arm, multi-center, phase II clinical trial.

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Background: High recurrence rate after curative resection significantly compromised the long-term survival of patients with HCC patients (pts). TACE as adjuvant therapy was proved to improve clinical outcomes for pts with high recurrence risk factors after hepatectomy. However, the prognosis of TACE reminded unsatisfactory and controversial. Therefore, this study aimed to identify the efficacy and safety of anlotinib as maintenance adjuvant treatment following inductional TACE in pts with HCC. The preliminary results had been reported in 2024 ASCO GI (Poster 513) and the update results were reported here with longer follow-up duration. **Methods:** HCC pts with one of the following high risk factors after hepatectomy: ≥5 cm and <10 cm of tumor diameter, tumor number ≥3, tumor microvascular invasion M1 or M2, portal vein tumor thrombus resection (Cheng's classification I/II) were recruited. Eligible pts received cTACE treatment within 1-2 months after hepatectomy, on the 3-5th day after cTACE, oral anlotinib (12mg, qd, d1-d14, q3w) was given until disease recurrence or unacceptable toxicity. The predefined sample size was 30. The primary endpoint was disease free survival (DFS). Secondary endpoints included 1-year DFS rate, time to recurrence and safety. **Results:** The data cut-off date was April 2024, 29 pts were enrolled and 27 pts received at least once tumor assessment. The median follow-up was 34.1 (IQR, 32.0-40.9) months. Of 27 with assessable efficacy, mDFS was 40.9 (95% CI, 11.4-70.3) months, with 1-, 2-, and 3-year DFS rates of 72.1% (95% CI, 50.2%-85.6%), 63.0% (95% CI, 40.7%-78.9%), and 56.7% (95% CI, 33.6%-74.5%), respectively. 21 of 27 pts (77.8%) experienced treatment-related adverse events (TRAEs). Grade 3 TRAEs occurred in 5 pts (18.5%) included hypertension (7.4%), leukocytopenia (7.4%) and ascites (3.7%), and blood bilirubin increased (3.7%). No grade 4 or 5 TRAEs occurred. **Conclusions:** Anlotinib combined with TACE as adjuvant therapy in HCC pts at high recurrence risk exhibited encouraging efficacy and tolerable safety profile. And the conclusions needed to be confirmed in large scale clinical trials subsequently. Clinical trial information: NCT04213118. Research Sponsor: None.

Niraparib in patients with *BRCA*-mutated unresectable or recurrent biliary tract, pancreatic and other gastrointestinal cancers: An investigator-initiated phase 2 trial (NIR-B trial).

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Background: It has been reported that there are *BRCA1/2*-mutated patients in biliary tract, pancreatic, and other gastrointestinal cancers. Niraparib is a poly(ADP-ribose) polymerase (PARP) inhibitor, and PARP inhibitors exert their cytotoxicity against cancer cells in the context of homologous recombination deficiency, such as *BRCA* mutations. **Methods:** Main eligibility criteria are unresectable, advanced or recurrent biliary tract cancers (BTC; cohort A), pancreatic cancers (PC; cohort B), and other gastrointestinal cancers (cohort C) with *BRCA1/2* gene mutations identified by germline test or genomic profiling test with either circulating tumor DNA (ctDNA) or tumor tissue, refractory or intolerant to previous treatments, and adequate organ function. Patients with body weight of 77 kg or more and a platelet count of 150,000 /μL or more receive 300 mg of niraparib, and less than 77 kg or having a platelet count less than 150,000 /μL received 200 mg of niraparib, orally once daily, until disease progression or intolerable adverse events occurred. Primary endpoint was the investigator-assessed objective response rate (ORR) in each cohort with a threshold response rate of 10% and an expected response rate of 35%. Key secondary endpoints were progression-free survival (PFS), overall survival (OS), disease control rate (DCR), and safety. Furthermore, pre-treatment ctDNA was collected and analyzed by Guardant360. **Results:** A total of 62 pts, 26 in cohort A, 26 in cohort B, and 10 in cohort C were enrolled between March 2021 and April 2023. Because one patient in cohort C did not receive protocol treatment, 61 pts were identified for primary efficacy and safety analysis. Median number of prior regimens was 1/2/2 (range, 1–2/1–2/1-6). In cohort A/B/C, the confirmed ORR was 15.4/15.4/0% [95% highest posterior density credible interval, 8.4–27.9/8.4–27.9/0.0–25.9]. The DCR, median (m) PFS, mOS were 57.7/53.8/22.2% [95% confidence interval (CI), 36.9–76.6/33.4–73.4/2.8–60.0], 2.7/1.8/1.4 months (95% CI, 1.5–4.1/1.4–9.3/1.0–2.8), 7.8/9.5/7.4 months (95% CI, 5.9–9.8/3.5–NE/2.9–8.6), respectively. In the pts with BRCA mutations by Guardant360 (cohort A/B/C were 18/21/8), ORR was 11.1/19.0/0% (95% CI, 2.0-30.2/6.5–38.3/0.0–28.3). The most common treatment-emergent adverse events were thrombocytopenia (31.1%), anemia (27.9%), neutropenia (14.8%), nausea (36.1%), fatigue (29.5%), anorexia (24.6%). **Conclusions:** Although niraparib had signs of clinical activity in pts with *BRCA*-mutated BTC and PC, the primary endpoint was not achieved statistically. No new safety signal was observed. Alternative approaches, such as evaluation in biomarker-selected patients or in combination with other agents, may demonstrate greater clinical activity of niraparib in this setting. Clinical trial information: jRCT2011200023. Research Sponsor: Japan Agency for Medical Research and Development; Takeda Pharmaceutical Co., Ltd.

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Neoadjuvant immune checkpoint blockade in resectable hepatocellular carcinoma: A systematic review and meta-analyisis.

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Background: Hepatocellular carcinoma (HCC) presents a significant therapeutic challenge due to its high recurrence rates after curative-intent resection. Neoadjuvant immune checkpoint blockade (ICB) has emerged as a promising strategy to improve tumor control by modulating the immune microenvironment prior to surgery. This systematic review and meta-analysis aim to assess the safety and efficacy of neoadjuvant ICB in improving clinical outcomes for patients with resectable HCC. **Methods:** A systematic search of PubMed, EMBASE, and Cochrane databases was conducted to identify clinical trials examining neoadjuvant ICB in resectable HCC. Primary endpoints included pathological complete response (pCR), major pathological response (mPR), overall response rate (ORR), and grade 3-4 treatment-related adverse events (TRAE). Prespecified subgroup analyses were conducted for studies combining ICB with tyrosine kinase inhibitors (TKIs) or dual ICB therapies. Pooled proportions were calculated using a random-effects model via the R package "meta." Inter-study heterogeneity was assessed using the I^2 statistic and test for subgroup differences were conducted. **Results:** Out of 1,321 studies screened, 14 clinical trials involving 252 patients with resectable HCC were included. Most studies were phase II trials (57%), and patient median age ranged from 57.5 to 68.0 years, with 88.3% of patients being male. The pooled prevalence of mPR was 32% (95% CI, 24–42%; I^2 = 9%)while pCR was achieved in 24% (95% CI, 17–34%; I^2 = 29%)of patients. The ORR was 27% (95% CI, 18–39%; I^2 = 49%), and severe adverse events (grade 3-4 TRAE) had a pooled prevalence of 24% (95% CI, 17–34%; I^2 = 16%). Subgroup analysis of trials combining ICB with TKIs showed an ORR 34% (95% CI, 20–55%; I^2 = 60%), an mPR rate of 27% (95% CI, 19–40%; I^2 = 0%), and a pCR rate of 20% (95% CI, 14–30%; I^2 = 0%). Dual ICB therapies demonstrated a higher mPR rate of 42% (95% CI, 26–67%; I^2 = 16%) and a pCR rate of 41% (95% CI, 29–59%; I^2 = 29%), but were associated with a higher incidence of severe TRAE (41%; 95% CI, 29–60%; I^2 = 0%). **Conclusions:** Neoadjuvant ICB demonstrates promising efficacy in resectable HCC, particularly when combined with TKIs or dual ICB therapies. While dual ICB enhances pathological response rates, it is associated with higher rates of severe TRAE. These results highlight the need of further studies to optimize combination strategies that balance efficacy and safety for patients with resectable HCC. Research Sponsor: None.

Long-term outcomes and surgical conversion following immunotherapy in MSI-H biliary tract cancers.

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Background: Deficient mismatch repair/microsatellite instability (MSI-H) in biliary tract cancers (BTC) is rare, estimated to be 1-5%. MSI-H across solid tumors has shown unprecedented durable response to immune checkpoint inhibitors (ICI) regardless of anatomic location. Here we describe the clinical outcomes of unresectable BTC patients with MSI-H treated with ICIs and their conversion to surgery. **Methods:** We conducted a multicenter, retrospective analysis of patients (pts) with advanced BTC who had been identified with MSI-H by molecular analysis performed between 2017 and 2024 at Mayo Clinic and University Hospitals Seidman Cancer Center. The outcomes of interest were overall survival (OS), defined as time from initiation of first-line systemic treatment to death; Time-to-treatment failure (TTF), defined as time from initiation of treatment to cessation due to any cause; and response. All analyses were done descriptively. **Results:** We identified 22 MSI-H pts with BTC, 19 with cholangiocarcinoma and 3 with gallbladder cancer. Median age at diagnosis was 61 (range 26-86), 20 (90.9%) were Caucasian, 9 (40.9%) were male gender, and 5 (22.7%) had etiology as Lynch syndrome. Twelve (54.5%) pts had metastatic disease at diagnosis. Twenty pts (90.9%) received first line systemic therapy, of whom 11 (55%) continued onto second line. Among the 20 pts that received first line therapy, median TTF was 7.6 months (95% CI: 4.8-14.3), objective response was observed in 7 (35%) pts, and disease control was observed in 15 pts (75%). Among the 11 pts that continued onto second line systemic treatment, objective response was observed in 4 pts (36%) and disease control was observed in 6 pts (55%). Among the 20 pts that received first line therapy, 19 (95%) ever received ICI +/- chemotherapy as a part of their treatment plan. Two-year OS for pts who ever received ICI was 63% (95% CI: 41%-96%). In total, 4 pts (20%) were restaged to resectable following treatment with an ICI, underwent resection, and currently have no evidence of disease. **Conclusions:** Administration of ICI during treatment plan in pts with MSI-H confirmed durable response and even downstage pts to resectable stage for curative intent. Testing for MSI-H should be performed upfront to guide treatment decision.
Research Sponsor: None.

Surufatinib combined with locoregional therapies and immune checkpoint inhibitor (ICI) for treating unresectable or metastatic intrahepatic cholangiocarcinoma.

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Background: Advanced metastatic ICC, characterized by poor survival and limited therapeutic options, may benefit from the combined use of surufatinib—a selective tyrosine kinase inhibitor targeting VEGFR 1, 2, and 3, FGFR1, and CSF-1R—and ICIs, alongside locoregional treatments like TACE, DEB-TACE, and HAIC. **Methods:** Eligible pts who were 18 -75 years old with histologically confirmed unresectable or metastatic intrahepatic cholangiocarcinoma were enrolled. Pts received surufatinib (250mg, orally daily), ICI, and locoregional therapies until surgery, disease progression, death, intolerable toxicity, or withdrawal of consent. The primary endpoint was objective response rate (ORR). Secondary endpoints included progression-free survival (PFS), disease control rate (DCR), overall survival (OS), conversion to surgical resection rate and safety. **Results:** By July 31, 2024, 20 pts were enrolled and 10 pts were evaluated. Median age was 56.4 years (range: 36-69), with a majority being male (70%). 9 pts were pathologically confirmed as adenocarcinoma, 1patient was small bile tract type. Treatment included DEB-TACE (20%), TACE (30%), and HAIC (50%). 50% (5/10) pts achieved partial response (PR), 20% (2/10) pts accepted conversion surgery, 30% (3/10) pts achieved stable disease (SD), and the confirmed ORR was 50%, DCR was 80%. The median PFS (95%Cl: 6.9-NA months) had not yet matured, with no significant differences in survival benefits based on age or gender, though TACE showed potential benefits. The most common AEs of all grades were nausea (30%), dizziness (30%), and headache (10%), No grade ≥ 3 TEAEs or new safety signals occurred. **Conclusions:** Surufatinib plus immune checkpoint inhibitor, along with locoregional therapies showed preliminary anti-tumor activity and manageable toxicity for the 1L treatment of ICC, providing an additional treatment option for pts with ICC. Clinical trial information: NCT05236699. Research Sponsor: None.

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Phase I/Ib study of afatinib with capecitabine in patients with refractory solid tumors and pancreaticobiliary cancers.

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Background: The epidermal growth factor receptor (EGFR) and ErbB2/HER2 signaling is overactive in many solid tumors. Afatinib, an irreversible inhibitor of ErbB signaling, down-regulates thymidine synthase and synergizes with capecitabine in preclinical models. This phase I trial evaluated the safety, maximum tolerated dose (MTD) and preliminary efficacy of afatinib plus capecitabine in patients (pts) with refractory solid tumors and pancreatic ductal adenocarcinoma (PDA) and biliary tract cancers (BTC). **Methods:** The phase I study had a 3+3 design with capecitabine 1000 mg/m^2 twice daily (BID) on days 1-14 and afatinib 20 mg, 30 mg or 40 mg daily (QD) in 21-day cycles. In phase Ib, 15 pts each with PDA and BTC were treated at MTD. **Results:** 41 pts enrolled from November 2016 to October 2021, who received a median of 2 (range 1-5) prior therapies. No dose-limiting toxicities were observed. The MTD was 40 mg afatinib QD plus 1000 mg/m^2 capecitabine BID. Grade 3 treatment related adverse events were diarrhea (12.2%) and nausea (4.9%). Among 36 response evaluable pts, best response was 1 partial response (PR, 3%) in a pt with *KRAS* wild type (WT)/EGFR amplified BTC, and 8 stable diseases (SD, 22%), with disease control rate (DCR) of 25%. PDA pts (n=20) had DCR of 24%, median progression-free (PFS) and overall survival (OS) of 1.9 months (95% CI 1.05, 2.03) and 3.2 months (95% CI 2.01, 5.82) respectively. BTC pts (n=18) had DCR of 31%, median PFS and OS of 1.9 months (95% CI 1.55, 3.39) and 4.6 months (95% CI 1.88, 6.09), respectively. Eight pts with cancers with EGFR/HER2/HER3 pathway alterations, most $KRAS^{WT}$ (n=6), had DCR of 29%, and median PFS and OS of 2.1 months (95% CI 1.05, 4.14) and 3.5 months (95% CI 1.64, 6.09), respectively. **Conclusions:** Afatinib plus capecitabine is tolerable but does not have significant efficacy in pts with refractory PDA/BTC, including in $KRAS^{WT}$ cancers with EGFR/HER2/HER3 alterations. Clinical trial information: NCT02451553. Research Sponsor: None.

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Efficacy and safety of durvalumab and gemcitabine-based chemotherapy combined with or without lenvatinib in advanced biliary tract cancer.

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Background: Immunotherapy in combination with gemcitabine-based chemotherapy, has now become the standard first-line treatment for advanced biliary tract cancer (BTC). Some small-sample studies have shown that immune checkpoint inhibitors (ICIs), when used in combination with lenvatinib and chemotherapy as first-line therapy, exhibit high anti-tumor activity in BTC. Therefore, we conducted a retrospective study to assess the efficacy and safety of durvalumab and gemcitabine-based chemotherapy with or without lenvatinib, in combination for advanced BTC in the real world. **Methods:** Between January 2021 and September 2024, patients with advanced BTC who received either durvalumab combined with or without lenvatinib, along with gemcitabine-based chemotherapy, were enrolled in this study. The study endpoints were overall survival (OS), progression-free survival (PFS), objective response rate (ORR), disease control rate (DCR), and safety. **Results:** Thirty patients with advanced BTC were included in this study. Fourteen (46.7%) patients had received durvalumab combined with lenvatinib plus chemotherapy. For all the patients, the median OS was 9.5 months(95% CI: 4.98-14.02), and the median PFS was 5.9 months(95% CI: 4.56-7.24). ORR was 43.4% and DCR was 76.7%. Patients with lenvatinib treatment had a longer median PFS (7.1 vs. 4.6 months, P = 0.06) compared with the non-lenvatinib treatment group. However, the median OS was shorter in patients with lenvatinib treatment group (9.0 vs. 9.5 months, P = 0.915). ORR and DCR in patients with lenvatinib were 57.1% and 92.9%, respectively. However, in patients without lenvatinib treatment group, ORR and DCR were only 31.2% and 62.5%, respectively. All patients experienced adverse events (AEs), addition of lenvatinib does not increase the risk of AEs. The data is still being updated. **Conclusions:** Durvalumab and gemcitabine-based chemotherapy combined with or without lenvatinib has shown to be effective and safe in routine practice. Regarding lenvatinib, its addition has only improved the DCR, ORR, and PFS, but has not been able to prolong OS. Research Sponsor: None.

Adjuvant capecitabine in resected biliary tract cancers: Real life data from a multicenter study (PRODIGE83-ACABi-PRONOBIL).

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Background: Biliary tract cancers (BTCs) are rare and heterogeneous tumors with a poor survival even in early-stage patients treated with a curative surgery. Since 2019 the current standard of care in the adjuvant setting is oral chemotherapy (CT) with capecitabine for 6 months, as an option following BILCAP phase III trial results. Nevertheless, outside of this trial, real-world data are lacking. **Methods:** We conducted a French, retrospective, multicenter study from the ACABi-PRONOBIL cohort. Eligible patients had intrahepatic, perihilar, distal cholangiocarcinoma or gallbladder carcinoma treated by surgery with curative intent without prior therapy. Two groups were compared, patients who received capecitabine (Cape group) as adjuvant CT and patients without adjuvant treatment before 2017 (surveillance group). The primary endpoint was overall survival (OS). Secondary endpoints were disease-free survival (DFS) and toxicity. **Results:** 324 patients were included, 200 in the Cape group and 124 in the surveillance group. The median OS was 43.5 months (CI95%= 35.4-48.0) and 44.6 months (CI95% 37.7-54.8) in the Cape and surveillance group, respectively. ECOG performance status ≥ 2, pT3 stage, vascular and node invasion were associated with a poorer OS. The DFS was 15.8 months (CI95%= 14.1-20.3) in the Cape group and 13.3 months (CI95%= 10.1-17.5) in the surveillance group. In inverse probability of treatment weighting (IPTW) analysis we observed a significant statistical association between Cape group and DFS (IPTW HR (CI95%) Cape group vs surveillance group = 0.76 (0.60-0.96), pvalue=0.0224) but not with OS. In the Cape group, 49% of patients had at least one dose reduction, 35% had to discontinue treatment due to toxicity (mainly gastrointestinal,13.3% of grade 3-4; and cutaneous, 27% of grade 3-4). **Conclusions:** In this retrospective analysis, adjuvant capecitabine appears to increase DFS without statistically significant impact on OS in patients undergoing surgery for BTC, confirming in real life the BILCAP study results. Research Sponsor: None.

Comparative restricted mean survival time (RMST) analysis of survival in advanced hepatocellular carcinoma (aHCC) from pivotal phase III trials: IMbrave-150, ORIENT-32, CARES-310, HIMALAYA, and CM-9DW.

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Background: In advanced hepatocellular carcinoma (aHCC), the evaluation of overall survival (OS) and progression-free survival (PFS) through restricted mean survival time (RMST) provides a nuanced understanding of treatment efficacy. The RMST analysis serves as a valuable complement to the hazard ratio (HR) analysis, offering insights into the average survival time over a specified period. This study compares RMST analyses of OS at 24 and 36 months, and PFS at 12 and 18 months, across five pivotal phase III trials: IMbrave-150, ORIENT-32, CARES-310, HIMALAYA, and CM-9DW. **Methods:** Data from the experimental arms of the five phase III trials were analyzed. The RMST was calculated using the area under the Kaplan-Meier survival curves up to the specified time points. Specifically, OS was evaluated at 24 and 36 months, and PFS at 12 and 18 months. The RMST provides an estimate of the average time that patients survive (for OS) or remain progression-free (for PFS) within a specific timeframe. Kaplan-Meier survival curves were digitized if not available in raw form, and the area under the curve (AUC) was computed using numerical integration methods. RMST values were extracted from the AUC for the specified periods. This approach accounts for censored data and provides a robust comparison of survival times across different treatment groups. **Results:** The results for OS RMST at 24 and 36 months and PFS RMST at 12 and 18 months are summarized in the table. **Conclusions:** Angiogenesis inhibitor + immune checkpoint inhibitor combinations (Atezolizumab + Bevacizumab; Sintilimab + IBI305) provide the most favorable RMST outcomes in terms of OS and PFS. Research Sponsor: None.

Restricted mean survival time (RMST) comparison of OS and PFS (months).

aHCC 1L Phase III Trials	Treatment Arm	OS 24m	OS 36m	PFS 12m	PFS 18m
IMbrave-150 [1, 2]	Atezolizumab + Bevacizumab	17.50	22.79	9.69	13.29
ORIENT-32 [3]	Sinitilimab + IBI305	17.01	21.09	9.18	12.54
CARES-310 [4]	Camrelizumab + Rivoracenib	16.34	19.75	9.29	12.14
HIMALAYA [5, 6]	Tremelimumab + Durvalumab	15.36	18.76	8.95	12.02
CM-9DW [7, 8]	Nivolumab + Ipilimumab	15.87	20.08	8.74	11.78

[1]Finn et al. N Engl J Med 2020;382:1894-905; [2]Cheng et al. J Hepatol 2022;76:862-873; [3]Ren et al. Lancet Oncol 2021;22:977–90; [4]Qin et al. Lancet 2023;402:1133–46; [5]Abou-Alfa et al. NEJM Evid 2022;1(8): EVIDoa2100070; [6]Sangro et al. Ann Oncol 2024;35:448-457; [7]Galle et al. J Clin Oncol 2024;42(suppl 17): abstr LBA4008; [8]Decaens et al. Ann Oncol 2024;35:S657.

598 **Poster Session**

Durvalumab (DUR) effectiveness in advanced biliary tract cancer (aBTC) patients based on real-world data evidence.

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Background: Based on TOPAZ-1 trial, Durvalumab (DUR) plus gemcitabine and cisplatin significantly improved overall survival (OS) in advanced biliary tract cancer (aBTC) patients. Real-World data evidence can evaluate the benefit of this combination outside controlled studies comparing patients treated with standard chemotherapy with or without DUR. **Methods:** Using the TriNetX Global Collaborative Network, a platform that operates globally based on anonymized and aggregated clinical data, a sample of aBTC patients from 128 healthcare organizations (HCOs) who met the initial criteria was selected. 5-year overall survival (OS) was analyzed between these cohorts using a Kaplan-Meier analysis. Hazard Ratio (HR) and its 95% confidence interval (95%CI) were calculated to evaluate the difference between cohorts. Propensity Score Matching (PSM) was used to balance the cohorts based on age, gender, and race mitigating possible cofounding variables. All statistical analyses were conducted utilizing the TriNetX Analytics function in the online research. **Results:** A total of 1099 patients met the study criteria and were included in our study. 399 received DUR plus chemotherapy and 700 received only chemotherapy. Patients treated with DUR showed a significant better OS compared to patients treated with only chemotherapy, both previous to PSM and before PSM (post-PMS median OS 418 days vs 359 days, HR 0.862, 95%CI 0.759-0.979, p=0.022). **Conclusions:** This is the largest World-Real Data study exploring the effectiveness of Durvalumab in aBTC patients outside clinical trials, confirming a similar benefit than TOPAZ-1 study. Research Sponsor: None.

Efficacy and safety of HER2-targeted therapies in biliary tract cancer: A systematic review and meta-analysis of early phase clinical trials.

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Background: Biliary tract cancers (BTC) are a rare type of gastrointestinal malignancy, often presenting in advanced stages and associated with a dismal prognosis. BTC are frequently resistant to conventional chemotherapy, with novel targeted therapies playing a pivotal role in overcoming this challenge. *HER2* has recently emerged as an important actionable target in patients with BTC. We aim to evaluate the efficacy and safety of *HER2*-directed monoclonal antibodies (MOAs), bispecific antibodies, antibody-drug conjugates (ADCs), and their combinations in BTC. **Methods:** A systematic literature search was conducted on PubMed, Embase, and Cochrane Central Register of Controlled Trials for clinical trials investigating *HER2*-directed MOAs, bispecific antibodies, ADCs, and their combinations in BTC. The study was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Guidelines. A random effects model was used to pool the outcomes along with 95% confidence intervals (CI). Statistical analyses were performed using program R version 4.4.1. **Results:** We included 12 clinical trials with three Phase I and nine Phase II trials. The pooled ORR (objective/overall response rate) was 35.5% (95% CI: 26.2-45.3, I^2: 69%), and the DCR (disease control rate) was 72.2% (95% CI: 65.3-78.8, I^2: 37%). The median follow-up period for DCR was four months. Trastuzumab, in combination with gemcitabine-cisplatin as first-line therapy, had the highest median progression-free survival of 7 months (95% CI: 6.2-7.8), whereas the combination of trastuzumab and tucatinib had the longest median overall survival 15.5 months (90% CI: 6.5-16.7) reported. The most common treatment-related ≥grade 3 adverse events reported across the cohort were anemia and neutropenia. Three cases of confirmed grade 5 interstitial lung disease were reported in patients who received trastuzumab deruxtecan. Clinical trials included in this study are summarized in the table. **Conclusions:** Advances in precision oncology and the emergence of novel targeted therapies have revolutionized cancer management. *HER2*-directed therapies have shown promising results for patients inflicted with BTC having *HER2* alterations in the early phase clinical trials. Further larger studies are awaited to explore the efficacy of *HER2*-directed therapies in BTC. Research Sponsor: None.

Clinical trials of *HER2*-directed therapies in BTC.

HER2-directed therapy	Number of clinical trials	Total number of BTC participants	ORR, in % (95% CI)	DCR, in % (95% CI)
Trastuzumab + chemo	2	124	43.2 (19.4-68.8)	80 (72.3-86.7)
Trastuzumab + pertuzumab	2	45	20.9 (9.1-35.3)	53.4 (37.8-68.8)
Trastuzumab deruxtecan	4	84	27.2 (6.9-52.6)	71.9 (58.4-84)
Trastuzumab + tucatinib	1	30	46.7 (28.5-65.5)	76.7 (57.7-89.7)
Zanidatamab	2	101	40.5 (30.9-50.4)	67.6 (57.9-76.6)
SHR-A1811	1	16	56.3 (29.9-80.2)	81.3 (54.4-96)

Long-term survival outcomes of atezolizumab plus bevacizumab treatment in patients with advanced hepatocellular carcinoma.

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Background: Immunotherapy combinations such as atezolizumab plus bevacizumab (Atezo-Bev) or STRIDE have been established as standard therapies for patients with advanced hepatocellular carcinoma (HCC). After dual immunotherapy treatment, durable responders and long-term survivors had been reported. However, the longevity of such outcomes following Atezo-Bev treatment remains to be evaluated. **Methods:** We analyzed medical records of four medical centers in Taiwan for patients with Child-Pugh class A liver reserve who received first-line Atezo-Bev treatment for advanced HCC from January 2018 to May 2021, to evaluate their survival outcomes after a long-term follow-up. **Results:** We enrolled 54 patients, predominantly male (90.7%) with the median age of 65 years. The main hepatitis etiology was viral hepatitis (92.6%), including 68.5% chronic hepatitis B. After a median follow-up of 61.9 months, the median overall survival (OS) was 21.0 months (95% confidence interval 10.4 – 31.6 months). The 3-year, 4-year and 5-year OS rate was 36.4%, 25.7% and 25.7%, respectively. For patients with tumor response (N = 19, 34.5%) or disease control (N = 44, 80.0%), the 3-year, 4-year, and 5-year OS rates are listed at Table 1. In univariate analysis, younger age ($\leq$ 70 years), absence of intrahepatic tumor, lack of macrovascular invasion (MVI), and ALBI grade 1 were associated with an OS $>$ 3 years. In multivariate analysis, after adjusting sex, viral hepatitis, MVI, extrahepatic spread, and initial alpha-fetoprotein $>$ 400 ng/ml, the absence of intrahepatic tumor (Odds ratio [OR] = 5.36, p = 0.047) and ALBI grade 1 (vs grade 2, OR = 10.02, p = 0.037) remained independent predictors for OS $>$ 3 years. **Conclusions:** In patients with advanced HCC, Atezo-Bev treatment led to a notable proportion achieving long-term survival, especially in patients with good response to the treatment. Absence of intrahepatic tumors and preserved liver function are predictive of prolonged survival. Research Sponsor: Ministry of Science and Technology, Taiwan; National Science and Technology Council; Ministry of Health and Welfare; National Taiwan University Hospital.

3-year, 4-year and 5-year OS rate of patients who receive Atezo-Bev treatment, stratified by treatment response.

	OS rate (%) by treatment response	
Landmark survival	**Disease control**	**Responder**
3 years	35.6	52.2
4 years	30.3	52.2
5 years	30.3	47.7

Responder: patients who achieved a best response of complete response or partial response after treatment with Atezo-Bev.
Disease control: includes responders and patients who achieved a best response of stable disease following Atezo-Bev treatment.

601 Poster Session

Updated results of ALTER-H006: A phase II study of benmelstobart (PD-L1 inhibitor) plus anlotinib as adjuvant therapy in hepatocellular carcinoma (HCC) with high risk of recurrence after radical resection.

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Background: Although radical resection offers one of the most favorable prognosis for hepatocellular carcinoma (HCC), the rate of postoperative recurrence remains high, which is the primary cause of mortality in HCC patients. The optimal adjuvant treatment strategy for patients is still under active investigation. Herein we evaluated the efficacy and safety of the anti-angiogenic tyrosine kinase inhibitor, anlotinib, plus benmelstobart, a novel programmed death-ligand 1 (PD-L1) inhibitor as an adjuvant treatment for HCC with high risk of recurrence after radical resection. **Methods:** This study enrolled patients diagnosed with HCC confirmed through histological or cytological examination, whose age was 18-75, with ECOG 0-1, Child-Pugh class A, 4~8 weeks after R0 resection with any of the following high-risk factors for recurrence: a) tumor nodules ≥4; b) portal vein tumor thrombus (PVTT): vp1 or vp2; c) hepatic vein tumor thrombus (HVTT): vv1 or vv2. Enrolled patients received anlotinib (12 mg, p.o., qd, d1-14, q3w) plus benmelstobart (1200 mg, i.v., d1, q3w) until disease recurrence or unacceptable toxicity or up to 18 cycles (~1 year), whichever occurred first. Tumor assessment was performed at the end of the second cycle and every four cycles thereafter, according to RECIST v1.1. The predefined sample size was 37. The primary endpoint was 1-year recurrence-free survival (RFS) rate. Secondary endpoints included RFS, 1-year overall survival (OS) rate, and safety. **Results:** As the cutoff date of September 13,2024, a total of 38 patients (pts) were enrolled and 35 pts included in per-protocol set analysis. The majority of the 38 pts were male (94.7%, n = 36), and the median age was 56.5 years (range: 33-75). 18 pts (47.3%) had CNLC Stage IIb and 20 (52.6%) had CNLC Stage IIIa HCC. Among them, 35 pts received at least once tumor assessment. According to RECIST 1.1, of the 35 pts, 18 had no recurrence, 15 experienced relapse, 1 dropped out and 1 discontinued due to serious adverse events. The 1-year RFS rate was 59.97% (95%CI: 39.12-79.68) and the primary median RFS was 16.89m(95%CI: 6.82-26.96). The 1-year OS rate was 91.47% (95%CI: 69.23-97.86). 34 of 38 pts (89.5%) experienced treatment-related adverse events (TRAEs). The most common grade 3 TRAEs occurred in 17 pts (44.7%) including hypertension (32%) and neutropenia(5.3%). No grade 4 or 5 TRAEs were observed. **Conclusions:** The present study indicated that anlotinib plus benmelstobart as adjuvant treatment for HCC with high risk of recurrence after radical resection exhibited promising efficacy and tolerable safety profile. The conclusion should be validated in more participants included subsequently. Clinical trial information: NCT05111366. Research Sponsor: None.

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Pembrolizumab and stereotactic radiotherapy combined in subjects with advanced HCC: A phase II study (PEMRAD).

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Background: Combination approaches of immune-checkpoint inhibitors and locoregional therapies have an increasing role in advanced HCC. **Methods:** The PEMRAD phase II trial investigated the efficacy of combination pembrolizumab and stereotactic body radiation (SBRT) following progression on sorafenib. Patients with advanced HCC, ECOG 0-1 and Childs-Pugh A were eligible. Patients received pembrolizumab on day 1 followed by SBRT starting Day 2 and delivered in five fractions. Pembrolizumab was continued every 21 days until progression, unacceptable toxicity or patient withdrawal. The primary endpoint was objective response rate (ORR) as measured by RECISTv 1.1. hypothesizing an increase to 40% with the combination. Tissue and liquid biopsies were collected for correlative analyses including multiplex IHC, longitudinal CyTOF and dynamic cytokine changes. **Results:** The trial was stopped early due to a shift in the treatment landscape. Between March 2018 and March 2023 18 patients of a planned 22 patients were enrolled and treated with combination therapy. Of these 10 (55%) had evidence of macrovascular invasion (MVI) and 15 (83%) had extrahepatic disease; viral hepatitis was the underlying etiology in 50%. The ORR was 41% meeting the proposed endpoint. Median number of cycles of pembrolizumab was 7.5 (2-35) .The median number of lesions treated by SBRT was 3 (1-5) and the median dominant gross tumour volume was 184 cc (IQR 69-285). The median PFS was 5.4 (2.8-9.9) months and median OS 12.6 (5.7-25.8) months. Of the 10 patients with macrovascular invasion 5 (50%) had evidence of a vascular thrombus response by RECIST criteria including 1 complete response. Treatment related adverse events ≥ grade 3 were reported in 5 patients (28%). One death due to myocarditis was attributed to treatment. **Conclusions:** In a poor prognostic group of patients in the second line setting, the combination of SBRT and pembrolizumab demonstrated high ORR and may have a specific role in patients with MVI. Correlative analyses will be presented at the meeting. Clinical trial information: NCT03316872. Research Sponsor: Merck.

Clinical outcome of radiofrequency ablation in elderly hepatocellular carcinoma patients over 80 years old.

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Background: The SURF trial showed the efficacy and safety of radiofrequency ablation(RFA). However, patients over 80 years old were excluded from the study. Aims: To evaluate clinical outcomes of RFA for largest hepatocellular carcinoma (HCC) diameter ≤3 cm, and ≤3 HCC nodules in patients over 80 years. **Methods:** Patients who underwent RFA at our institution for initial treatment of largest HCC diameter ≤3 cm, and ≤3 HCC nodules from January 2011 to December 2023. Treatment outcome and prognosis were examined in the elderly group for cases 80 years or older and in the non-elderly group for cases under 80 years. The Cox proportional hazards model was used to analyze the factors associated with treatment outcome and prognosis. **Results:** Of the 518 eligible patients, 136 patients were 80 years or older. Median overall Survival (OS) was 80 (95%CI 60-96) vs. 123 (95%CI 101-NA) months (p=0.021) in the elderly vs. nonelderly group. The median recurrence-free survival was 16 (95%CI 14-22) vs. 26 (95%CI 19-30) months (p=0.023), better in the non-elderly group, Interestingly, for liver disease-related deaths, median OS was 97 (95% CI 80-NA) vs. NR (95% CI NA-NA) months (p=0.62) in the elderly vs. non-elderly group. In the multivariate analysis, factors associated with OS were ALBI grade 2 or 3 (HR 1.67, 95%CI 1.07-2.60), DCP ≥ 40mAU/ml (HR 2.08, 95%CI 1.42-3.05), persistent HCV infection (HCV non-SVR) (HR 5.45, 95%CI 3.07-9.67), nonviral liver disease (HR 4.18, 95%CI 2.31-7.55). HCC recurrence was significantly associated with male (HR 1.51, 95%CI 1.18-1.94), elderly group (HR 1.46,95%CI 1.10-1.95), PT ≥80% (HR 0.69, 95%CI 0.53-0.91), DCP ≥40mAU/ml (HR 1.37, 95%CI 1.07-1.76), and HCV non-SVR (HR 1.73, 95%CI 1.35-2.21). The factors associated with liver disease-related death were ALBI grade 2 or 3 (HR 2.16, 95% CI 1.25-3.73), DCP ≥ 40mAU/ml (HR 2.34, 95% CI 1.48-3.70) and HCV non-SVR (HR 2.22, 95% CI 1.39-3.56). **Conclusions:** In RFA for diameter ≤3 cm, and ≤3 HCC nodules, over 80 years was not a significant factor associated with OS or liver disease-related death. The results support that RFA would be a promising treatment option for HCC patients over 80 years. Research Sponsor: None.

Mediation of the liver protective effect of the anti-cancer drug candidate namodenoson via adiponectin.

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Background: Namodenoson (CF102) is a small, orally available, anti-cancer drug candidate, currently in pivotal phase 3 clinical study for the treatment of hepatocellular carcinoma. Adiponectin is a positive cytokine released by adipocytes and endothelial cells, known to induce liver-, cardio-, and neuro-protective effects. We evaluated the potential role of adiponectin as the mediator of the observed namodenoson-induced protective effects. **Methods:** Proteins extracted fromN1S1 tumor-bearing mice and from murine adipocyte culture treated with namodenoson or vehicle were analyzed using Western blot. Apoptosis was determined in the N1S1 tumor-bearing mice using the TUNEL assay. The effect of namodenoson on hepatic ischemia/reperfusion (IR) was evaluated in Wistar rats by temporarily clamping the hepatic vasculature; the effect on hepatic inflammation was evaluated in Con. A-induced hepatitis in mice. The STAM murine in-vivo model was used to study the effect of namodenoson on metabolic dysfunction-associated steatohepatitis (MASH). Adiponectin levels were determined in the serum of patients in the phase 2 trial evaluating namodenoson vs placebo in patients with metabolic dysfunction-associated steatotic liver disease (MASLD)/MASH. **Results:** The anti-cancer effect of namodenoson was attributed to its capability to bind to the A3 adenosine receptor, inducing inhibition of PI3K and subsequent de-regulation of the Wnt/β-catenin and NF-kB signal transduction pathways, resulting in apoptosis of liver tumor cells. At the same time, namodenoson demonstrated a liver protective effect manifested by prevention of hepatic IR injury, suppression of liver inflammation, and induction of anti-fibrotic and anti-steatotic effects. In adipocyte cell culture, namodenoson stimulated adiponectin production. An increase in adiponectin levels was also shown in the murine MASH model (STAM). These in-vitro and in-vivo results were consistent with translational data from the phase 2 study of namodenoson in MASLD/MASH where upregulation of adiponectin was observed with namodenoson (mean change from baseline of 539 ng/mL for namodenoson 12.5 mg BID vs -78 ng/mL for placebo, P = 0.032). **Conclusions:** The anti-cancer effect of namodenoson combined with its protective effects, position it as a unique drug inducing a dual effect in patients with advanced liver cancer.
Research Sponsor: Can-Fite BioPharma.

605 Poster Session

Results from a phase 2 study of triplet blockade of the IL-27, PD-(L)1, and VEGF pathways with casdozokitug (casdozo, CHS-388) in combination with atezolizumab and bevacizumab in patients with unresectable, locally advanced or metastatic hepatocellular carcinoma (uHCC).

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Background: Casdozo is the first in class and only clinical-stage IL-27 targeting antibody, which induces increased serum IFN-γ and NK cell gene activation, leading to reversal of IL-27-mediated immune suppression. A phase (Ph) 1 study demonstrated a favorable safety profile and antitumor activity of casdozo alone and in combination with PD-1 blockade in indications with known high levels of IL-27 pathway activation, including HCC (NCT04374877). This open-label Ph 2 trial examined the potential antitumor activity and safety of casdozo with PD-L1 and VEGF blockade in uHCC (NCT05359861). Updated clinical and biomarker data are presented. **Methods:** Patients (Pts) with untreated uHCC received casdozo 10 mg/kg, atezolizumab (atezo) 1200 mg, and bevacizumab (bev) 15 mg/kg IV Q3W. Primary endpoints were safety and tolerability, with key secondary endpoints of investigator-assessed ORR by RECIST1.1 and mRECIST, PFS, and OS. **Results:** As of Sept 4, 2024, 30 pts received casdozo/atezo/bev. Most pts were male (77%), Asian (67%), had viral etiology (53% HBV, 17% HCV), and poor-prognosis disease assessed by metastatic spread (67%) and/or macrovascular invasion (13%). Median follow-up was 15 months (mo). Median time on therapy was 8 mo. The most common (>20%) adverse events related to any study drug were proteinuria (43%), hypertension (27%), fatigue (23%), and diarrhea (20%); Grade ≥3 events occurred in 67%, with only hypertension (20%), arising in >10% of pts. Treatment-emergent adverse events (TEAEs) resulting in any study drug discontinuation occurred in 30% of pts. No treatment-related deaths were reported. In the 29 RECIST1.1 response-evaluable pts, ORR was 38% with 17% of pts achieving a CR (5 CRs, 6 PRs); mPFS (95% CI) was 8.1 mo (1.9, 13.6); landmark PFS rates at 6, 12 and 18 mo were 58.6%, 36.1% and 27.1%, respectively; mDoR (95% CI) was NR (6.1, -). In the 28 pts evaluable for mRECIST, ORR was 43% (5 CRs, 7 PRs); mPFS (95% CI) was 8.4 mo (2.0, 14.5); mDoR was NR (12.7, -). mOS (95% CI) was 19.9 mo (14.2, -); the 12-mo survival rate (95% CI) was 85% (65, 94). More than 40% of pts remain on study, and 16.7% remain on study treatment. Available archival tissue of responders expressed IL-27+ TAMs by IHC, including tumors of viral and nonviral etiologies. **Conclusions:** Triplet blockade of IL-27, PD-(L)1, and VEGF pathways with casdozo/atezo/bev continues to show a manageable safety profile with promising antitumor activity in uHCC that warrants continued exploration. Toxicity was consistent with the known profiles of the agents, with no new safety signals identified. Biomarker analysis to evaluate immune response and associations of IL-27 pathway and clinical outcomes is ongoing. Additional clinical studies of casdozo in combination with toripalimab are planned. Clinical trial information: NCT05359861. Research Sponsor: Coherus BioSciences; Roche.

606 **Poster Session**

First-in-human phase 1/2 study (MYCHELANGELO I) of first-in-class epigenomic controller OTX-2002 targeting MYC oncogene in patients with hepatocellular carcinoma (HCC) and other solid tumors.

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Background: The c-MYC (MYC) oncogene is a master transcription factor of tumor cell and microenvironment regulation; it is often dysregulated in cancer, including HCC. OTX-2002 is a MYC-targeted Epigenomic Controller (MYC-EC), an mRNA drug substance encapsulated in a clinical lipid nanoparticle (LNP), designed to downregulate MYC expression pre-transcriptionally with high specificity and durability, inhibiting tumor cell viability while sparing normal cells. **Methods:** A first-in-human dose-escalation study (NCT05497453) investigates OTX-2002 monotherapy in HCC and other solid tumors. Key outcomes include safety, tolerability, pharmacokinetics (PK) and the recommended dose for expansion. In addition to standard safety measures, LNP-associated safety measures including cytokine profiling and an assessment of the presence of anti-PEG antibodies (ADA) were included. Initial pharmacodynamic activity (PD) was assessed via measurement of DNA methylation at the MYC locus in cell-free DNA [cfDNA] from whole blood. **Results:** As of August 23, 2024, 24 patients were treated at 6 doses (19 HCC, 1 colorectal, 1 cervical, 1 pancreatic, 2 sarcoma). OTX-2002 was generally well-tolerated; most adverse events (89%) were low grade, with infusion-related reactions being the most common (30%). Consistent with known LNP profiles, OTX-2002 adverse events were transient and manageable, including cytokine and ADA expression levels. Plasma PK exposures of OTX-2002 were linear across the first five doses analyzed. Significant, on-target dose-dependent changes in MYC gene methylation were seen in cfDNA for all patients post-dose and were durable for at least 15 days, consistent with the designed durability of the MYC-EC. Updated data will be presented. **Conclusions:** OTX-2002 is well-tolerated and induces rapid and durable epigenomic changes to the MYC locus. These results provide clinical proof of mechanism of the epigenomic controller platform and support continued development of OTX-2002. Clinical trial information: NCT05497453. Research Sponsor: Omega Therapeutics.

607 **Poster Session**

Real-world data of durvalumab plus gemcitabine and cisplatin in patients with advanced biliary tract cancer: A Thailand multicenter observational study.

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Background: Currently, durvalumab plus gemcitabine and cisplatin is a standard of care for patients with advanced biliary tract cancer (BTC). We aimed to evaluate efficacy and tolerability of durvalumab plus gemcitabine and cisplatin in real world experience in Thailand where there is high incidence of liver-fluke related cholangiocarcinoma (CCA). **Methods:** This is an observation study of patients with advanced BTC participating in the early access program of durvalumab since August 2023. Data were collected and analyzed for efficacy and toxicity. **Results:** Fifty patients participated the early access program in 14 centers across Thailand. Median follow-up time was 8.5 months. Median number of treatment cycles was 8.8 (1-16) cycles. Thirty six (72%) patients were alive and 8 (16%) patients were still on treatment. The mean age was 63 years, 60% were male and 92% were ECOG 0-1. There were 68% intrahepatic cholangiocarcinoma (CCA), 28% extrahepatic CCA and 8% gallbladder cancer. There were 29.5% objective response rate and 79.5% disease control rate. Median progression free survival (PFS) was 6.0 (95%CI 4.49-7.50) months. Eight (16%) patients were still on treatment with PFS more than 10 months. Grade 3 or 4 adverse events (AEs) were reported in 50% of patients. Most AEs were related to chemotherapy, there were 10% immune related AEs. **Conclusions:** In this real-world experience, durvalumab plus gemcitabine and cisplatin was efficacious and tolerable in Thai patients with advanced BTC. Research Sponsor: None.

Tinengotinib in patients with advanced, metastatic cholangiocarcinoma: Overall survival results and biomarker correlative analysis from a phase 2 clinical trial.

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Background: Tinengotinib, a spectrum-selective multi-kinase inhibitor with unique binding properties to *FGFR*, potently inhibited *FGFR2* fusion/rearrangement and acquired resistant/gatekeeper mutations in pre-clinical models and exhibited antitumor activity in cholangiocarcinoma (CCA) patients (pts) in phase 1/2 trials. Here we present the overall survival (OS) data and biomarker correlative analysis from a phase 2 clinical trial of tinengotinib in CCA. **Methods:** Eligible pts with advanced/metastatic CCA who had received ≥ 1 prior systemic chemotherapy with ECOG PS 0-1 were treated with tinengotinib 10 mg QD across four Cohorts: A1: *FGFR2* fusion(s) with primary progression on previous *FGFR* inhibitor (*FGFR*i); A2: *FGFR2* fusion(s) with progression after prior response to *FGFR*i; B: *FGFR* alteration(s) without *FGFR2* fusion(s); C: *FGFR* wild-type. Efficacy was evaluated per RECIST v1.1 and safety was assessed by CTCAE v5.0. Genomic alterations in circulating tumor DNA (ctDNA) were assessed by Foundation Medicine NGS panel at baseline, C3D1 and EOT. Planned correlative analyses of response (PR+SD>6 months) and PFS in pt subgroups based on ctDNA mutational status were performed. **Results:** 55 eligible pts were enrolled (18 in A1, 11 in A2, 13 in B, 13 in C) with median age 61.0 [range 24-81] years, ECOG PS 0 in 50.9% pts, 56.4% female, and 60.0% with ≥ 3 lines of prior therapy. Among 42 pts with *FGFR* alterations, 78.4% had ≥ 1 prior *FGFR*i, and 97.6% had prior chemotherapy. Median follow up time was 8.6 months (0.4-30.5). The median OS (months) was 17.1 (95%CI 7.5-19.5) in A1; not reached (95%CI 9.6, -) in A2; 18.0 (95%CI 9.6, -) in A1+A2; not reached (95%CI 8.0, -) in B; 6.5 (95%CI 4.8-16.4) in C. The OS rate in A, B and C was 100%, 91.7% and 75% at month 3; 83.7%, 83.3% and 66.7% at month 6, and 65.8%, 55.6% and 23.8% at month 12. Partial response was observed in 26% (10/39) of evaluable FGFR-altered pts (A1+A2+B). The safety profile was consistent with previous reports. The most common any-grade treatment-related AEs were hypertension (63.6%) and diarrhea (47.3%). Baseline ctDNA results were available in 46 pts, 35 in cohorts A+B and 11 in C. A positive correlation was shown in MED12 mutation vs response (P=0.03476). BCOR and ARID1A mutation were found to be negatively associated with PFS (P=0.01366 and P=0.0437). Acquired/secondary mutations at baseline were observed in cohort A, including V564F/I (3/24 and 5/24) and N549K/H (5/24 and 5/24); favorable anti-tumor activity was noted despite these resistant mutations, consistent with preclinical data. **Conclusions:** Tinengotinib has shown promising anti-tumor efficacy in CCA pts with FGFR fusion after prior *FGFR*i and in those with primary FGFR mutations. An ongoing randomized phase III study will investigate tinengotinib vs. chemotherapy in FGFR inhibitor refractory CCA (NCT05948475). Clinical trial information: NCT04919642. Research Sponsor: None.

609 **Poster Session**

Safety and efficacy of combining PD-L1 and CTLA-4 blockade with transarterial chemoembolization in intermediate-stage hepatocellular carcinoma: A phase II study.

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Background: Transarterial chemoembolization (TACE) is the standard treatment for intermediate-stage hepatocellular carcinoma (HCC) defined as Barcelona Clinic Liver Cancer (BCLC) B, but the benefit of adding systemic therapy remains unclear. While durvalumab and tremelimumab are approved for advanced HCC, the safety and efficacy in combination with TACE for intermediate-stage disease is undefined. Combining these PDL1 and CTLA4 inhibitors with TACE may enhance anti-tumor activity by leveraging TACE's proinflammatory effects while blocking immune evasion and reducing immune suppression. This phase IIb study evaluates the safety and efficacy of this combination in BCLC-B HCC. **Methods:** We enrolled patients with intermediate-stage HCC that are also unresectable and ineligible for transplant. All patients received drug-eluting bead TACE (DEB-TACE) followed by durvalumab (1500 mg Q4W) and tremelimumab (300 mg once). Durvalumab monotherapy was continued until disease progression or unacceptable toxicity, for a maximum of 13 cycles. Primary endpoints include objective response rate (ORR) based on mRECIST. Secondary endpoints were overall survival (OS), and safety/adverse events (AEs). The trial is designed to rule out ORR of 30%. We obtained 3 serial biopsies and plasma for participating patients prior and one week after TACE and after the administration of immunotherapy for correlative studies. **Results:** 21 patients were enrolled. Of these, 20 patients were considered evaluable after receiving at least one dose of TACE and immunotherapy. ORR was 55% (95% CI: 32%-77%, CR 10% (n=2), PR 45% (n=9), meeting the primary endpoint. Five patients had PD (25%) per mRECIST, of which two were unconfirmed on repeat imaging and continued treatment beyond progression. Three patients received curative intent treatment as the result of treatment: surgical resection (n=1) and transplant (n=2). At a median follow-up of 18.4 months, median OS was 28.8 mo (95% CI: 15.1, NA) and mPFS was 6.1mo (95% CI: 3.3, NA). Grade 3 or higher immune related adverse events were rash (n=2), diarrhea, myalgia, nausea, elevated AST/ALT. One patient developed multiple irAEs after first dose (myocarditis, myositis, pericarditis) and was taken off study. **Conclusions:** The combination of DEB-TACE with PD-L1/CTLA-4 blockade is safe and shows promising activity in patients with BCLC-B HCC. The ongoing immune analysis and biomarker studies on serial blood and tissue biopsies will provide further insights into the mechanistic effects of this combination approach and inform future therapeutic strategies. Clinical trial information: NCT03638141. Research Sponsor: AstraZeneca.

610 Poster Session

Modified Frailty Index (mFI-5) as a prognostic tool of mortality, complications, and readmissions in patients undergoing hepatic ablation for hepatocellular carcinoma.

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Background: Hepatocellular carcinoma (HCC) is a leading cause of cancer deaths, often diagnosed late, limiting curative options. Transarterial chemoembolization, Y-90 radioablation, radiofrequency ablation, and thermoablation are commonly used for downstaging, transplant bridging, and palliation in unresectable HCC. Frailty, linked to postoperative complications, has not been studied in hepatic ablation patients. This study evaluates the predictive value of the modified frailty index (mFI-5) for mortality, readmissions, complications, redo procedures, and discharges. **Methods:** We utilized the National Surgical Quality Improvement Program (NSQIP) database from 2015 to 2019, querying cases of hepatic ablation for CPT codes: 47370 (closed hepatic ablation, single site) and 47380 (open hepatic ablation, single site). Selected cases were analyzed for frailty using the mFI-5 criteria. A multivariate analysis was performed using mFI-5 frailty scores, stratified into non-frail (0), pre-frail (1), frail (2), and severely frail (3+). The outcomes analyzed included mortality, discharge to a destination other than home, major complications, minor complications, readmission, and reoperation. **Results:** We identified 1921 patients as: 737 non-frail, 701 pre-frail, 442 frail and 41 severely frail. The rates of readmission (5.4%), reoperation (1.4%), major (5.3%) and minor (3.5%) complications were low. Eight patients died following hepatic ablation, and 79 patients were discharged to a destination other than home. In multivariate analysis, most outcomes were not significantly associated with increasing frailty. However, discharge to a destination other than home was significantly more likely in patients with higher frailty scores. In frail patients, the odds ratio (OR) was 2.325 (95% CI 1.228–4.402), and in severely frail patients, this increased to 8.720 (95% CI 3.389–22.435). **Conclusions:** Hepatic ablation is generally well tolerated with low complication rates; however, frailty significantly influences postoperative outcomes. Frail and severely frail patients are more likely to be discharged to care facilities rather than returning home, suggesting that even in the absence of major complications, these patients require a higher level of post-procedural care. This underscores the need for tailored discharge planning and resource allocation based on patient frailty. Research Sponsor: None.

Rates and risk factors for recurrence of hepatocellular carcinoma after initial complete remission in response to transcatheter arterial chemoembolization: A systematic review and meta-analysis.

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Background: Transcatheter arterial chemoembolization (TACE) is used for palliative treatment of patients with inoperable hepatocellular carcinoma (HCC). Despite remarkable initial remission rates, frequent recurrences of HCC after TACE have resulted in questionable survival benefit. Our investigation aimed to study the rates and risk factors for HCC recurrence following remission after TACE. **Methods:** Comprehensive search of 5 databases (EMBASE, Web of Science, PubMed, CINAHL, ClinicalTrials.gov) was conducted to yield 4263 studies, which were screened based on prespecified inclusion and exclusion criteria. Outcomes assessed were total recurrence rate; 6-month, 1, 2, and 3-year cumulative recurrence rates; and 1 and 3-year survival rates for patients with and without recurrence. Meta-analysis was conducted by the random-effects model and heterogeneity was assessed by the I^2 statistic and 95% prediction interval. Systematic review of risk factors for recurrence was performed. **Results:** Twelve studies and 1,152 patients who underwent complete remission in response to TACE were included. The pooled rate of total recurrence was 60% (95% CI 53-67%). The pooled 6-month, 1-year, 2-year, and 3-year cumulative recurrence rates were 26% (95% CI 18-33%), 33% (95% CI 22-43%), 49% (95% CI 34-64%), and 62% (95% CI 53-71%), respectively. The pooled 1-year survival rates for patients with and without recurrence were 94% (95% CI 89-99%) and 99% (95% CI 99-100%), respectively. The pooled 3-year survival rates for patients with and without recurrence were 61% (95% CI 31-90%) and 91% (95% CI 81-100%). Heterogeneity was considerable based on the 95% prediction interval. Predominant risk factors for recurrence were larger tumor size, multinodularity, portal vein thrombosis, and higher AFP levels. **Conclusions:** HCC frequently recurs after initial remission achieved through TACE, with differences in survival rates between patients with and without recurrence. Identifying high-risk patients based on tumor size, multinodularity, portal vein thrombosis, and AFP levels could improve long-term outcomes. Research Sponsor: None.

613 Poster Session

Development and regulatory approval of a new systemic targeted therapy for advanced hepatocellular carcinoma.

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Background: There are no commercially available devices for the treatment of patients with advanced hepatocellular carcinoma who have failed 1st line and 2nd line therapy. We have identified tumor-specific modulation frequencies in patients with advanced hepatocellular carcinoma. We exposed patients to radiofrequency electromagnetic fields, which are amplitude-modulated from 0.1 Hz to 100 kHz and measured changes in pulse pressure. Frequencies eliciting measurable changes in pulse pressure were selected as tumor-specific frequencies. **Methods:** Treatment is administered to patients with advanced hepatocellular carcinoma by means of a battery-operated portable device emitting 27 MHz radiofrequency electromagnetic fields, which are amplitude-modulated at hepatocellular carcinoma frequencies. The device is connected to a coaxial cable ending with a spoon-shaped antenna placed on the anterior part of the patient's tongue during treatment. Treatment is administered three times a day for one hour. **Results:** A total of 69 patients with advanced hepatocellular carcinoma received treatment with the TheraBionic device until progression or death. The median overall survival of patients who received treatment with the TheraBionic device after failing 1st line and 2nd line therapy was 9.15 (95% CI 1.6-34.8) months and the median overall survival of Child-Pugh A patients was 9.5 months, which are comparatively longer than the 7.8 months pooled estimated medians of the placebo arms of 11 randomized 1st line and 2nd line placebo-controlled studies assessing the safety and efficacy of new agents for the treatment of Child-Pugh A advanced HCC (Llovet, Montal et al., J Hepatol 2019). We also analyzed the survival of patients who had received two lines of systemic therapy and did not receive any additional cancer treatment while and after receiving treatment with the TheraBionic device. The median OS of these patients was 9.8 months and the median OS of the Child-Pugh A patients was 17.2 months, which are comparatively longer than the 7.74 months pooled estimated medians of the placebo arms of the above cited 11 randomized studies. To assess unknown device-related adverse events, we assessed patient reported symptoms using the same scale in the SHARP and Asian Pacific Sorafenib studies. The incidence of any grade adverse events among Child-Pugh A who received treatment with the TheraBionic device was not different from the Placebo group of the SHARP and Asian Pacific Sorafenib studies. There were no NCI Grade 2, 3 or 4 toxicities. One patient developed grade 1 mucositis and one patient developed grade 1 fatigue. **Conclusions:** The data presented here provide reasonable assurance of safety and probable benefit in patients with advanced hepatocellular carcinoma who have failed 1st line and 2nd line therapy: https://www.fda.gov/medical-devices/recently-approved-devices/therabionic-p1-h220001. Research Sponsor: None.

Evaluating the safety of immune checkpoint inhibitors prior to liver transplant.

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Background: Liver transplantation (LT) offers a 5-year survival exceeding 70% for selected patients with hepatocellular carcinoma (HCC). Immune checkpoint inhibitors (ICIs) may be used to downstage or bridge to transplantation. This study aims to evaluate the safety of ICIs prior to LT. **Methods:** Multicenter retrospective study, involving 9 centers, included adults who received ICIs and, subsequent LT, between 2019 and 2023. The ICI cohort was matched by age, sex, liver diseases and transplant date (1: 3) with a similar cohort of HCC transplanted patients that did not receive ICIs. **Results:** The ICI cohort included 45 patients (Table part 1). The most common underlying liver diseases were viral. 15.5% of patients had macrovascular invasion and, 62.2% multifocal disease. The most commonly used ICI was nivolumab (75.6%), followed by atezolizumab-bevacizumab (17.8%). The median follow up was 32.5 months (17.4 - 47.6). The median ICI duration was 147 days (129.5-276.5), with a median interval of 58 days (22–193) between the last ICI dose and LT. There were 8 (17.8%) cases of graft rejection, 2 steroid-resistant leading to 1 graft failure. There were 4 (8%) HCC recurrences. During follow-up there were 6 deaths, 2 related to HCC recurrence and 4 non liver related. When compared with 135 non-ICI cohort patients, there were no differences within the groups (Table part 2). Importantly, we found no significant differences in crude rejection rates (p=0.5). **Conclusions:** Our study suggest that ICI treatment prior to LT is safe. ICI rejection primarily occurs in the acute post-transplant period (within 3 months), with a low risk of death due to graft failure. Further clinical trials are needed. Research Sponsor: None.

Part 1: ICI cohort descriptive analysis. N (%); Median (IQR); Part 2: Comparison between cohorts. N (%); Mean (SD).

(part 1)		ICI-cohort n=45	
General data	Male	38 (84.4)	
	Age	61.2 (7.6)	
	Etiology of liver diseases	Hepatitis C 17 (37.8)	
		Hepatitis B 6 (13.3)	
		MASLD 6 (13.3)	
		Viral +Alcohol 5 (11.1)	
		Alcohol 3 (6.7)	
		Autoimmune 3 (6.7)	
		Other 3 (6.7)	
		Combined viral 2 (4.4)	
	Largest tumor (mm)	39 (22)	
	Macrovascular invasion	7 (15.5)	
	Multifocal	28 (62.2)	
	Locoregional therapy	36 (80.0)	
	Number of locoregional therapies	2 (2)	
Immunotherapy	Nivolumab	34 (75.6)	
	Atezolizumab/bevacizumab	8 (17.8)	
	Pembrolizumab	1 (2.2)	
	Nivolumab+pembrolizumab	1 (2.2)	
	PD1 inhibitor clinical trial	1 (2.2)	
	Days of ICI treatment	147 (203)	
Radiological tumor response ICI	No	17 (37.8)	
Transplant	Deceased donor	43 (95.6)	
	Washout period (days)	58 (171)	
	Induction therapy (anti-thymoglobulin)	8 (17.8)	
Outcomes	Rejection	8 (17.8)	
	Biopsy proven	8 (100)	
	Time to rejection (days)	43 (102)	
	Graft failure	2 (4.4)	
	Rejection	1 (2.2)	
	HCC recurrence	4 (8)	
	Deaths	6 (13.3)	
Follow up	Since transplant (months)	32.5 (30.2)	
(part 2)			
	ICI-cohort (n=45)	**Non-ICI cohort (n= 135)**	**P**
Post transplant rejection rate	8 (17.8)	32 (23.7)	0.53
Time to rejection (days)	180.1 (SD 218)	62.2 (SD 62.8)	0.25
Overall deaths	6 (13.3)	10 (7.4)	0.36

615 Poster Session

MDM2 amplification in patients with biliary tract cancer in RETUD registry.

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Background: Antagonism of Mouse double minute 2 homolog (*MDM2*) can restore p53 activity and represents a novel therapeutic strategy in biliary tract cancer (BTC). We aimed to characterize the epidemiology of total population versus that harbouring *MDM2* amplifications; potential associations of *MDM2* to other genetic alterations and survival outcomes. **Methods:** We evaluated a large real-world cohort of patients (pt) diagnosed with BTC from 1st January 2017 to 31st December 2022 from the Spanish RETUD registry. Next generation sequencing (NGS) test was performed in all patients. Data collected included demographic and clinical characteristics, molecular profile, systemic oncologic treatment and effectiveness (best response and survival outcomes). Progression free survival (PFS) outcomes and best response were calculated for pts who received first line therapy. Descriptive statistic was used to analyse sociodemographic data, tumor characterization, molecular analysis, and best response. PFS and overall survival (OS) were estimated by the Kaplan-Meier method. **Results:** A total of 301 evaluable pt were included in 24 sites. *MDM2* amplifications were reported in 19 pt (6.3%) and 2 pt (10.5%) had TP53 mutations. At primary diagnosis, in the *MDM2* amplified population the tumour was mostly intrahepatic (63.2%); followed by gallbladder cancer (21.0%). Half of the pt had advanced disease at diagnosis (52.6%) and 21.0% resectable disease. No significant differences were observed in clinical characteristics (age, sex, tumor location, stage, ECOG performance status and metastasis location) and most common actionable genomic alterations (BRAF, HER2, IDH, FGFR2) in pt with *MDM2* amplification compared to overall population. Median overall survival (95% Confidence Interval [CI]) in pt with *MDM2* amplification was 18.4 months (12.3-19.9), and 17.8 months (95% CI, 12.3-19.9, p=0.247 in pt without amplification. Median PFS in first-line was 5.3 months (95% CI, 2.7-8.9) in *MDM2* amplified group, as compared with 6.0 months in non-*MDM2* amplified group (95% CI, 5.3-6.8; p=0.4233. The objective response rate in first-line was 21.4% in the *MDM2* amplified group versus 29.6% in non-amplified group. The proportion of pt who received systemic therapy in first, second and third line was similar in both groups. The main first-line regimen in *MDM2* amplified pt was cisplatin/gemcitabine (n=9, 47.4%), followed by clinical trial (n=3, 15.8%), GEMOX (n=2, 10.5%) and gemcitabine (n=2, 10.5%). The median OS and PFS of patients treated with cisplatin/gemcitabine in first line was similar in both groups. **Conclusions:** This analysis provides insights into the characterization of *MDM2* amplified BTC pt. Similar incidence of *MDM2* amplification was observed compared to other cohorts of BTC. No significant differences were observed in clinical characteristics, molecular profile and survival in *MDM2* amplified pt compared to non-amplified pt. Research Sponsor: Boehringer Ingelheim.

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The relationship between the tumor microenvironment of hepatocellular carcinoma and apparent diffusion coefficient.

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Background: The tumor microenvironment is critical for the acquisition of tumor malignancy in various cancer types. The objectives of this study were to investigate whether the levels of cancer-associated fibroblasts (CAFs) and tumor-associated macrophages (TAMs) reflect the prognosis of patients with hepatocellular carcinoma (HCC) after hepatectomy (Hx) and to determine whether the apparent diffusion coefficient (ADC) from diffusion-weighted imaging (DWI) reflects CAF and TAM expression. **Methods:** The study cohort comprised 109 patients who underwent initial curative resection for HCC. Alpha smooth muscle actin (αSMA) was selected as a CAF marker and CD204 as a TAM marker. Protein expression was immunohistochemically evaluated in the intratumoral regions of resected specimens. Clinicopathological factors, including the long-term prognosis after Hx, were investigated between αSMA-negative and -positive tumors and between CD204-negative and -positive tumors. The correlation between CAF/TAM marker expression and the calculated minimum ADC using DWI was also evaluated. **Results:** αSMA-positive expression was correlated with tumor number, invasive growth pattern, and advanced stage. CD204-positive expression was correlated with the presence of venous invasion. Both αSMA-positive expression and CD204-positive expression were significant prognostic factors in the univariate analysis of overall survival and disease-free survival. αSMA/CD204 double positivity was associated with an extremely poor prognosis after Hx and was a significant independent prognostic factor for overall survival (p=0.02, hazard ratio: 3.27). Patients with double positivity also showed a significantly higher ADC^{low} rate (83%). **Conclusions:** Expression of both CAF and TAM markers reflected a poor prognosis after Hx. Furthermore, the preoperative ADC could be a clinical surrogate marker in the tumor microenvironment in patients with HCC. Research Sponsor: None.

617 Poster Session

Safety, efficacy, and outcomes of 90-yttrium radioembolization in combination with immune checkpoint inhibitors as a definitive first-line treatment approach in non-surgical hepatocellular carcinoma.

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Background: Immune checkpoint inhibitors (ICI) have emerged as first line therapy for advanced-stage hepatocellular carcinoma (HCC). However, response rates remain modest (<20%). Combination treatments offer potential to improve initial response rates while extending duration of response. While early studies of ^{90}Yttrium transarterial radioembolization (^{90}Y)-ICI showed fewer adverse events (AE) and improved objective responses (OR) (30-40%), there were no improvements in time to progression (TTP) and overall survival (OS). This study evaluates safety, efficacy, and outcomes of first line ^{90}Y-ICI while evaluating the role of ICI regimen and treatment sequencing. **Methods:** A retrospective, multi-center study was conducted in BCLC A-C stage HCC patients (ECOG 0-1) receiving first line ^{90}Y-ICI with atezolizumab/bevacizumab (A+B) or tremelimumab/durvalumab (STRIDE). Treatment response was recorded at ^{90}Y follow-up. TTP, progression-free survival (PFS), and OS were primary endpoints, with AE and safety as secondary outcomes. **Results:** The study included 40 patients receiving ^{90}Y-ICI from 2021–2024, with a median follow-up of 12 months. The cohort was predominantly Child-Pugh A (96%) with HCV-related cirrhosis (76%), BCLC-C stage (51%), and a median target HCC size of 7.7 cm (IQR, 6-11 cm). Grade 3-4 AEs occurred in 15 patients (37%), leading to steroid use (12, 29%), treatment delays (13, 31%), or discontinuation (4, 10%). The first cycle target OR rate was 82% (32/39), with a 51% (20/39) complete response (CR) rate. The first cycle overall OR was 59% (23/39), with a 41% (16/39) CR rate. No significant differences were found in adverse events, delays, or discontinuations based on sequencing or regimen. Sequencing had no significant effect on target or overall OR with no downstream effect on TTP ($P = 0.968$) or PFS ($P = 0.906$). Similarly, there was no effect of ICI regimen on OR, TTP ($P = 0.813$) or PFS ($P = 0.904$), although the data trended toward higher response rates with the STRIDE regimen. Log-rank analysis revealed target CR was associated with improved TTP ($P = 0.004$) and PFS ($P = 0.009$), but not OS ($P = 0.237$). Achieving an overall CR was associated with further improvements in TTP ($P < 0.001$) and PFS ($P < 0.001$). The cohort 2-year OS was 55% with median OS not yet reached. **Conclusions:** First line ^{90}Y-ICI therapy in BCLC A-C stage HCC is an effective treatment strategy with AEs in the expected range of systemic therapy and low rates of treatment discontinuation. Sequencing and ICI regimen did not significantly impact AEs or treatment outcomes. However, patients achieving CR showed significantly improved TTP and PFS. Although follow-up is limited, these findings suggest achieving CR is key to optimal long-term outcomes. We anticipate long-term follow-up will demonstrate an OS benefit linked to first line CR rate. Research Sponsor: None.

618 Poster Session

Association between ex vivo pharmacotyping of patient-derived tumor organoids and personalized therapeutic options for patients with biliary tract cancer.

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Background: Biliary tract cancers (BTC), including cholangiocarcinomas and gallbladder adenocarcinomas, present significant therapeutic challenges due to limited treatment options and poor prognoses. We report findings from the CLIA-certified PARIS assay, evaluating drug sensitivities of patient-derived tumor organoids (PDTOs) to a panel of oncology drugs. **Methods:** PDTOs were successfully cultured from 27 out of 46 live tumor samples obtained from 43 BTC patients. The majority of patients presented with advanced metastatic disease (60% stage IV, 13% stage III, 10% stage II, 4.4% stage I, 10% unknown) and had exhausted standard therapeutic options. Each PDTO culture underwent testing against an average of 50 drugs, encompassing chemotherapeutic agents and targeted therapies. **Results:** Comparative analysis of PDTO drug sensitivities with patients' prior treatment histories revealed non-responsiveness to >80% of drugs associated with clinical progression, including gemcitabine, cisplatin, and targeted therapies for FGFR translocations. In contrast, 100% (26/26) of the samples demonstrated significant sensitivity to one or more targeted agents, particularly inhibitors of EGFR, MEK, ERK, mTOR, PI3K, MDM2, BCL2, and BET families. Despite these shared sensitivities, each individual PDTO culture exhibited unique responses to targeted therapies, highlighting the genetic and phenotypic diversity between BTC patients. Comparison of *ex vivo* drug sensitivities with tumor genomic profiles validated oncogenic drivers such as HER2 amplification, *KRAS* and *PIK3CA* pathogenic mutations, correlating with sensitivities to EGFR/HER2 inhibitors, MEK inhibitors, and PI3K inhibitors, respectively. While mutations in *KRAS*, *BRAF*, *BRCA1*, *BRCA2*, *ERBB2*, *ERBB3*, or *MET* are present in less than 8% of cases, their role as biomarkers in BTCs requires further validation. PDTOs offer a promising tool to expedite this validation process. The PARIS assay results guided treatment decisions for five patients with metastatic disease, two of whom maintained treatment for over 4 weeks. Notably, two patients showed clinical benefit with everolimus (5 weeks) and dasatinib (11 weeks), experiencing symptom relief and reduced ascites. Intriguingly, 5 out of 7 PDTOs with *FGFR* alterations exhibited exceptional PARIS test responses to dasatinib, suggesting potential efficacy linked to FGFR activation. **Conclusions:** In conclusion, our study demonstrates that ex vivo drug testing of PDTO cultures can inform treatment selection and potentially accelerate drug approvals for BTC, leveraging therapies approved for other cancers. Early integration of such assays in patient management, possibly at diagnosis, holds promise for improving outcomes in this challenging disease. Research Sponsor: None.

619 Poster Session

The Oncopig hepatocellular carcinoma model: An innovative large animal platform for evaluating treatment effects.

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Background: Hepatocellular carcinoma (HCC) is an aggressive disease associated with poor prognosis and limited treatment options. Systemic therapies for advanced HCC include sorafenib, lenvatinib, and atezolizumab plus bevacizumab. However, given the limited efficacy of systemic treatment options, novel treatment approaches including combination of systemic and locoregional therapies (LRTs) is required to improve patient outcomes. These approaches require advanced large animal models to prove effectiveness. Pigs represent an ideal platform that provide both regulatory acceptable and clinically relevant large animal cancer models due to their similar physiology, size, genetics, immunity, and metabolism in relation to humans. The Oncopig HCC model is an ideal tool for testing LRT and other oncolytic based treatments. **Methods:** The Oncopig cancer model develops tumors following induced expression of $KRAS^{G12D}$ and $TP53^{R167H}$ driver mutations utilizing an adenoviral vector encoding Cre recombinase (AdCre). Oncopig HCC cell lines were developed by isolating hepatocytes from Oncopig liver biopsies and transformed in vitro via exposure to AdCre. Transgene expression and hepatocyte cell origin were validated by RT-PCR and arginase-1 staining, respectively. Comparison of in vitro Oncopig, human, and murine HCC cell line phenotypes (proliferation, migration, and chemotherapeutic response) was performed. Oncopig HCC tumor formation was performed via autologous intrahepatic injection of Oncopig HCC cells. Tumor formation was evaluated using ultrasound and CT imaging. **Results:** Oncopig and human HCC cell lines displayed similar cell cycle lengths and migration rates. Oncopig HCC cells were consistently more predictive of human HCC responses than murine cells when treated with standard chemotherapeutic agents (doxorubicin, cisplatin, mitomycin C, and sorafenib). Importantly, consistent with human HCC, Oncopig HCC cells were non-responsive to 5-fluorouracil, while murine HCC cells were responsive. Intrahepatic injection of Oncopig HCC cells resulted in tumors visible on ultrasound and CT scan within 2-4 weeks. Resulting tumors were consistent with human tumors on imaging. Histological analysis of tumor biopsies confirmed the identity of the resulting masses as HCC tumors based on positive arginase-1 and $KRAS^{G12D}$ staining. **Conclusions:** The Oncopig HCC model may be more predictive of human HCC chemotherapeutic responses than currently employed murine models. Tumors develop rapidly (within 2-4 weeks) and are readily identified on ultrasound and CT imaging, making this an ideal model for evaluation of novel therapeutic approaches. Further exploration and development of additional tumor models including further translational characterization will be important for broader utility. Research Sponsor: None.

620 Poster Session

Analysis of gut and mucosal microbiomes in patients with extrahepatic cholangiocarcinoma.

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Background: Cholangiocarcinoma is an aggressive malignancy with a poor prognosis. While gut microbial dysbiosis is linked to various diseases, including biliary disorders and cancer, its association with cholangiocarcinoma remains unclear. **Methods:** This study compared the mucosal and gut microbiomes of patients with extrahepatic cholangiocarcinoma (n = 21) to those with cholecystitis (n = 21) and healthy individuals (n = 28). Mucosal microbiomes were collected from cancer lesions and gallbladders, with fecal samples obtained from 31 participants. The 16S rRNA gene (V3 and V4 region) was sequenced using the Illumina MiSeq platform and analyzed with QIIME 2. **Results:** 16S rRNA sequencing revealed Proteobacteria as the dominant taxon in the cancer cohort, with *Klebsiella, Aeromonas, Staphylococcus, Fusobacterium, and Haemophilus* also enriched. In contrast, *Firmicutes* dominated in the cholecystitis and control cohorts. β-diversity analysis highlighted significant differences between cohorts. Elevated levels of bile acids, particularly chenodeoxycholic acid and ursodeoxycholic acid, were observed in the cancer cohort and positively correlated with *Proteobacteria, Aeromonas, Enterobacter, and Enterococcus.* **Conclusions:** These findings suggest that microbiome dysbiosis contributes to cholangiocarcinoma development. Research Sponsor: None.

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Evaluating immune cell abundance across hepatocellular carcinoma disease stages I-III: Findings from TCGA analysis.

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Background: Hepatocellular carcinoma (HCC), the predominant form of primary liver cancer, is the sixth most prevalent cancer globally and a leading cause of cancer-related mortality. The tumor microenvironment (TME) in HCC, characterized by complex interactions between immune and tumor cells, significantly influences tumor progression and immune suppression. Understanding the immune landscape across the different stages of HCC can inform prognosis and therapeutic strategies. **Methods:** We analyzed RNA seq data from 333 HCC patients (disease stages I-III), from The Cancer Genome Atlas (TCGA), using the immune deconvolution algorithms CIBERSORT and xCell. Differential gene expression analysis was performed using the DESeq2 package in R. Immune cell abundance across various HCC stages was imputed, and survival analyses was conducted using Kaplan-Meier curves and log-rank tests. **Results:** Of the 19,514 genes analyzed, 2,774 were up-regulated and 947 were down-regulated in stage II and III in comparison to stage I. CIBERSORT and xCell analyses revealed significant variability in the immune cell composition between HCC stages. Notable differences were observed in the T cells (CD4 and CD8 subsets), B cells, plasma cells, and stromal components. ANOVA and post hoc tests identified significant stage-specific differences in several immune cell types. For the survival analysis, we categorized patients into low and high groups based on median cell abundances. Overall survival showed no statistically significant differences among the various groups for common myeloid and lymphoid progenitor cells. In contrast, disease-free survival demonstrated statistically significant differences across the groups for both cell types. **Conclusions:** This study emphasizes the tumor microenvironment's pivotal role in HCC, revealing that immune cell composition significantly influences disease-free survival, despite limited effects on overall survival. These findings advocate for further exploration into targeted immunotherapies and the development of predictive models that integrate immune landscape characteristics, ultimately aiming to enhance patient outcomes in HCC. Research Sponsor: None.

Cell type	xCell		Cibersort	
	Mean	SD	Mean	SD
T cell CD4 naive	0.00466	0.01994	1.31e-5	1.69e-4
T cell CD8	0.01368	0.03253	0.03161	0.04947
Eosinophil	3.96e-5	3.41e-4	1.67e-4	8.69e-4
Macrophage	0.02323	0.02441	0.02924	0.04345
Macrophage.M1	0.00914	0.01526	0.01806	0.02054
Macrophage.M2	0.02303	0.01697	0.05439	0.04815
B cell memory	0.00196	0.01418	0.00107	0.00733
B cell naive	0.00163	0.01034	0.01319	0.02148
Neutrophil	2.14e-4	0.00137	0.00112	0.00539
Immune score	0.04621	0.06817	-	-

Table presents various cell type abundances for each algorithm used.

Association of GPR120 and NGF with perineural invasion in cholangiocarcinoma: Clinical and experimental insights.

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Background: Cholangiocarcinoma (CCA) is a highly malignant biliary tumor characterized by frequent perineural invasion (PNI), which is associated with a poor prognosis; however, the underlying mechanisms remain unclear. G-protein-coupled receptor 120 (GPR120) has been implicated in the development of various tumors, while nerve growth factor (NGF) has a strong correlation with PNI. This study aims to elucidate the role of GPR120 in the production of NGF and to investigate the mechanisms by which PNI is induced in CCA. **Methods:** We conducted a retrospective review of medical records for 386 patients with cholangiocarcinoma (CCA), including both extrahepatic (ECC) and intrahepatic (ICC) types, who underwent curative resection at the Department of General Surgery, First Affiliated Hospital of Soochow University, between January 2015 and December 2021. NGF and GPR120 expression were assessed in 60 paraffin-embedded archived CCA tissue samples via immunohistochemical staining (IHC), with an additional 60 normal bile duct samples serving as controls. All tissues were sourced from the aforementioned 386 patients, none of whom received preoperative treatment. Tumors were staged according to AJCC classification. Western blot analysis was employed to evaluate NGF and GPR120 expression in CCA cell lines (QBC 939, RBE, HUCCT-1, and HCCC-9810). To assess the invasion and migration capabilities of CCA cell lines, QBC 939 and HCCC-9810 were treated with TUG-891 (a GPR120 agonist) and AH7614 (a GPR120 inhibitor). **Results:** Both GPR120 and NGF were found to be upregulated in CCA tissues, correlating with aggressive clinicopathological features. IHC staining revealed cytoplasmic localization of GPR120 and NGF, with significantly higher expression in cancerous tissues compared to adjacent non-cancerous samples. Among the 386 patients, PNI was observed in 339 (87.8%). PNI correlated with tumor diameter, CA 19-9 levels, TNM stage, preoperative bilirubin levels, tumor differentiation, and preoperative drainage. Notably, GPR120 expression increased in poorly differentiated tumors and advanced clinical stages, indicating a clinical association with CCA progression. In vitro studies demonstrated that GPR120 promoted invasion and migration in the QBC 939 and HCCC-9810 cell lines. Western blot analysis confirmed elevated GPR120 levels in these lines, which also exhibited higher NGF expression. The GPR120 inhibitor AH7614 significantly reduced invasion in QBC 939 and HCCC-9810 cells, while the GPR120 agonist TUG-891 showed no effect on any of the CCA cell lines tested. **Conclusions:** GPR120 and NGF are implicated in perineural invasion (PNI) in CCA, with a notable correlation between GPR120 and NGF in the context of PNI occurrence. Research Sponsor: None.

623 **Poster Session**

Evaluation of EpiSwitch in predicting immunotherapy response in hepatocellular carcinoma and gastrointestinal tumors.

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Background: The EpiSwitch Checkpoint Inhibitor Response Test (CiRT) is a blood-based assay that analyzes DNA conformations in immune cells to predict responses to immune checkpoint inhibitors (ICIs) targeting PD-L1/PD-1. CiRT has demonstrated superior predictive accuracy compared to traditional biomarkers, such as tumor mutational burden (TMB) and PD-L1 immunohistochemistry (IHC), for predicting responses in urothelial cancer (ESMO 2022). However, its role in hepatocellular carcinoma (HCC) and other gastrointestinal (GI) tumors remains unclear. As immunotherapy becomes more common in these cancers, identifying reliable predictive biomarkers is essential to optimize treatment outcomes. **Methods:** This retrospective study evaluates CiRT-predicted immunotherapy responses and clinical outcomes in patients with HCC and GI tumors, including cholangiocarcinoma, pancreatic adenocarcinoma, and gastric cancer. All patients received one or more lines of immunotherapy. CiRT responses were categorized into high probability (HP) and low probability (LP). Treatment response was assessed per RECIST 1.1 criteria. Sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), overall survival (OS), and progression-free survival (PFS) were analyzed. **Results:** 43 patients (24 HP, 19 LP) were included. HCC patients (n = 33) received varied immunotherapy regimens, such as combinations of bevacizumab with atezolizumab, tremelimumab with durvalumab, and nivolumab with ipilimumab. Non-HCC patients (n = 10) received combinations of immunotherapy and chemotherapy such as durvalumab with cisplatin and gemcitabine and nivolumab with oxaliplatin and capecitabine. Across all patients, CiRT revealed a sensitivity of 73.91%, a specificity of 65%, a PPV of 70.83%, and an NPV of 68.42%. In HCC, sensitivity was 70.59% and specificity was 68.75%. In non-HCC, sensitivity was 83.33% and specificity 50%. HP responders showed better treatment outcomes compared to LP responders, with higher rates of complete response (CR: 12.5% vs 0%), partial response (PR: 29.2% vs 21.1%), and stable disease (SD: 29.2% vs 10.5%), and a lower rate of progressive disease (PD: 29.2% vs 68.4%) ($p = 0.0467$). Overall, 70.83% of HP achieved CR, PR, or SD vs. 31.58% of LP ($p = 0.0098$). PFS was significantly better in HP ($p = 0.044$, Log-rank test; median PFS = 2.0 months for the LP group, mPFS not reached for the HP group). No statistically significant OS difference was noted due to the lack of events in the HP group. **Conclusions:** CiRT HP status was associated with improved response rates to immunotherapy, longer PFS, and better treatment outcomes in HCC and GI tumors. HP responders experienced notably better outcomes, suggesting that CiRT could serve as a promising predictive biomarker for immunotherapy response, addressing a critical unmet need for reliable biomarkers to treat HCC and other GI tumors. Research Sponsor: None.

Influence of MRP4 levels on clinical outcomes in biliary tract cancer.

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Background: Biliary tract cancer (BTC) are rare tumors with a poor prognosis.Multidrug resistance protein 4 (MRP4) is a transmembrane efflux transporter that influences on cellular growth and differentiation. Recent studies have linked high MRP4 levels to increased aggressiveness in pancreatic cancer and other tumor types. This study aimed to evaluate the role of MRP4 in BTC and its association with clinical outcomes in a real-world cohort from endemic areas in Argentina. **Methods:** We conducted a retrospective analysis of clinical outcomes and MRP4 expression in paraffin-embedded tissue samples from 78 patients with BTC. Immunohistochemistry staining was performed, and MRP4 expression was scored based on staining intensity and the percentage of positive cells. A score exceeding 100 was classified as high MRP4 levels. Statistical analyses included Chi-square, Kaplan-Meier survival curves, and log-rank tests, with significance set as $p<0.05$ (two-tailed). **Results:** High MRP4 levels were observed in 46 out of the 78 samples (59%). Patients with gallbladder cancer (n=64) and elevated MRP4 levels had significantly poorer overall survival (OS) compared to those with low MRP4 levels (median 11.3 and 14.5 months, high and low, $p=0.038$). In the broader BTC cohort evaluable for survival (n=73; 64 gallbladder, 7 cholangiocarcinoma, 1 ampulla of Vater, and 1 biliary undefined) patients with elevated MRP4 levels had non-significant shorter OS (median 12.4 and 13.5 months, high and low, $p=0.17$). Poorly differentiated tumors were more likely to express high MRP4 levels ($p=0.06$). High MRP4 expression was found in 59% of localized stages (0-II) and in 57% of advanced stages (III, IV and relapsed). **Conclusions:** High MRP4 expression is common in BTC, particularly gallbladder cancer, and could be a marker of poor prognosis. MRP4 levels appear more elevated in poorly differentiated tumors, and are similar in both, early and advanced stages, suggesting its involvement in early tumor progression. Further prospective studies are needed to confirm MRP4 as a biomarker. If MRP4 is a driver gene, targeting with inhibitors could represent a promising therapeutic strategy for these challenging tumors.
Research Sponsor: None.

625 Poster Session

Analysis of clinically actionable alterations in baseline tumor versus plasma samples in participants of the TOPAZ-1 study of durvalumab plus gemcitabine and cisplatin in advanced biliary tract cancer.

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Background: In the TOPAZ-1 study (NCT03875235), durvalumab + gemcitabine and cisplatin (D+GC) significantly improved overall survival (OS) versus placebo + GC (P+GC) in participants (pts) with advanced biliary tract cancer. Updated results have shown a clinically meaningful long-term OS benefit for D+GC versus P+GC at 3 years. This exploratory analysis tested circulating tumor DNA (ctDNA) in plasma samples for blood-based detection of clinically actionable alterations (CAAs) and investigated the potential of this method to guide treatment decisions. **Methods:** Baseline genomic alterations were retrospectively assessed in evaluable tumor (FMI biomarker evaluable population [BEP], n=441) and plasma samples (GH BEP, n=643) using FoundationOne (Foundation Medicine Inc., Cambridge, MA) and Guardant INFINITY (Guardant Health, Redwood City, CA) assays, respectively. Mutation prevalence and association with outcomes were compared in the FMI BEP and GH BEP. Positive and negative percent agreement of CAAs detected in tumor versus ctDNA were assessed in 419 pts with both tumor and plasma samples (FMI-GH BEP). **Results:** The FMI BEP and GH BEP represented 64% and 94% of the TOPAZ-1 final analysis set (685 pts), respectively. The relative prevalence and overall mutational landscape detected in plasma ctDNA was consistent with that observed by tumor profiling, with the notable exception that genes harboring complex alterations (e.g. gene amplification, rearrangements, or homozygous deletions) were less frequently detected in ctDNA (e.g. *ERBB2*, *FGFR2*, and *CKDN2A/2B/MTAP*). The most common alterations (>15% prevalence in both BEPs) observed were in *TP53* (49%/52%), *KRAS* (24%/17%), and *ARID1A* (21%/16%) (in the FMI/GH BEPs, respectively). The relative prevalence of alterations within geographic and anatomic subgroups was similar in tumor versus plasma for most CAAs. The overall percent agreement in CAAs was ≥93%, and negative percent agreement was ≥97%. However, positive percent agreement was notably low for *ERBB2* amplification (52%) and *FGFR2* fusions (47%). OS hazard ratios for D+GC versus P+GC in the GH BEP were <1 for both CAAs and wild-type, except for *ERBB2* amplification, as previously reported in the FMI BEP. **Conclusions:** The overall concordance and relative prevalence of simple mutations (e.g. single nucleotide variants) were similar using the FoundationOne tumor assay and Guardant INFINITY ctDNA assay, suggesting that plasma ctDNA testing has potential utility in clinical practice. However, negative status by ctDNA for the complex alterations found in *FGFR2* and *ERBB2* would require further testing of tumors, based on their low detectability in plasma. Clinical trial information: NCT03875235. Research Sponsor: AstraZeneca.

Clinical landscape of cell free DNA alterations in patients with advanced biliary tract cancer treated with targeted therapies.

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Background: Although 40% of patients (pts) with advanced biliary tract cancer (BTC) harbor actionable alterations amenable to targeted therapy (TT), tumor genotyping can be challenging due to limited tissue and other factors. Next generation sequencing (NGS) of cell free DNA (cfDNA) may aid in prognostication, identification of TT opportunities, and characterization of acquired resistance mechanisms to TT. **Methods:** cfDNA was collected prospectively in pts with BTC treated at MSK from 2016-2023. Samples were analyzed using a CLIA-approved, cfDNA targeted NGS assay (MSK-ACCESS). When available, matched tumor samples were analyzed using an FDA-authorized targeted NGS assay (MSK-IMPACT). Objectives included evaluating association of cfDNA genomic alteration variant allele frequency (VAF) with outcomes and description of genomic features identified on cfDNA at resistance to TT. **Results:** N=170 BTC pts (intrahepatic cholangiocarcinoma [CCA], n=121; extrahepatic CCA, n=29, gallbladder cancer, n=20) underwent cfDNA genotyping, comprising 270 samples overall. A majority of pts (146/170, 86%) had locally advanced or metastatic disease. OncoKB level 1/2 alterations were identified in 19% of pts overall. Pts with treatment-naïve locally advanced or metastatic BTC whose baseline detected cfDNA genomic alterations were VAFhigh (defined as > median VAFmax across all samples, n=60) had significantly worse median PFS (4.5 (95% CI 2.2-6.5) vs. 11.8 (95% CI 5.4-18.4) months, (HR 2.7, p=0.002) and OS (14.1 (95% CI 5.4-19.9) vs. 32.0 (95% CI 13.8-39.9) months, HR 3.1, p=0.002)) to first line treatment. Serial cfDNA was available from 28 pts who received TT. Emergent RAS isoform alterations were identified at resistance in 7/25 (28%) pts who received BRAF-, FGFR-, or HER2-directed TT. These included 3/5 pts who received BRAF/MEK TT for BRAF V600E+ BTC (emergent KRAS G12D in 1, NRAS Q61K in 1, polyclonal KRAS G12D, G12V, G13D in 1), 2/13 pts who received FGFR TT for FGFR2 fusion+ BTC (KRAS Q61H in 1, NRAS Q61R in 1), and 2/7 pts who received HER2 TT for ERBB2-amplified BTC (KRAS G12C in 1, KRAS amp in 1). Serial cfDNA also revealed canonical FGFR2 resistance mutations in 3 pts who received FGFR TT, as well as loss of index ERBB2 amp in 2, acquired ERBB2 resistance mutation in 1, acquired MYC amp in 2, and MET amp in 1 pt who received HER2 TT. **Conclusions:** cfDNA NGS in pts with BTC may assist with prognostication in advanced disease and can identify novel patterns of resistance, including frequent emergence of RAS alterations at resistance across multiple TTs. These may identify therapeutic opportunities for KRAS- and NRAS-directed therapeutics in development, and underscore the importance of serial cfDNA profiling. Research Sponsor: None.

Positive homologous recombination signature (HRDsig+) and the genomic landscape of intrahepatic cholangiocarcinoma (iCCA).

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Background: Defective DNA repair has not been a major focus in iCCA clinical research. In this comprehensive genomic profiling (CGP) study, we queried whether the presence of a scar-based HRDsig biomarker could identify a subset of patients with unique genomic alterations (GA) and potential for responsiveness to PARP inhibitor-based treatment regimens. **Methods:** 6,271 cases of clinically advanced iCCA underwent hybrid capture based CGP to evaluate all classes of GA. Microsatellite instability (MSI) status, tumor mutation burden (TMB) levels, genomic ancestry and genomic trinucleotide signatures were determined from the sequencing data. HRDsig status was calculated using a broad set of genome-wide copy number features (PMID 37769224). PD-L1 was determined by IHC using the Dako 22C3 tumor proportional score (TPS). Results were compared using the Fisher exact system with the Benjamini-Hochberg procedure. **Results:** 288 (4.6%) of the sequenced iCCA featured an HRDsig+ status. Gender distribution (46-49% male) and median age (67 years) were similar in both HRDsig+ and HRDsig- iCCA. More pathogenic GA per tumor were found in HRDsig+ vs HRDsig- iCCA (5 vs 4; P<0.0001). Genomic ancestry distribution was similar; European ancestry ranged from 73% to 76% in the two cohorts. The APOBEC signature was uncommon but slightly more frequent in the HRDSig+ cases (1.7% vs 0.4%; P=.034). MSI High status was slightly more frequent in the HRDsig- group (1.8% vs 0.0%; P=0.034). Median TMB was higher in the HRDsig+ cases (3.6 vs 1.2 mut/Mb; P<0.0001) as was the frequency of TMB > 10 mut/Mb (9.1% vs 3.5%; P<0.0001). PD-L1 expression was similar with low level (1-49% TPS) ranging from 19% to 23% in the 2 groups. As anticipated, GA in genes associated with HRD including *BRCA1* (6.6% vs 0.8%; P<0.0001), *BRCA2* (25.0% vs 1.3%; P<0.0001), *PALB2* (8.7% vs 0.3%; P<0.0001), and *ATM* (6.3% vs 3.4%; P=0.033) were all more frequent in the HRDsig+ iCCA. In contrast, GA in genes associated with iCCA targeted therapies were less frequent in the HRDsig+ cases including *FGFR2* (8.0% vs 12.5%; P=0.033), *IDH1* (3.5% vs 14.0%; P<0.0001) and *ERBB2* (3.5% vs 5.7%; NS). In the HRDsig- group, 83.3%/74.1% of *BRCA1*/*BRCA2* mutated iCCA were mono-allelic likely non-driver GA respectively. MTOR pathway activating GA were more frequent in the HRDsig+ cases including *PTEN* (7.3% vs 3.3%; P=0.003) and *NF1* (8.0% vs 2.8%; P<0.0001). *MTAP* deletion, an emerging iCCA target, was more frequent in the HRDsig+ iCCA (23.3% vs 16.9%; P=0.024). **Conclusions:** Nearly 5% of advanced iCCA feature HRDsig positive status which is associated with both significant differences in the frequencies of therapy associated genomic targets and the potential for introducing PARP inhibitors for these patients. Research Sponsor: None.

	HRDsig- iCCA N = 5983	HRDsig+ iCCA N = 288	P-value
ATM	3.4%	6.3%	0.033
BRCA1	0.8% *(83.3% mono-allelic)*	6.6%	<.0001
BRCA2	1.3% *(74.1% mono-allelic)*	25.0%	<.0001
PALB2	0.3%	8.7%	<.0001

A global comparative study on real-world clinical and genomic differences in advanced biliary tract cancer.

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Background: Biliary tract cancers (BTC) are aggressive malignancies with poor survival outcomes and limited treatment options. This study compared the epidemiological and molecular differences of advanced BTC patients through a global collaboration involving the Mayo Clinic and National Cancer Center Hospital East. **Methods:** We included patients with advanced BTC who underwent comprehensive tumor molecular profiling who received at least 30 days of systemic therapy. Patients were grouped based on specific alterations. Overall survival (OS) was calculated from the start of first-line therapy until death and compared among subgroups using Cox regression models. **Results:** 332 (36%) out of 932 patients had at least one actionable mutation*: somatic BRCA1/2* mutations N=106, *FGFR2* fusions N=61, *HER2* amplifications N=65, *KRAS G12C/D* mutations N=46, *IDH1* mutations N=38, and TMB >10 mut/Mb N=16. *BRCA1/2*-mutated BTC were more commonly found in older Asian males with intrahepatic tumors and often presented with peritoneal and liver metastases upon initial diagnosis. *FGFR2* fusions were more prevalent in younger White females with intrahepatic tumors, while *HER2* amplifications were mainly seen in Asian females with gallbladder tumors. *KRAS* mutations were primarily observed in Asian males, and *IDH1* mutations were common in White males with intrahepatic tumors, frequently presenting with liver metastasis. Survival outcomes varied significantly based on the alteration type: patients with *FGFR2* fusions had the longest median OS (mOS) at 23.3 months (mo) (95%CI 18.4-34.9), followed by those with *IDH1* mutations (19.3 mo; 95% CI 17.4-NE) and TMB >10 mut/Mb (17.8 mo; 95% CI 13.6-NE). *BRCA* mutations had a mOS of 16 mo (95% CI 14.6-19.8), while *HER2* amplifications and *KRAS* mutations had mOS of 15.5 (95% CI 11.9-20.2) and 11.6 mo (95% CI 10.0-20.9), respectively. **Conclusions:** These findings underscore the heterogeneity in clinical and genomic characteristics among BTC patients, highlighting the importance of tailored therapeutic approaches. Research Sponsor: None.

Baseline patient characteristics stratified by mutation status.

	KRAS G12C/D (N=46)	*BRCA 1/2* (N=106)	*IDH1* (N=38)	*HER2* amplification (N=65)	*FGFR2* fusion (N=61)	TMB>10 mut/Mb (N=16)	P-value
Median Age, year	67	68	64	66	52	73	<0.001
Female, n (%)	19 (41)	34 (32)	18(47)	34 (52)	45(74)	6 (38)	<0.001
Race, n (%)							<0.001
Asian	31 (67)	97 (92)	13 (34)	51 (79)	10 (16)	6 (38)	
White	14 (30)	8 (8)	24 (63)	13 (20)	48 (79)	10 (63)	
Primary tumor location, n (%)							<0.001
Intrahepatic	28 (68)	45 (45)	33 (100)	19 (31)	54(95)	10 (67)	
Extrahepatic	9 (22)	27(27)	0 (0)	10 (16)	2 (4)	4 (27)	
Gallbladder	4 (10)	28 (28)	0 (0)	32 (53)	1 (2)	1 (7)	
Sites of Metastasis at diagnosis, n (%)							
Abdominal Wall / Peritoneal	9 (24)	15 (19)	4(13)	7 (12)	6 (10)	4 (27)	0.309
Bone	33 (92)	70 (90)	26 (87)	2 (3)	9(15)	0 (0)	0.212
Liver	20 (49)	52 (61)	28 (82)	45 (73)	45 (76)	6 (38)	0.001
Lung	26 (70)	65 (82)	22 (69)	36 (58)	27(45)	11 (69)	0.006

Impact of HER2-positivity on prognosis and targeted therapeutic outcomes in advanced biliary tract cancer.

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Background: HER2 overexpression and amplification have been identified as a promising druggable target in biliary tract cancer (BTC). This study aims to characterize the clinical and molecular outcomes associated with HER2 positivity in BTC and evaluate the prognostic implications of HER2-targeted therapies. **Methods:** A retrospective study was conducted involving patients with advanced BTC diagnosed at Yonsei Cancer Center (YCC) and the University of Texas MD Anderson Cancer Center (MDACC) between 2009 and 2023. Patients were classified as HER2-positive either by immunohistochemistry (IHC 3+ or 2+/ISH+) or by next-generation sequencing (NGS, *ERBB2* amplification). Overall survival (OS) and palliative first-line progression-free survival (PFS) were analyzed using Kaplan-Meier and Cox regression methods. **Results:** A total of 389 advanced BTC patients were initially screened (310 from YCC; 79 from MDACC). Among the 310 BTC patients from YCC with available HER2 IHC results, 78 (25.2%) were HER2-positive and 232 (74.8%) were HER2-negative. HER2 positivity was significantly more common in gallbladder cancer (55.1%) compared to its prevalence in intrahepatic or extrahepatic cholangiocarcinoma (22.8%, $p<0.001$). Of the 201 patients with available NGS data, 186 (92.5%) were *ERBB2* non-amplified; among these, 27 (14.9%) were HER2-positive by IHC. For survival analysis, 310 patients were eligible: 239 from YCC (both HER2-positive and negative) and 71 from MDACC (all HER2-positive). Kaplan-Meier survival analysis indicated a trend towards shorter OS in HER2-positive patients compared to HER2-negative patients (median OS, 13.7 vs. 17.1 months, p=0.084, HR=1.246, 95% CI=0.97–1.60) regardless of exposure to HER2-targeted therapy, while first-line PFS on chemotherapy only was significantly shorter in the HER2-positive group (5.1 vs. 7.4 months, $p<0.001$, HR=1.906, 95% CI=1.46-2.48). Among HER2-positive patients, 56.8% received HER2-targeted therapy in the second line setting. HER2-positive patients who did not receive HER2-targeted therapy had a significantly poorer prognosis compared to HER2-negative patients (median OS 8.1 vs. 17.1 months, HR=2.46, 95% CI=1.73-3.48), whereas those receiving HER2-targeted therapy had comparable OS to HER2-negative patients (median OS 18.2 vs. 17.1 months, HR=0.95, 95% CI=0.71-1.27). Multivariable analysis identified a history of surgical treatment, radiation therapy, HER2 positivity, and HER2-targeted therapy as independent prognostic factors for OS. **Conclusions:** HER2 positivity is associated with a poorer prognosis in advanced BTC. However, HER2-targeted therapies improve outcomes, suggesting their potential inclusion in standard care for HER2-positive subgroups. Discrepancies between IHC and NGS in HER2 assessment emphasize the need for optimized testing guidelines in BTC. Research Sponsor: None.

Extra-hepatic (ehCCA) and intra-hepatic cholangiocarcinomas (iCCA): A genomic landscape comparison of peri-hilar (phCCA), common bile duct (cbdCCA) and gallbladder (gbCCA) with iCCA.

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Background: Metastatic CCAs have poor prognosis. Comprehensive genomic profiling (CGP) was used to compare genomic alterations (GA) in phCCA, with cbdCCA, gbCCA and iCCA. Frequencies of GA in therapy targets such as FGFR2, IDH, ERBB2 & BRAF, & Immuno-oncology (IO) drug response predictors (MSI status, TMB & PDL1) were examined. **Methods:** DNA extracted from 110 cases of phCCA, 743 cases of cbdCCA, 2983 cases of gbCCA and 9178 cases of ihCCA underwent hybrid capture based CGP to evaluate classes of genomic alterations (GA), MSI status, TMB levels, genomic ancestry, genomic signature, HRD score and germline status. phCCA cases were confirmed at surgery and pathology as originating from either left, right or common hepatic ducts. cbdCCA cases were confirmed by cbd biopsies or Whipple procedures. PD-L1 expression was determined by IHC using the Dako 22C3 assay using tumor proportional score (TPS) system. **Results:** The pt age distributions were similar; female gender was more frequent in gbCCA. gbCCA featured higher GA/tumor (P=.0001). The frequencies of FGFR2 & IDH1 targetable GA in phCCA, cbdCCA and gbCCA were uncommon especially when compared iCCA (P<.0001). GA in BRAF, CDKN2A, MTAP and PIK3CA in gbCCA were like phCCA and cbdCCA. At 16.5%, the frequency of GA in ERBB2 were highest in gbCCA and in all ehCCA when compared with iCCA (P<.0001). The frequencies of KRAS GA including KRAS G12C were greater in both phCCA and cbdCCA when compared with gbCCA (P=.0015) and iCCA (P.0001). TP53 GA were highest in gbCCA (P=.0007). IO biomarker frequencies were similar in all. The frequencies of HRD positive (gLOH) status and MMR genomic signature were similar in all. **Conclusions:** ehCCA subtypes feature characteristic GA profiles that have potential to influence therapy selection with lower frequencies of FGFR2 & IDH1 when compared with iCCA. In contrast, targetable ERBB2 & KRAS G12C are more frequent in ehCCA than iCCA, and highest in gbCCA & cbdCCA. Research Sponsor: None.

	peri-hilarCCA	cbdCCA	P-value (phCCA vs cbdCCA)	gbCCA	P-value (phCCA vs gbCCA)	ihCCA	P-value (phCCA vs iCCA)
BRAF	3.6%	4.7%	NS	2.4%	NS	5.1%	NS
ERBB2	14.5%	10.6%	NS	16.5%	NS	5.1%	0.002
FGFR2	0.9%	2.0%	NS	1.5%	NS	11.7%	0.001
IDH1	3.6%	0.8%	NS	0.6%	NS	13.9%	0.006
KRAS	29.1%	46.2%	0.015	14.2%	0.0015	19.8%	0.071
TP53	45.5%	63.8%	0.011	65.7%	0.0007	33.9%	0.061
MSI-high	1.8%	1.3%	NS	1.2%	NS	1.8%	NS
TMB≥10 mut/mb	3.6%	4.2%	NS	5.2%	NS	3.6%	NS
PD-L1 low positive	21.1%	31.3%	NS	23.5%	NS	23.7%	NS

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Exclusive liver invasion as prognostic indicator in T3 gallbladder cancer.

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Background: T3 gallbladder cancer (GBC) contains a diverse range of invasive tumors. However, distinct prognostic outcomes may be observed among T3GBC confined to localized range despite sharing the same T3GBC. There is uncertainty regarding whether there are noticeable prognostic variances between tumors that invade the liver, or serosa only and those that extend to adjacent organs. This study aims to investigate prognostic differences according to the extent of invasion in patient with T3GBC. **Methods:** The patients with T3GBC who underwent curative intent surgery from February 2002 to December 2021 were analyzed. The patients were divided into those who had tumor invaded liver only, serosa only, or extended beyond. Their clinico-pathologic characteristics, perioperative details and prognostic data were compared. Survival analyses were carried out using the log-rand test including subgroups considering nodal metastasis, and other T stages. Cox-proportional hazard model including other factors were used to identify the independent risk factors for survival. **Results:** We identified 194 patients with T3GBC. The patients with tumor invaded liver only, serosa only, and adjacent organs were 86, 33, and 35, respectively. The median survival was 23 months. The patients with invasion limited to the liver showed better survival than those who involve adjacent structures (26 months vs 9 months, $p < 0.001$). In multivariate analysis, it was demonstrated that the concomitant presence of invasion in structures other than the liver serves as independent prognostic factors for survival. (hazard ratio [HR], 6.8, 95% confidence interval [CI] 1.746 – 26.803, p = 0.006) as well as lymphovascular invasion (HR 2.5, 95% CI 1.093 – 4.193, p= 0.026). **Conclusions:** In the T3 gallbladder cancer, exclusive liver invasion is a better prognostic indicator compared to concurrent invasion around adjacent structure. It is crucial to recognize that the extent of invasion in T3GBC patients before surgery. Multicenter studies are needed for the definite results. Research Sponsor: None.

Exploring Nectin-4 expression in hepatocellular carcinoma.

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Background: Nectin cell adhesion protein 4 (Nectin-4) is overexpressed in various malignancies, contributing to cancer progression and poor prognosis. We aimed to assess Nectin-4 expression in hepatocellular carcinoma (HCC). **Methods:** Patients who underwent primary liver tumor resection for HCC upon initial diagnosis at National Taiwan University Hospital in 2019 were included. Exclusion criteria consisted of prior systemic or loco-regional therapy, or death within three months post-surgery. Immunohistochemical (IHC) staining for Nectin-4 (EPR15613-68, Abcam) was conducted and independently reviewed by two pathologists. Nectin-4 expression patterns (membranous, cytoplasmic, and nuclear) and intensity were scored on a 0–3 scale, with normal eccrine glands (2+) serving as a positive control. Nectin-4 expression was further assessed in primary and paired progressive or metastatic tissue samples from patients with available paired samples. **Results:** Of the 193 patients, 175 had specimens available for IHC analysis. Nectin-4 expression was detected in 22.9% of these patients, with intensity scores of 0 in 77.1%, 1 in 22.3%, and 2 in 0.6%. The majority of staining pattern was cytoplasmic (94.3%), with membranous staining observed in 5.1% of cases. Nectin-4 expression was significantly associated with higher pathological grades (p = 0.005) but showed no correlation with recurrence-free survival (p = 0.771) or overall survival (p = 0.396) after surgery. Importantly, Nectin-4 expression intensity significantly increased in paired progressive or metastatic samples compared to initial HCC specimens (p = 0.046; n = 27). **Conclusions:** Nectin-4 expression is associated with higher pathological grades but not related to clinical outcomes in HCC. Increased Nectin-4 expression in progressive or metastatic disease highlights its potential as a therapeutic target. Research Sponsor: National Science and Technology Council.

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Use of pathological response versus RECIST assessment in predicting relapse-free survival following neoadjuvant immunotherapy for hepatocellular carcinoma.

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Background: Immune checkpoint inhibitors (ICI) administered prior to liver resection (LR) lead to pathological responses in patients with hepatocellular carcinoma (HCC). However, the relative value of pathological versus radiological response in predicting relapse-free survival (RFS) remains unclear. **Methods:** We pooled patient-level data from 111 patients (pts) with HCC receiving ICI prior to LR as part of 5 phase I/II trials and observational clinical studies conducted in 12 centres in the United States, United Kingdom and Asia, as part of an academic consortium (NeoHCC). Pathological response was measured as the percentage of non-viable tumour in the resected specimen, with major (pMR) and complete pathological response (pCR) corresponding to ≥70% and 100% tumour regression. Radiological overall response rate (ORR) was assessed by RECIST v1.1 and modified RECIST (mRECIST) criteria. We correlated pathological response and ORR with RFS using Cox regression. **Results:** Pts received preoperative ICI between Oct 5, 2017, and Nov 15, 2023, mostly ICI combinations (69%, n=76). Most pts were male (78%, n=87) with viral chronic liver disease (66%, n=73), BCLC stage A HCC (55%, n=61). ORR was 28% per RECIST v1.1 and 32% per mRECIST criteria (available for n=81). Out of the 104 pathologically evaluable pts, pMR and pCR rates were 32% (n=33) and 18% (n=19). Radiological response by RECIST v1.1 showed a significant correlation with pathological response (R^2 = 0.43, p<0.001). When using RECIST v1.1, 74% of pts with ORR achieved pMR vs 14% of those without ORR (n=23/31 vs 10/73, p<0.001). However, 30% of pMR were not predicted by ORR (n=10/33). The discrepancy between pMR and ORR decreased using mRECIST. ORR per mRECIST was 83% in pMR pts (n=19/23), whereas it was 10% in non-pMR (n=5/51). After a median follow-up of 27.2 mos (95%CI 22.3-32.1), median RFS for the whole cohort was 43.6 mos (95%CI 28.3-NE). Achievement of pMR, pCR and ORR was associated with improved RFS (Table). However, reduction in the risk of relapse and/or death was higher in pts achieving pMR or pCR than ORR, regardless of the use of RECIST v1.1 or mRECIST. Conversely, pts without pMR had a lower mRFS than pts without ORR (28.3 mos [95%CI 12.8-43.8] and 32.8 mos [95%CI 13.7-51.9], respectively). **Conclusions:** Whilst radiological responses to neoadjuvant ICI are associated with improved RFS in pts with HCC, achievement of pMR is more accurate than RECIST and mRECIST-based responses in predicting for RFS, with lack of pMR identifying pts at a higher risk of relapse or mortality. Research Sponsor: None.

Cox regression for RFS	HR (95% CI)	p value
pCR	0.19 (0.05-0.78)	0.02
pMR	0.25 (0.10-0.66)	0.005
RECIST 1.1	0.34 (0.13-0.86)	0.02
mRECIST	0.38 (0.13-1.11)	0.08

RFS, relapse-free survival; HR, hazard ratio; CI, confidence interval; pCR, complete pathological response; pMR, major pathological response; mRECIST, modified RECIST.

Genomic alterations as predictive biomarkers for cancer-associated thrombosis (CAT) in biliary tract cancer (BTC).

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Background: CAT is the second leading cause of cancer-related death, yet the genetic underpinnings of VTE risk in BTC remain underexplored. The evaluation of CAT risk is based on Khorana score (KS), which is limited to clinical parameters. Incorporation of a genomic signature in CAT risk assessment may allow for individualized risk assessment. **Methods:** We collected data from consecutive patients (pts) with BTC treated at Mayo Clinic Hospitals (Phoenix, Rochester, Jacksonville) between 1998-2024. Genomic profiling was conducted on FFPEs and blood samples through next-generation sequencing (NGS) (NextSeq, Tempus xT CDx, FoundationOne CDx). Association between CAT and clinic-pathologic features (age, gender, race, ethnicity, BMI, tumor site (TS), surgery, stage, radiotherapy, systemic therapy, history of thromboembolism (hVTE)) were assessed with multivariate Cox regression ($p<0.05$). A separate Cox regression adjusting for age, hVTE, radiotherapy, systemic therapy, metastatic disease stratified for TS was used for gene associations. Benjamini–Hochberg method was applied ($q<0.1$) based on previous publications. Median overall survival (mOS) was calculated from the date of diagnosis to the date of death or last follow-up. **Results:** 740 adult pts were included, median age 62 years (IQR: 56-71), male/female (51%/49%), median BMI 26.6 (IQR: 22.8-31.1) with intrahepatic (496, 67%) and extrahepatic BTC (173, 23%), gallbladder cancer (71, 10%). 244 (33%) had metastatic disease. 216 (29%) experienced CAT. For those with CAT, 31 (14%) scored 1, 9 (4%) scored 2 and 1 (<1%) scored 3 on KS. mOS was 33 months (mos) (95% CI 28.9-38.4) vs. 55 mos (95% CI 25.1-39.1) ($p=0.2$) for pts with and without CAT, respectively. hVTE was associated with CAT (HR 3.14, 95% CI 1.98-4.98, $p<0.001$) along with systemic therapy (HR 1.19, 95% CI 1.11-2.89, $p<0.01$) and metastatic disease (HR 1.65, 95% CI 1.11-2.44, $p<0.01$). 305 pts (41%) underwent NGS. The most common altered genes were TP53 (126, 41%), KRAS (64, 21%), CDKN2A (63, 21%). The most common altered currently targetable genes were FGFR2 (43, 14%), IDH1 (30, 10%) and HER2/ERBB2 (16, 5%). MYC (4, 1%) (HR 11.60, 95% CI 3.04-44.25, $p<0.001$, $q=0.058$) and HER2/ERBB2 (HR 2.45, 95% CI 1.26-4.77, $p=0.008$, $q=0.087$) alterations were significantly associated with an increased risk of CAT. MYC alterations were all copy number gains (100%). The most frequent HER2/ERBB2 alterations were copy number gains (10, 62%). FGFR2 fusions (31, 10%) were linked to a lower CAT risk (HR 0.46, 95%CI 0.22-0.93, $p=0.032$, $q=0.075$) with FGFR2-BICC1 being the most frequent (5,16%). **Conclusions:** MYC and HER2/ERBB2 alterations are novel predictive biomarkers for an increased risk of CAT in BTC, while FGFR2 fusions confer lower risk. The knowledge of mutational status lends insight into guidance for anticoagulant prophylaxis management for this subgroup and integration with KS. Research Sponsor: None.

Expanded biomarker analysis of an international, open-label phase 2 study of regorafenib plus pembrolizumab in patients with advanced hepatocellular carcinoma (HCC) previously treated with immune checkpoint inhibitors (ICI).

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Background: The optimal treatment for patients (pts) with advanced HCC who progress on an ICI-containing regimen has not been defined. In pts who progressed on one prior ICI regimen, regorafenib plus pembrolizumab (rego + pembro) showed modest anti-tumor activity (primary completion analysis: overall objective response rate 7.4%, all partial responses [PR]) and an expected safety profile (NCT04696055; El-Khoueiry, ASCO 2024). We conducted additional exploratory analyses of tumor markers to gain insights into the clinical results. **Methods:** Overall, 95 adults with unresectable advanced HCC (C-P Class A, BCLC stage B/C, ECOG PS 0/1) received oral regorafenib 90 mg once daily 3 weeks on/1 week off plus pembrolizumab 400 mg i.v. every 6 weeks. Prior treatment: Cohort 1 = atezolizumab + bevacizumab (n=68); Cohort 2 = any other ICI alone or in combination (n=27). Biomarker analyses in subsets of pts explored potential correlations with clinical benefit (defined as tumor response or stable disease [SD] ≥ 4 months). Immune cell subtypes in blood were analyzed by flow cytometry. Tumor immune markers were assessed by immunohistochemistry. Gene expression profiling in tumor samples was by RNAseq. **Results:** Twenty-one pts had baseline samples for RNAseq, 7 of whom had clinical benefit (n= 3 Cohort 1 [including 1 PR], n=4 Cohort 2). Pts with clinical benefit had differential gene expression at baseline including enriched expression of genes associated with Notch, Hedgehog, fibroblast, TGF-β, and insulin-like growth factor binding protein signaling; in contrast, there was higher expression of genes associated with myeloid derived suppressor cells in pts without clinical benefit. Of note, tumor samples from pts in this study had increased Tregs and T-cell exhaustion signatures compared with pts treated with rego + pembro in a first-line study (NCT03347292).Analyses of available paired tumor samples from 4 patients (n=2 SD, n=2 progressive disease) found reductions in activated fibroblast, TGF-β, and collagen gene expression signatures from baseline to Week 6 Day 1 of treatment. **Conclusions:** Rego + pembro had modest activity in pts who progressed on one prior ICI-based regimen. Exploratory biomarker analyses suggest different gene expression in pts with or without clinical benefit. Clinical benefit was associated with enriched expression of genes in Notch, Hedgehog, fibroblast, and TGF-β pathways. Pre- and post-treatment biopsies showed reductions in fibroblast, TGF-β, and collagen gene expression, which are known features of HCC. In the context of the limitations of small sample size and absence of a control arm, these findings are hypothesis generating and consistent with HCC biology and regorafenib mechanism of action. Clinical trial information: NCT04696055. Research Sponsor: Bayer Healthcare Pharmaceuticals.

Radiomics-based prediction of HCC response to atezolizumab/bevacizumab.

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Background: Advanced hepatocellular carcinoma (HCC) treatment has evolved with the introduction of atezolizumab/bevacizumab, showing improved outcomes over sorafenib. However, the response varies among patients, particularly between viral and non-viral etiologies. This study aimed to develop and evaluate multimodal prediction models combining quantitative imaging and clinical markers to predict treatment response in HCC patients. **Methods:** From March 2020 to May 2023, patients with advanced HCC treated with atezolizumab/bevacizumab were retrospectively identified from six centers in Germany and Austria. Patients underwent baseline contrast-enhanced liver MRI and follow-up imaging to assess therapy response. Machine learning models, including RandomForestClassifier, were developed for radiomics, clinical, and combined datasets. Hyperparameter tuning was performed using RandomizedSearchCV, followed by cross-validation to evaluate model performance. **Results:** The study included 103 patients, with 70 achieving disease control (DC) and 33 experiencing disease progression (PD). Key findings included significant differences in treatment response and progression-free survival between DC and PD groups. The radiomics model, using 14 selected features, achieved 73.1% accuracy and a ROC AUC of 0.635 on the test set. The clinical model, with 4 selected features, achieved 73% accuracy and a ROC AUC of 0.649 on the test set. The combined model showed improved performance with 69% accuracy and a ROC AUC of 0.753 on the test set. Hyperparameter tuning further enhanced the combined model's accuracy to 80.1% and ROC AUC to 0.771 on the test set. **Conclusions:** The hybrid model combining clinical and radiological data outperformed individual models, providing better predictions of response to atezolizumab/bevacizumab in HCC patients. Research Sponsor: None.

Expression of Trop2, Nectin-4, and FOLR1 cell surface markers in biliary tract cancers: Is it time to repurpose drugs?

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Background: Biliary tract cancers (BTC), including gall bladder cancer, intrahepatic and extrahepatic cholangiocarcinoma are aggressive malignancies, with poor prognosis. Despite advances in frontline setting, and in biomarker selected population (IDH1, FGFR2, Her2, BRAF, HRD), there is a dire need to advance therapy in subsequent lines. **Methods:** We identified BTC patients (pts) at Roswell Park Cancer Center who underwent curative intent surgery between 01/2006-01/2020. After excluding duplicate samples and those with insufficient tissue, we found 78 pts with adequate follow up for analysis. Expression of trophoblast cell surface antigen 2 (Trop2), nectin4, folate receptor alpha (FOLR1) was assessed by immunohistochemistry and correlated with clinical outcomes. H score, a weighted index score that factor in, the % of cells across different intensity was calculated. H scores were classified as negative (0) and positive (>1). Positive H scores were grouped as weak (1-100), intermediate (101-200) and strong (201-300). Descriptive statistics were conducted, and multivariate analyses performed (R studio) to study any correlation between expression of markers and survival outcomes. **Results:** The median age of our cohort was 67.5 years (range 40-87) with 58% male; 24.4%, 30.8%, 20.5%, 7.7% had stages I, II, III, IV disease. Staging information was missing in 16.7% of pts. Of 73 analyzable samples for Trop2, all were positive for Trop2, with a mean H score of 86, SD of 68.9. 56.4% of patients had a weak H score, 30.8% had an intermediate H score and 6.4% had a strong H score for Trop2. Median overall survival (mOS) among low, intermediate and high expressing group of pts with Trop2 was 48.9 months (m), 23.9 m, and 41.9 m with median Progression Free Survival (mPFS) of 37.8m, 15.3 m, and 37.4 m, respectively. Of 75 analyzable samples for FOLR1 and Nectin4, FOLR1 was expressed in 51.3% with a mean H score of 6.97 (SD 21.9). Nectin4 was expressed in 78.2%, with a mean H score of 13.7 (SD 26.3). Pts with Trop2 expression had a mOS of 39.5 m and a mPFS of 28.7 m. There were no significant differences in mOS (48.9 m vs 33.3 m, p=0.14) and mPFS (37.8 m vs 26.2 m, p=0.35) among patients with positive vs negative FOLR1 respectively. There was no difference in mOS (41.6 m vs 38.3 m, p=0.92) and mPFS (32.8 vs 26.4 m, p-value=0.5) in patients with positive vs negative Nectin4. Multivariate analyses showed that pts with intermediate Trop2 (H score=101-200) had a high risk of death (HR 21.44, CI 2.81-163.4, p= 0.003). **Conclusions:** Though no difference was seen in clinical outcomes in pts with strong Trop2 expression, FOLR1, and Nectin4 expression, this could be explained by the small sample size. Trop2 is frequently expressed in BTCs and patients with Intermediate Trop2 H score had lower mOS and mPFS, with a higher risk of mortality. This warrants further investigation into potential use of Antibody Drug conjugates against Trop2 in BTC. Research Sponsor: Roswell Park Start-Up Fund.

Tumor vascularity as a predictor of FGFR inhibitor response in FGFR2-fused intrahepatic cholangiocarcinoma.

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Background: Fibroblast growth factor receptor 2 (FGFR2) fusion is a well-established mutation in intrahepatic cholangiocarcinoma (iCCA), accounting for approximately 10% of cases. Current treatments, including pemigatinib, infigratinib, and futibatinib, have shown promising results with response rates of less than 40%. Most patients experience progression within 6-8 months, and reliable biomarkers to predict treatment response are still lacking. This study aimed to assess the clinical outcomes of FGFR inhibitors in patients (pts) with FGFR2-fused iCCA, while investigating tumor vascularity as a potential predictor of treatment efficacy. **Methods:** We retrospectively analyzed data from 134 pts treated at MD Anderson Cancer Center between 2009 and 2024, focusing on 73 pts with comprehensive clinical data. The cohort included pts treated with futibatinib (n=16), infigratinib (n=16), pemigatinib (n=36), and derazantinib (n=5). Tumor vascularity was assessed via Hounsfield units (HU) from CT images, including a mathematical parameter to calculate a ratio of hypervascular tumor rim thickness to tumor diameter. Transcriptomic analysis of 11 pts was performed to explore mRNA expression related to tumor vasculature. Kaplan-Meier analysis for progression-free survival (PFS) and ANOVA for statistical comparisons were applied. **Results:** The median age was 60 years, and 41 pts were female. For FGFR2-fused iCCA, median PFS for first-line gemcitabine-based chemotherapy with or without immunotherapy was 4.07 months [3.0-5.1]. Among FGFR inhibitors, mPFS was 7.07 months [95% CI: 5.10-8.16], with no significant difference in outcomes between FGFR inhibitors. 59 pts had comparable CT images and were grouped into 3 groups based on K-means clustering: Group 1 (n=30) had mPFS of 3.07 months [2.03-4.10]; Group 2 (n=22), mPFS of 9.10 months [8.2-11.2]; Group 3 (n=7), mPFS of 34.03 months [18.2-37.6]. Group 1 had the most hypovascular tumors (the lowest HU in the periphery of tumors), while Group 3 had the most hypervascular tumors (p=0.017). The HU ratio of the hypervascular tumor rim thickness to tumor diameter showed a trend, with the highest ratio in Group 3 and the lowest in Group 1 (p=0.09). Transcriptomic analysis revealed a trend toward higher VEGF/VEGFR gene expression in hypervascular tumors. Eight pts received second-line FGFR inhibitors, with mPFS of 5.0 months [2.1-8.3]. **Conclusions:** Pts with FGFR2-fused iCCA have a shorter PFS on first-line chemotherapy, and PFS outcomes for FGFR inhibitors were consistent with existing data. Tumor vascularity, as assessed by CT imaging, may be a valuable predictor of response to FGFR inhibitors. This study suggests that hypervascular tumors may have better outcomes, warranting further investigation in a larger cohort to confirm these findings and establish tumor vascularity as a predictive biomarker for FGFR inhibitor therapy. Research Sponsor: None.

639 **Poster Session**

Safety and efficacy of anti-HER2 agents in the treatment of biliary tract cancers: A systematic review.

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Background: Limited treatment options exist for patients with locally advanced or metastatic biliary tract cancers (BTCs). Recently, several clinical trials provided preliminary evidence for human epidermal growth factor receptor-2 (HER2) as a new target for patients with HER2 expressing BTC. We conducted a systematic review and pooled analysis of the safety and efficacy of anti-HER2 agents in patients with advanced BTC. **Methods:** A comprehensive search of PubMed/MEDLINE and EMBASE was performed to identify phase 1, 2, or 3 clinical trials published between January 2019 and March 2024 that evaluated anti-HER2 therapy in locally advanced or metastatic BTC. Primary endpoints included objective response rate (ORR), disease control rate (DCR), and progression-free survival (PFS). Secondary endpoints included the incidence of treatment-related adverse events (TRAEs), rate of treatment discontinuation, and death. Forest plots were generated to summarize the results and demonstrate the effect size estimates for each study, along with their confidence intervals and the pooled estimate from the random-effects model. The random-effects models were configured to use the Restricted Maximum Likelihood method for estimating between-study variance for all analyses. **Results:** After excluding duplicate and irrelevant articles, data extraction and analysis were performed on eight selected clinical trials. The analysis included 368 patients with a median age of 64 (range 49-68), with 55% being female. Patients were treated with several anti-HER2 agents, including zanidatamab, pertuzumab plus trastuzumab, tucatinib plus trastuzumab, trastuzumab deruxtecan, trastuzumab plus chemotherapy, trastuzumab-pkrb plus chemotherapy and neratinib. The pooled ORR and DCR were 34% (95% CI 24-44) and 64% (95% CI 51-77) respectively. The pooled weighted PFS and median OS survival were 4.8 and 9.4 months, respectively. The pooled duration of response was 5.0 months. In our subset analysis, which included only patients who had received ≥1 prior therapy, the pooled ORR was 30% (95% CI 22-39), the pooled DCR was 62% (95% CI 48-75), and the pooled median PFS was 4.4 months. In the study cohort, 82.6% of patients experienced any adverse event, and 32.1% experienced a grade 3-4 adverse event. Only 5.7% of the patients discontinued treatment secondary to TRAEs. **Conclusions:** In patients with HER2 expressing BTCs, anti-HER2 therapies demonstrated preliminary signals of efficacy with manageable toxicities, particularly in the second-line setting. Research Sponsor: None.

640 **Poster Session**

Economic burden of hepatocellular carcinoma recurrence after curative treatment in Spain.

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Background: The literature on burden of disease studies of hepatocellular carcinoma is scarce. The aim of this analysis was to quantify the economic burden of disease recurrence in very early or early-stage HCC patients who underwent liver resection, ablation or transplantation in Spain. **Methods:** A decision-tree model was developed to estimate the cost of a recurrence after curative treatment with an initial complete response of very early or early-stage primary HCC tumor over a 5-year time horizon. Clinical parameters were collected by a multidisciplinary group of experts who also validated the assumptions and results of the analysis. According to the Barcelona Clinic Liver Cancer (BCLC) recommendations, patients with recurrent HCC were classified into five HCC stages (0, A, B, C, and D) and allocated to different treatment options. A payer perspective was adopted, so only direct costs (€2024) were considered (local/locoregional treatments costs, drug acquisition and administration costs, adverse events management costs, diagnostic and follow-up costs and end-of-life costs). Sensitivity analyses were performed to assess uncertainty. **Results:** A total of 1,250 HCC recurrences were estimated in Spain, resulting in a total cost of €32,817,142 in the first year of recurrence management. After 5-years follow-up, an additional €10,1M was estimated, mainly due to patient surveillance (€2,8M) and 228 new recurrences (€5,7M). The average cost of a relapse was €26,253 euros in the first year of recurrence management, rising to €34,342 euros for 5-years follow-up. A hypothetical patient experiencing a second relapse 5 years after their first relapse would incur costs of €58,758. **Conclusions:** This study is the first in Spain to explore the economic burden of hepatocellular carcinoma. Estimating the cost of recurrence after curative treatment and complete response of a primary HCC tumor is complex and depends on multiple factors, so further studies exploring both efficacy and costs in hepatocellular carcinoma are needed. Research Sponsor: Roche.

641 Poster Session

Alteration of gut microbiota in patients with advanced hepatocellular carcinoma.

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Background: Hepatocellular carcinoma (HCC) is the most common primary liver cancer and the third leading cause of cancer-related death worldwide. Despite advancements in treatment, such as novel combinations of immunotherapy and targeted therapy or dual immunotherapy, the prognosis remains poor due to a lack of predictive biomarkers for treatment response. Gut microbiota alterations have been proposed as both potential diagnostic biomarker and predictive biomarker. However, data in this field are not well-established and are mostly derived from studies conducted in China and Western countries, which may not fully reflect the situation in Thailand. **Methods:** Fecal samples were collected from pre-treated advanced HCC patients who visited medical oncology outpatient clinic at Srinagarind hospital, Khon Kaen University (HCC group = 27) and analyzed using 16S rRNA sequencing of gut microbiota. Additional sequenced data from a healthy population (Control group = 31) who had no underlying disease and had normal liver ultrasonography were retrieved from a previous study conducted by Khon Kaen University (Cholangiocarcinoma Screening and Care Program, CASCAP). The datasets were compared using the Wilcoxon rank-sum test to analyze the alteration of gut microbiota between HCC group and control group. **Results:** The HCC group exhibited significantly lower biodiversity index ($p < 0.001$) than control group in terms of richness, Shannon diversity index and Simpson's index. The HCC group had significantly higher relative abundance of the phylum Proteobacteria ($p < 0.001$), Firmicutes ($p < 0.001$) and a lower abundance of the phylum Actinobacteria ($p < 0.001$) compared to the control group. Additionally, the HCC group had significantly higher relative abundance of Stenotrophomonas, Granulicatella, Ruminococcus, Blautia, Phascolarctobacterium, Butyricoccus, Streptococcus, Escherichia-Shigella, Flavonifractor, Haemophilus and Lachnospira at the genus level. Further analysis of relative abundance of gut microbiota at the genus level and clinical prognostic parameters showed positive correlation between Streptococcus and 6-month survival status. **Conclusions:** Gut microbiota profile in patients with advanced HCC was significantly altered from healthy control group. This could be the evidence of microbial dysbiosis and may potentially be biomarker for prognosis. Research Sponsor: Faculty of Medicine, Khon Kaen University, Thailand.

Total cost of care of immune-oncology treatments for unresectable hepatocellular carcinoma: A Singapore healthcare perspective analysis.

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Background: Tremelimumab + durvalumab (STRIDE) and atezolizumab + bevacizumab are recommended first-line immune-oncology treatment options for unresectable hepatocellular carcinoma (uHCC). This study aimed to quantify the total economic burden to the Singapore healthcare system of treating patients with uHCC using STRIDE *versus* atezolizumab + bevacizumab. **Methods:** A model was developed to estimate the total cost of care of using STRIDE or atezolizumab + bevacizumab, which followed patients with uHCC eligible for first-line systemic therapy, considering direct medical costs. Costs considered in the model included treatment acquisition and administration, endoscopy, healthcare resource use, and cost for treating grade ≥3 treatment-emergent adverse events (TEAEs). Data on treatment regimens and TEAEs were taken from the HIMALAYA (STRIDE) and IMbrave150 (atezolizumab + bevacizumab) trials. Resource use rates were based on an average of 168 days treatment for both regimens. All costs were obtained from Singaporean sources. **Results:** The total cost of care for STRIDE and atezolizumab + bevacizumab was estimated at SGD$40,277 (USD$30,932) and SGD$43,605 (USD$33,487) *per* patient, respectively. This translates into an average cost savings of SGD$3,328 (USD$2,555]) *per* patient when treated with STRIDE compared to atezolizumab + bevacizumab. The main driver of cost savings was treatment acquisition and administration (-SGD$1,917 [-USD$1,472], -6%) though lower costs were also associated with resource use (-SGD$927 [-USD$712], -8%), endoscopy (-SGD$342 [-USD$263], -77%) and TEAE management (-SGD$141 [-USD$108], -74%). Assuming an incident population of 260 patients eligible for treatment, the total burden was estimated at SGD$10,489,604 (USD$8,055,696) when all patients were treated with STRIDE, and SGD$11,356,273 (USD$8,721,271) when treated with atezolizumab + bevacizumab (Table). The incremental cost offset to the health plan was SGD$866,669 (USD$665,575) when treated with STRIDE. **Conclusions:** This study suggests that STRIDE offers cost savings over atezolizumab + bevacizumab for uHCC treatment in the Singapore healthcare system. Additional real-world evidence studies are warranted to capture the long-term survival benefits and associated treatment costs. Research Sponsor: AstraZeneca.

Annual cost of care based on 260 patients receiving first-line immune-oncology treatment.

Category	STRIDE	Atezolizumab + Bevacizumab	Incremental difference
Endoscopy costs	$26,358	$115,555	-$89,197
Resource Use costs*	$2,607,000	$2,848,307	-$241,307
Adverse events costs	$12,899	$49,716	-$36,817
Treatment acquisition and administration	$7,843,347	$8,342,696	-$499,349
TOTAL	**$10,489,604**	**$11,356,273**	**-$866,669**

All costs in SGD$. *Includes: physician visits; laboratory and radiological tests; and hospitalisation.

Obesity and biliary tract cancers: Changing epidemiology.

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Background: The incidence of biliary tract cancers (BTC) continues to rise (Sung et al. 2024). The reasons remain ill-defined. Over 70% of patients are diagnosed without identifiable predisposing factors. Obesity and overweight have emerged as putative contributors. We examined the epidemiologic and molecular characteristics of patients with BTC, with a focus on body mass index (BMI) at presentation and overall survival (OS). **Methods:** A retrospective cohort study of patients diagnosed with BTC at Memorial Sloan Kettering was conducted from January 2022 to December 2023. Patients were divided based on BMI into: obese (BMI≥30 kg/m^2) and non-obese (BMI<30 kg/m^2). We collected patients' baseline demographics, and results of somatic next generation sequencing using the MSK-IMPACT platform and compared between obese and non -obese groups using Fisher's exact test for categorical and Wilcoxon rank sum test for continuous variables. OS was calculated from date of diagnosis, estimated using Kaplan-Meier methods and compared between groups using log rank test. **Results:** A total of 142 patients were identified. The table summarizes the baseline characteristics including age, BMI, sex, stage at diagnosis, family history, history of malignancy, and race. Obesity was significantly associated with family history of malignancy and white race. Among surviving patients (n=93), with a median follow-up of 13.8 months, 6-month, 1-year, and 2-year overall survival (OS) rates were 85% [95%CI: 80%-92%], 71% [95%CI: 64%-79%], and 54% [95%CI: 43%-68%] respectively. OS was not statistically different among obese and non-obese patients. However, when stratifying OS by BMI and gender, obese males were noted to have inferior OS compared to obese females (6-mo OS 75% and 90% respectively), though not statistically significant. Genetic sequencing did not reveal a specific molecular pattern in either group. **Conclusions:** Obesity appears not to be associated with young age, stage, or particular somatic alterations in this limited study. OS was not statistically different among obese and non-obese patients. Males with obesity appeared to have a worse prognosis. This observed gender-based difference needs to be further studied, and if confirmed further research is needed to explore potential factors that may explain the clinical observation. Research Sponsor: None.

Baseline characteristics.

Characteristics	Overall n=142	Obese (BMI≥30 kg/m^2) n=54	Non-Obese (BMI<30 kg/m^2) n=88	p-value
Age	68 (35-91)	69 (35-87)	67 (36-91)	>0.9
BMI (kg/m^2)	28.1 (16.6-48.5)	33.5 (30-48.5)	24.7 (16.6-29.6)	
Sex female/male	79 (56%)/63 (44%)	32 (59%)/22 (41%)	47 (53%)/41 (47%)	0.6
Stage Localized/advanced	21 (15%)/121 (85%)	11 (20%)/ 43 (80%)	10 (11%)/ 78 (89%)	0.3
Family history of cancer	91 (65%)	41 (76%)	50 (57%)	0.030
History of second malignancy	26 (18%)	12 (22%)	14 (16%)	0.4
Race White	99 (70%)	44 (81%)	55 (63%)	0.004
African	12 (8.5%)	4 (7.4%)	8 (9.1%)	
Asian	20 (14%)	1 (1.9%)	19 (22%)	
Other	11 (7.7%)	5 (9.3%)	6 (6.8%)	

Utilization of circulating tumor DNA (ctDNA) testing in biliary tract cancers.

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Background: There is limited data to guide surveillance in biliary tract cancers (BTCs) following resection. Minimal residual disease (MRD) assays utilize circulating tumor DNA (ctDNA) to detect early recurrence. Current data is limited on the use of ctDNA in biliary tract cancers in conjunction with surveillance imaging. In this study, we aim to elucidate the role of ctDNA monitoring in patients with biliary tract cancers who have undergone curative resection. **Methods:** We conducted a retrospective search at a tertiary care and several satellite settings to identify patients with BTCs who underwent curative surgery and subsequently had ctDNA monitoring. We collected demographic variables, disease characteristics, treatment regimens, and follow-up data. The primary endpoint of the analysis is recurrence-free survival (RFS), comparing patients with detectable ctDNA to those with negative ctDNA post-surgery. Kaplan-Meier analysis and Log-Rank tests were performed to evaluate RFS, and Cox proportional hazards(PH) models were employed for both univariate and multivariate analyses. **Results:** We identified 36 pts with a median age of 72 (34-87) yrs at diagnosis. In this study cohort, 22 (59%) were female, 32 (88%) were white, and 3 (8%) were African American. 12 (33%) pts were ctDNA positive at any point during surveillance, while the remainder were ctDNA negative. In the ctDNA-positive group, recurrences developed in 7 out of 12 (58%) pts. In the ctDNA negative group, only 2 out of 24 (8.3%) pts developed recurrences. The median time to recurrence following primary surgery was 10.7 months, while the time to ctDNA positivity following surgery was 7.8 months. The median time from ctDNA positivity to recurrence was 1.1 months. Pts with ctDNA positivity had a significantly lower 1-year RFS of 47.62% (95% CI: 7.5%-81%) compared to those with ctDNA negativity (85%, 95% CI: 60%-95%; p = 0.0008). Multivariate analysis demonstrated that ctDNA positivity at any point was associated with a significantly higher risk of recurrence (HR: 21.8, 95% CI: 2.3–203, $p<0.0001$). **Conclusions:** In pts with BTCs who underwent resection, ctDNA testing had independent prognostic value. Further prospective trials are needed to help define the role of ctDNA in this patient population. Research Sponsor: None.

Baseline demographics and clinical characteristics of patients who were ctDNA positive vs. negative.

Characteristics	Patients with Biliary Tract Cancers who were ctDNA positive	Patients with Biliary Tract Cancers who were ctDNA negative
N	12	24
Age in years, median (range)	72 (62 – 87)	72 (34 - 85)
Sex n (%)		
Male	7 (58)	8 (33)
Female	5 (42)	16 (67)
Race n (%)		
White	12 (100)	20 (83)
Black/African American	0 (0)	3 (13)
Asian	0 (0)	1 (4)
Primary Tumor Location n (%)		
ICC	4 (33)	11 (46)
ECC	4 (33)	5 (21)
GBC	2 (17)	6 (25)
AC	2 (17)	2 (8)
Margin Status n (%)		
Positive	2 (17)	3 (13)
Negative	10 (83)	21 (87)
Clinical Stage		
I	0 (0)	1 (4)
II	2 (17)	12 (50)
III	8 (66)	10 (43)
IV	2 (17)	0 (0)
Unknown	0 (0)	1 (3)

TPS645 **Trials in Progress Poster Session**

A phase IIa clinical trial of first-line cyclical therapy alternating gemcitabine, cisplatin, and durvalumab with pemigatinib for advanced biliary tract cancers with *FGFR2*-alterations.

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Background: Unresectable biliary tract cancer (BTC) has a poor prognosis. Frontline chemo-immunotherapy (C-IO) for advanced BTC leads to ~25% overall survival (OS) at 24-month, but 93-99% of patients reported adverse events (AEs), and 13-19% of patients stopped treatment drugs due to AEs. FGFR kinase inhibitors (FGFRi) are second-line therapy for 10% of BTC with *FGFR2* fusions. However, clinical trials of upfront FGFRi have not accrued well. Cumulative toxicities frequently interrupt treatment and limit feasibility of concurrent C-IO + FGFRi. Over time, side effects from each of therapies have a negative impact on patients' quality of life (QoL). We previously treated two *FGFR2*-altered BTC patients with cyclical chemotherapy (CT) and FGFRi and observed prolonged survival with better QoL. By monitoring serial circulating tumor DNA (ctDNA), we observed development of FGFRi-resistant subclones while on FGFRi and their subsequent clearance after resuming CT. Thus, we have developed a clinical trial to validate our hypothesis that upfront cyclical therapy alternating between C-IO and FGFRi can improve therapeutic efficacy and QoL. Alternating therapy is hypothesized to avoid accumulating toxicities and treat emerging resistance subclones. **Methods:** In this single-center, single-arm, Phase IIa trial, all treatment-naive patients with *FGFR2*-altered advanced BTCs, PS $\leq$ 1, are eligible. Patients will receive cyclical therapy alternating between two therapy-blocks, starting from C-IO block (2 cycles of gemcitabine, cisplatin and durvalumab, 28-day/cycle), then FGFRi block (3 cycles of pemigatinib on day 1-14 of 21-day/cycle). Block-switch continues until progressive disease (PD), and then we will continue alternated block until the next PD. A sample size of 30 is calculated based on an expected 12-month OS rate of $\geq$ 55%. The trial is anticipated to enroll the first patient in early 2025. **Results:** The primary objective is 12-month OS rate. Secondary objectives are overall response rate, progression-free survival, AEs. We will stratify patients to PD-L1 CPS $<$1 or $\geq$ 1 to assess if cyclical therapy can enhance IO. Exploratory objectives are (1) assessing dynamics of *FGFR2*-ctDNA variants throughout therapy correlating with clinical outcomes, using a custom liquid biopsy (FGFR-Dx) developed at the Ohio State University; (2) assessing QoL scores with EORTC-QLQ-C30 and -BIL21 questionnaires, correlating with clinical outcomes. Results will be compared to historical controls from TOPAZ-1 trial. **Conclusions:** Cyclical therapy will be one of the first to combine alternating C-IO and targeted therapy in cancer. This strategy can be validated in other tumors if it is proven effective in our trial. Cyclical therapy holds great promise to introduce FGFRi earlier into treatment for BTC and improve both efficacy and QoL. Research Sponsor: Incyte Corporation (pharmaceutical company).

TPS646 **Trials in Progress Poster Session**

A multicenter, open-label, first-in-human study of TYRA-200 in advanced intrahepatic cholangiocarcinoma and other solid tumors with activating *FGFR2* gene alterations (SURF201).

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Background: Approved Fibroblast Growth Factor Receptor (FGFR) inhibitors have demonstrated clinical benefit in locally advanced or metastatic cholangiocarcinoma harboring oncogenic *FGFR2* fusions or other rearrangements. However, the emergence of acquired resistance mutations limits the clinical activity and duration of responses to currently available inhibitors. TYRA-200 is an orally bioavailable FGFR1/2/3 inhibitor designed to specifically address these clinically observed acquired resistance alterations in *FGFR2.* **Methods:** SURF201 is a single arm, multicenter, open-label, first-in-human, dose escalation and expansion study designed to investigate TYRA-200 in patients (pts) with advanced intrahepatic cholangiocarcinoma and other solid tumors with primary activating *FGFR2* gene alterations and on-target acquired known *FGFR2* resistance mutations. The study is being conducted in two parts. Part A dose escalation uses an i3+3 design and is evaluating the safety, tolerability, and the pharmacokinetic profile of TYRA-200. Part B dose expansion will further characterize the safety profile and evaluate the preliminary anti-tumor activity of TYRA-200 by RECIST v1.1 in pts with unresectable locally advanced/metastatic intrahepatic cholangiocarcinoma who have previously received an FGFR inhibitor(s) and have developed acquired *FGFR2* kinase domain resistance mutations. The study is currently planned for approximately four centers in the US and is actively enrolling (NCT06160752). Clinical trial information: NCT06160752. Research Sponsor: Tyra Biosciences, Inc.

TPS647 **Trials in Progress Poster Session**

KIRROS: A phase II trial in progress of first-line atezolizumab (atezo) with or without bevacizumab (bev) in patients with unresectable hepatocellular carcinoma (HCC) and Child-Pugh B (CPB) cirrhosis.

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Background: HCC treatment is more complex for the sizable proportion of patients with CPB cirrhosis. These patients have poor survival due to both the cancer and underlying liver dysfunction. As such, they are typically excluded from pivotal clinical trials for first-line systemic therapy for unresectable HCC. The IMbrave150 trial established atezo + bev as a global standard of care for unresectable HCC, demonstrating a statistically significant improvement in overall survival (OS), progression-free survival (PFS) and confirmed objective response rate (ORR) of atezo + bev vs sorafenib (Finn N Engl J Med 2020; Cheng J Hepatol 2022). Real-world studies have consistently suggested benefit of atezo + bev in patients with HCC and CPB cirrhosis (e.g., Tanaka Hepatol Res 2022, D'Alessio Hepatol [Baltimore, Md] 2022). However, prospective clinical trial data are needed to comprehensively characterize safety and efficacy in this population. Single-agent immunotherapy is often used in patients with HCC and CPB cirrhosis given potential safety concerns of combination therapy. Safety data from real-world studies are limited by ascertainment and measurement biases. Therefore, a prospective study is warranted to address the unmet need for safety and efficacy data for first-line atezo ± bev in patients with CPB cirrhosis. **Methods:** KIRROS (NCT06096779) is a prospective, open-label, multicohort, multicenter phase II study in patients aged ≥18 years who have measurable and confirmed unresectable HCC, and CPB-7 or -8 cirrhosis who have not received prior systemic therapy. Eligible patients will be recruited into 2 cohorts for which endpoints will be non-comparatively assessed: Cohort A (n=60) patients will be treated with atezo 1200 mg intravenous (IV) + bev 15 mg/kg every 3 weeks (Q3W), and Cohort B (n=60) patients, those who do not meet Cohort A inclusion criteria (e.g., main portal vein tumor thrombus and proteinuria), will receive atezo 1200 mg IV Q3W only. Patients will receive treatment until unacceptable toxicity or loss of clinical benefit. The primary objective is to evaluate the safety of the treatment regimens as determined by incidence, nature and severity of adverse events and laboratory abnormalities graded per the National Cancer Institute Common Terminology Criteria for Adverse Events v5.0. Secondary objectives are to estimate efficacy with ORR, duration of response, and PFS per Response Evaluation Criteria in Solid Tumors (RECIST) 1.1 and HCC modified RECIST, as well as OS. Quality-of-life secondary endpoints include patient-reported tolerability of the treatment regimens assessed by PRO-CTCAE and EORTC IL46, and health-related quality of life per EORTC QLQ-C30 and QLQ-HCC18 scores. As of September 2024, the KIRROS study is recruiting patients at 40 sites in the US and Puerto Rico. Clinical trial information: NCT06096779. Research Sponsor: Genentech Inc.

TPS648 **Trials in Progress Poster Session**

HERIZON-BTC-302: A phase 3 study of zanidatamab with standard-of-care (SOC) therapy vs SOC alone for first-line treatment of human epidermal growth factor receptor 2 (HER2)-positive advanced/metastatic biliary tract cancer (BTC).

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Background: Zanidatamab is a humanized, IgG1-like, HER2-targeted bispecific antibody that simultaneously binds to 2 non-overlapping domains on HER2. Zanidatamab is being investigated for the treatment of HER2-expressing solid tumors, including BTC. In the phase 2 HERIZON-BTC-01 trial, zanidatamab monotherapy demonstrated promising antitumor activity in 80 patients with previously treated HER2-positive BTC. Confirmed objective response rate (cORR) was 41.3% with rapid and durable responses and a manageable safety profile. This phase 3 trial is assessing zanidatamab + SOC therapy vs SOC alone for first-line treatment of HER2-positive advanced/metastatic BTC. **Methods:** This ongoing, global, phase 3, randomized, open-label trial (NCT06282575) is investigating the efficacy and safety of zanidatamab with cisplatin and gemcitabine (CisGem) vs CisGem alone ± a programmed cell death protein-1/ligand 1 (PD-1/L1) inhibitor (pembrolizumab or durvalumab at physician's discretion if locally approved) as first-line treatment for patients with advanced HER2-positive BTC. Eligibility criteria include ≥18 years of age; locally advanced, unresectable, or metastatic HER2-positive BTC by immunohistochemistry and in situ hybridization assay (IHC 3+ or IHC 2+/ISH+); and Eastern Cooperative Oncology Group performance status ≤1. Patients may have received ≤2 cycles of a gemcitabine-based regimen ± pembrolizumab or durvalumab. Prior HER2-targeted therapy is not allowed except for patients who completed it for breast cancer >5 years prior to BTC diagnosis. Eligible patients will be randomized to receive zanidatamab (flat 2-tiered dosing: 1800 mg intravenous [IV; body weight <70 kg] or 2400 mg IV [body weight ≥70 kg] every 3 weeks) + a standard dose of CisGem ± a PD1/L1 inhibitor or CisGem alone ± a PD1/L1 inhibitor (≤8 cycles). The primary endpoint is progression-free survival (PFS) in patients with IHC 3+ tumors. Secondary/exploratory endpoints include overall survival (IHC 3+ subgroup; overall population), PFS (overall population), cORR, incidence and severity of adverse events, and patient-reported outcomes. The study is currently recruiting patients. Clinical trial information: NCT06282575. Research Sponsor: Jazz Pharmaceuticals.

TPS649 **Trials in Progress Poster Session**

A multicenter, open-label study investigating RP2 oncolytic immunotherapy in combination with second-line systemic atezolizumab plus bevacizumab in patients with locally advanced unresectable or metastatic hepatocellular carcinoma (HCC).

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Background: Despite advances in unresectable HCC treatment, long-term survival rates remain poor. The combination of atezolizumab (Atezo) plus bevacizumab (Bev) is approved as frontline therapy for advanced HCC, but only a minority of patients (pts) respond, and secondary resistance usually occurs within months. HCC has an immune-suppressed tumor microenvironment (TME) mediated by the expression of immune checkpoint signals and angiogenesis pathways, which may contribute to therapeutic resistance. RP2 is an enhanced potency oncolytic herpes simplex virus type 1 (HSV-1) that expresses GM-CSF, a fusogenic glycoprotein (GALV-GP-R−), and an anti–CTLA-4 antibody-like molecule. RP2 showed preliminary clinical activity alone or combined with anti–PD-1 in a phase 1 study in pts with advanced solid tumors. The direct oncolytic effect coupled with immune stimulation by RP2 in the TME is intended to provide systemic antitumor activity and synergize with anti–PD-1/PD-L1 agents, such as Atezo. Preclinical data have demonstrated improved distribution of oncolytic HSV within tumors when administered with Bev, supporting the clinical combination of RP2 with Bev. This study will evaluate the safety and efficacy of RP2 combined with Atezo plus Bev as second-line systemic therapy for unresectable advanced HCC (NCT05733598). **Methods:** This is an open-label, single arm, phase 2 trial. Up to 30 pts will be enrolled and receive RP2 in combination with Atezo plus Bev. Key inclusion criteria include advanced unresectable HCC with ≥1 measurable tumor of ≥1 cm in longest diameter, Child-Pugh class A, an Eastern Cooperative Oncology Group performance status of 0 to 1, and progression on 1 prior systemic treatment, which must have included a PD-1/PD-L1–directed agent. Key exclusion criteria include untreated/incompletely treated esophageal and/or gastric varices with bleeding or at high risk for bleeding and macroscopic invasion of the tumor into any major blood vessel(s) and/or main bile ducts. Pts will receive intratumoral RP2 Q2W for 4 doses, then Q3W for up to 4 doses. Bev will be given at 10 mg/kg Q2W starting with the first dose of RP2, then at 15 mg/kg Q3W starting with cycle 4; Atezo will be given at 840 mg Q2W for cycles 2 and 3, then at 1200 mg Q3W starting with cycle 4. Pts will receive treatment until confirmed progressive disease, loss of clinical benefit, or unacceptable toxicity. The primary endpoint is overall response rate (ORR), defined as the proportion of pts achieving a best overall response of complete response or partial response per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1 as modified for this study. Secondary endpoints include safety, ORR using HCC-modified RECIST, duration of response, complete response rate, and progression-free survival. Clinical trial information: NCT05733598. Research Sponsor: Replimune, Inc.

AMBER part 2F: Cobolimab in combination with dostarlimab in treatment-naïve patients with locally advanced/metastatic and/or unresectable hepatocellular carcinoma (HCC).

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Background: The introduction of targeted treatment (tx) and immune checkpoint inhibitors (ICIs) has transformed the 1L advanced/metastatic (A/M) HCC tx landscape.[1] Combinations of ICIs with anti-angiogenic agents or other ICIs have demonstrated synergism in patients (pts) with A/M HCC, and shown improved efficacy versus single agents or tyrosine kinase inhibitors.[1,2] Combinations with anti-T-cell immunoglobulin and mucin-domain containing-3 (TIM-3) and anti-programmed cell death protein-1 (PD-1) ICIs have shown promising anti-tumor activity in pts with solid tumors[3,4] including advanced HCC in US pts.[5] AMBER (NCT02817633) part 2F will evaluate the efficacy and safety of anti-TIM-3 cobolimab plus anti-PD-1 dostarlimab in tx-naïve pts with A/M and/or unresectable HCC. **Methods:** AMBER is a two-part, global, open-label, Phase Ib study assessing cobolimab as monotherapy or in combination with other drugs in pts with advanced solid tumors. For part 2F, ~45 pts are planned for enrollment. Key eligibility includes pts aged ≥18 years, with Eastern Cooperative Oncology Group (ECOG) performance score 0–1, histologically confirmed A/M HCC with measurable disease, Child-Pugh Class A liver function, no prior systemic tx, documented hepatitis B and C test at screening, and a pre-tx biopsy sample. Eligible pts will receive cobolimab (300 mg IV) plus dostarlimab (500 mg IV) every 3 weeks (Q3W) for up to 2 years or until progression, toxicity, discontinuation, or death. The primary endpoint is investigator-assessed objective response rate (ORR) per Response Evaluation Criteria in Solid Tumors version 1.1 (RECIST v1.1); secondary endpoints include investigator-assessed disease control rate (DCR), duration of response (DOR) and progression-free survival (PFS) per RECIST v1.1, and overall survival, alpha-fetoprotein response, and safety. Exploratory endpoints include immune-related (ir)-ORR, irDCR, irDOR and irPFS per irRECIST, and biomarker and pharmacokinetic assessments. Disease assessments will consist of CT/MRI evaluations of the chest, abdomen, and pelvis, Q9W from first dose then Q12W after 1 year of tx. Pts will undergo safety follow-up at 30 and 90 (±7) days after last study tx. Efficacy analysis will be based on the safety population (pts who receive ≥1 cobolimab dose) and will include summary statistics and point estimates with 2-sided 95% confidence intervals. Time-to-event analyses will be performed via Kaplan–Meier methods. After ~25 pts have been followed-up for ≥3 scans, an interim analysis may be performed. References: 1. Gordan JD, et al. *J Clin Oncol*. 2024:42:1830–50; 2. Finn RS, et al. *N Engl J Med*. 2020;382:1894–1905; 3. Curigliano G, et al. *Clin Cancer Res.* 2021;27: 3620–29; 4. Davar D, et al. *J ImmunoTher Cancer*. 2023:11(Suppl 1):596; 5. Acoba JD, et al. *J Clin Oncol*. 2023:41:580. Funding: GSK (213348). Clinical trial information: NCT02817633. Research Sponsor: GSK; 213348.

TPS651 **Trials in Progress Poster Session**

A phase II open-label study of sacituzumab govitecan in patients with previously treated locally advanced, recurrent, or metastatic cholangiocarcinoma (SIGNA).

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Background: Cholangiocarcinoma (CCA) is a highly aggressive cancer with limited treatment options. Trop-2, a transmembrane calcium signal transducer, plays a key role in cellular self-renewal, proliferation, and transformation, thereby promoting tumorigenesis. Its overexpression has been associated with disease progression and shorter survival in several epithelial tumors, including CCA. Sacituzumab govitecan (SG) is an antibody-drug conjugate composed of a humanized anti-trophoblast cell-surface antigen 2 (Trop-2) monoclonal antibody coupled to SN-38, the active metabolite of the topoisomerase inhibitor, irinotecan. Based on the results from the Phase III ASCENT trial, SG was approved by the FDA for treating relapsed or refractory triple-negative breast cancer and hormone receptor-positive breast cancer, regardless of Trop-2 expression. Given Trop-2's expression in CCA, SG is being investigated as a potential treatment option in this setting. **Methods:** This Phase II, non-randomized, open-label, single-arm study is being conducted at the University of Kansas Cancer Center and its affiliated sites. *Eligibility*: Key inclusion criteria include patients with locally advanced, recurrent, or metastatic CCA who have undergone at least one line of prior therapy, have adequate archival tissue for biomarker evaluation or are willing to undergo a biopsy, and have an ECOG performance status of 0-1. Key exclusion criteria include known homozygosity for the UGT1A1*28 allele, which is associated with irinotecan toxicity, and prior treatment with topoisomerase I inhibitors. *Treatment Plan*: SG is administered at 10 mg/kg intravenously on Days 1 and 8 of every 21-day cycle. Imaging will be performed every 6 weeks. Biopsies will be required both prior to treatment and on-treatment (between Days 15-21 of Cycle 1). *Objectives*: Primary objective: To determine the anti-tumor activity of SG as measured by the overall response rate (ORR) according to RECIST 1.1. Secondary objectives: To assess treatment safety in participants who have received at least one dose of SG, and to evaluate anti-tumor activity based on progression-free survival (PFS), disease control rate (DCR), and overall survival (OS). *Statistical Plan*: The sample size (n=22) was calculated using a modified Simon two-stage design, hypothesizing an improvement in ORR to 25%, compared to a baseline value of 8%. In the first stage, 14 patients will be enrolled. If ≤1 participant demonstrates an ORR, the study will be stopped for futility. If ≥2 participants show ORR, an additional 8 patients will be enrolled (total n=22). The study will be deemed successful if ≥4 participants achieve an ORR. This design provides 80.9% power with an 8.5% type I error rate. Stopping rules include halting the trial if the grade 4-5 treatment-related toxicity rate reaches 10%. Enrollment is ongoing. Clinical trial information: NCT06178588. Research Sponsor: Gilead.